Supplementary Material for "Model Averaging for Conditional Quantile Treatment Effects under Model Uncertainty"

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The Supplementary Material are organized as follows: Section A gives the additional notations and assumptions; Section B provides detailed proofs of the theorems; Section C presents additional simulation results; and Section D contains further results from the real data analysis.

A Notations and Assumptions

A.1 Notations and Assumptions for Theorem 1

To proceed, we first give some notations. Let $f_d(\cdot|\boldsymbol{X}_i)$ and $F_d(\cdot|\boldsymbol{X}_i)$ denote the conditional probability density function (PDF) and cumulative distribution function (CDF) of $\varepsilon_i(d,\tau)$ given $\boldsymbol{X}_i$, respectively for $d=0,1$. Further, denote $f_{\eta^*}(\cdot|\boldsymbol{X}_i)$ and $F_{\eta^*}(\cdot|\boldsymbol{X}_i)$ as the conditional PDF and CDF of η_i^* given $\boldsymbol{X}_i$, where $\eta_i^*(\tau)=\varepsilon_i(1,\tau)-\varepsilon_i(0,\tau)$. For simplicity, we suppress the dependence of $\eta_i^*(\tau)$ on τ. Denote $f_{Y(d)|\boldsymbol{X}}(\cdot|\boldsymbol{X}_i)$ as the conditional PDF of $Y_i(d)$ given $\boldsymbol{X}_i$ for $d=0,1$. Define the pseudo-parameter under treatment d as

$$\boldsymbol{\beta}_d^{*(m)}(\tau)=\arg\min_{\boldsymbol{\beta}\in\mathbb{R}^{pm}}\mathsf{E}\Big[\rho_\tau\Big(Y_i-\boldsymbol{X}_i^{(m)\top}\boldsymbol{\beta}\Big)\Big|D_i=d\Big]. \tag{S.1}$$

For $m=1,2,\ldots,M$, we define

$$\begin{aligned}A_{d,(m)}&=\mathsf{E}\Big[f_d(-u_{i,d,(m)}|\boldsymbol{X}_i)\boldsymbol{X}_i^{(m)}\boldsymbol{X}_i^{(m)\top}\Big|D_i=d\Big]\\&=\mathsf{E}\Big[f_{Y(d)|\boldsymbol{X}}(\boldsymbol{X}_i^{(m)\top}\boldsymbol{\beta}_d^{*(m)}(\tau)|\boldsymbol{X}_i)\cdot\boldsymbol{X}_i^{(m)}\boldsymbol{X}_i^{(m)\top}\Big|D_i=d\Big],\end{aligned}$$

and

$$B_{d,(m)} = \mathsf{E}\Big[\psi_\tau^2\left\{\varepsilon_i(d,\tau) + u_{i,d,(m)}\right\} \cdot \boldsymbol{X}_i^{(m)} \boldsymbol{X}_i^{(m)\top} \Big| D_i = d\Big],$$

where $u_{i,d,(m)} = Q_d(\tau; \boldsymbol{X}_i) - \boldsymbol{X}_i^{(m)\top} \boldsymbol{\beta}_d^{*(m)}(\tau)$ and $\psi_\tau(u) = \tau - \mathbf{1}(u \leq 0)$. Let $\overline{p} = \max_{1 \leq m \leq M} p_m$. For a matrix $\boldsymbol{A}$, let $\lambda_{\max}(\boldsymbol{A})$ and $\lambda_{\min}(\boldsymbol{A})$ denote its largest and smallest eigenvalues, respectively.

To characterize heterogeneity in individual treatment effects, we introduce a rank-based representation of the potential outcomes. Conditional on the observed characteristics $\boldsymbol{X} = \boldsymbol{x}$, the potential outcomes $Y(0)$ and $Y(1)$ satisfy $Y(d) = Q_d(U_d; \boldsymbol{x}), d = 0, 1$ for some $U_d \sim U(0,1)$. Moreover, the individual treatment effect $\alpha^* = Y(1) - Y(0)$ satisfies $\alpha^* = Q_{\alpha^*}(U_{\alpha^*}; \boldsymbol{x})$ for some $U_{\alpha^*} \sim U(0,1)$, where $Q_{\alpha^*}(\tau; \boldsymbol{x})$ denotes the conditional quantile function of α^* given $\boldsymbol{X} = \boldsymbol{x}$. We will refer to U_d and U_{α^*} as rank variables, which determine the relative ranking of individuals in terms of potential outcomes and treatment effects.

We make the following additional assumptions for Theorem 1.

Assumption S.1. *(Rank Invariance)*

(i) $Q_d(\tau; \boldsymbol{x}), d = 0, 1$, *and* $Q_{\alpha^*}(\tau; \boldsymbol{x})$ *are strictly increasing in* τ.

(ii) $U_1 = U_0 = U_{\alpha^*}$.

Assumption S.2. *(i)* $\boldsymbol{X}$ *is a* $p \times 1$ *vector of continuous covariates taking values in a compact and convex support* $\mathbb{X} \subset \mathbb{R}^p$, *with density bounded and bounded away from zero on* $\mathbb{X}$.

(ii) $\{Y_i, D_i, \boldsymbol{X}_i\}_{i=1}^n$ *are independent draws from the distribution of* $(Y, D, \boldsymbol{X})$.

(iii) For each $d = 0, 1$ *and* $\tau \in (0,1)$, *the conditional quantile function* $Q_d(\tau \mid \boldsymbol{x})$ *is Lipschitz continuous in* $\boldsymbol{x}$ *on* $\mathbb{X}$.

Assumption S.3. *(i)* $\{Y_i(d)\}_{i=1}^n$ *is i.i.d. for each* $d = 0, 1$.

(ii) For $d = 0, 1$, $P(\varepsilon_i(d,\tau) \leq 0 | \boldsymbol{X}_i) = \tau$, $\mathsf{E}[\varepsilon_i^2(d,\tau) | \boldsymbol{X}_i] = O(1)$. *In addition,* $\mathsf{E}[\rho_\tau(\eta_i^*)] > c_0 > 0$ *for some positive constant* c_0.

(iii) For $d = 0, 1$, $\mathsf{E}[Q_d^4(\tau; \boldsymbol{X}_i)] < \infty$, $\|\boldsymbol{\beta}_d^{*(m)}(\tau)\| = O(p_m^{1/2})$, *for* $m = 1, 2, \ldots, M$, *and* $\sup_{j \geq 1} \mathsf{E}(X_{ij}^4) \leq c_x$ *for some constant* $c_x < \infty$.

Assumption S.4. *(i) For* $d = 0, 1$, $f_{Y(d)|\boldsymbol{X}}(\cdot | \boldsymbol{X}_i)$ *is almost surely continuous over its support and bounded above by a finite constant* c_f.

(ii) *There exist positive constants* $\underline{c}_{A(m)}$ *and* $\overline{c}_{A(m)}$*, possibly depending on* p_m*, such that for* $d = 0, 1$*,* $0 < \underline{c}_{A(m)} \leq \lambda_{\min}(A_{d,(m)}) \leq \lambda_{\max}(A_{d,(m)}) \leq c_f \lambda_{\max}\left(\mathsf{E}\left[\boldsymbol{X}_i^{(m)} \boldsymbol{X}_i^{(m)\top}\right]\right) \leq \overline{c}_{A(m)} < \infty$.

(iii) *There exist positive constants* $\underline{c}_{B(m)}$ *and* $\overline{c}_{B(m)}$*, which may depend on* p_m*, such that for* $d = 0, 1$*,* $0 < \underline{c}_{B(m)} \leq \lambda_{\min}(B_{d,(m)}) \leq \lambda_{\max}(B_{d,(m)}) \leq \overline{c}_{B(m)} < \infty$.

(iv) $(\overline{c}_{A(m)} + \overline{c}_{B(m)})/p_m = O(\underline{c}_{A(m)}^2)$.

Assumption S.5. *Let* $\overline{c}_A = \max_{1 \leq m \leq M} \overline{c}_{A(m)}$.

(i) $n^{-1/2} \overline{p}\, \overline{c}_A \log n = o(1)$.

(ii) *There exists some* $\delta \in (0, 1)$*, such that* $n^{\frac{\delta - 1}{2}} (\overline{p} \log n) \log\{\overline{p} \log n\} = o(1)$.

Assumption S.1 requires that, conditional on the observed covariates, the rank variables for the treated outcome, the control outcome, and the individual treatment effect coincide. In other words, the same latent factor determines an individual' s relative position in the distributions of $Y(1)$, $Y(0)$, and α^*. Related assumptions have been widely discussed in Chernozhukov and Hansen (2005, 2006, 2008). Relative to their framework, we further impose a stronger form of rank invariance. In particular, while Chernozhukov and Hansen (2005) assumes rank invariance only for the potential outcomes, we extend this requirement to the treatment effect itself. This stronger condition is to identify the entire distribution of treatment effects, rather than only the marginal quantiles of the potential outcomes. Under Assumption S.1, the CQTE in (1) can be interpreted as the conditional quantile of the treatment effect, which is critical for the validity of our model averaging approach. The plausibility of Assumption S.1 may increase as the conditioning set becomes richer, since conditioning on more relevant covariates can make individuals more comparable in terms of unobserved heterogeneity.

Assumption S.2, which is standard in the matching literature (e.g., Abadie and Imbens, 2006, 2011; Li et al., 2025), is used to derive the asymptotic properties of the matching discrepancy. Assumptions S.3–S.5 are standard regularity conditions in the quantile regression literature (Lu and Su, 2015; Wang et al., 2021; You et al., 2024). Assumption S.3(ii) combines the conditional quantile restriction with a bounded conditional second-moment condition on the errors, while Assumption S.3(i) and S.3(iii) provide the independence and moment conditions needed in the proofs. Assumption S.4(i) allows for conditional heteroskedasticity, and Assumption S.4(ii)–(iii) are standard when regressors and quantile errors may be dependent. Assumption S.4(iv) is the same as that in Lu and Su (2015). It allows $\overline{c}_{A(m)}$ and

$\overline{c}_{B(m)}$ to diverge, and $\underline{c}_{A(m)}$ and $\underline{c}_{B(m)}$ to converge to zero with p_m, provided that these rates are slower than p_m. In addition, this condition ensures the usual $\sqrt{p_m/n}$-consistency of the quantile regression estimator. Assumption S.5 further restricts the growth of $\overline{p}$ and $\overline{c}_A$.

A.2 Notations and Assumptions for Theorem 2

Denote $R_n^*(\boldsymbol{w},\tau) = \mathsf{E}\Big[\rho_\tau\left\{\alpha_{n+1}^* - \Delta^*(\tau;\boldsymbol{X}_{n+1},\boldsymbol{w})\right\} - \rho_\tau(\eta_{n+1}^*)\Big]$, where $\Delta^*(\tau;\boldsymbol{X}_{n+1},\boldsymbol{w}) = \sum_{m=1}^{M} w_m \Delta^{*(m)}(\tau;\boldsymbol{X}_{n+1})$ and $\Delta^{*(m)}(\tau;\boldsymbol{X}_{n+1}) = \boldsymbol{X}_{n+1}^{(m)\top}\left\{\boldsymbol{\beta}_1^{*(m)}(\tau) - \boldsymbol{\beta}_0^{*(m)}(\tau)\right\}$. Let $\xi_n^* = \inf_{\boldsymbol{w}\in\mathcal{W}} R_n^*(\boldsymbol{w},\tau)$. We impose the following additional assumption to demonstrate the asymptotic optimality of the proposed method in the sense of minimizing $R_n(\boldsymbol{w},\tau)$.

Assumption S.6. *(i)* $\xi_n^* > 0$. *(ii)* $(\xi_n^* \log n)^{-1} = o(1)$.

Assumption S.6(i) is a standard assumption. Assumption S.6(ii) requires that ξ_n^* does not decay faster than $(\log n)^{-1}$. This assumption indicates that all candidate models are misspecified (i.e., $\xi_n^* \neq 0$), which is a common requirement in the model averaging literature (Hansen, 2007; Hansen and Racine, 2012; Zhang and Liu, 2023).

A.3 Notations and Assumptions for Theorem 3

We impose the following additional assumptions for Theorem 3.

Assumption S.7. *There exists some* $\lambda > 0$, *such that for any* $\widetilde{\boldsymbol{w}}_\tau \in \mathcal{W}$ *satisfying* $||\widetilde{\boldsymbol{w}}_\tau - \boldsymbol{w}_\tau^0|| \geq n^{-\lambda}$, $R_n^*(\widetilde{\boldsymbol{w}}_\tau,\tau)/R_n^*(\boldsymbol{w}_\tau^0,\tau) \to \infty$.

Assumption S.7 characterizes the local sharpness of the risk function $R_n^*(\cdot,\tau)$ in a neighborhood of the infeasible optimal weight vector $\boldsymbol{w}_\tau^0$, which is a standard separation condition. It states that, if a weight vector deviates from the optimum, then its relative risk $R_n^*(\widetilde{\boldsymbol{w}}_\tau,\tau)/R_n^*(\boldsymbol{w}_\tau^0,\tau)$ diverges to infinity. This assumption ensures that the risk function has a sufficiently steep minimum at $\boldsymbol{w}_\tau^0$, so that the risk incurred by such a deviation is non-negligible relative to that of the optimum. Hence, it guarantees the identifiability of the optimal weight vector.

A.4 Notations and Assumptions for Theorem 4

Let c_τ be a positive constant satisfying $c_\tau < 1$, and let $\mathcal{W}_S = \left\{\boldsymbol{w}\in\mathcal{W} : \theta_\tau = \sum_{m\in\mathcal{M}_c} w_m \leq c_\tau\right\}$ be the subset of $\mathcal{W}$. Denote $\widetilde{\xi}_n = \inf_{\boldsymbol{w}\in\mathcal{W}_S} R_n^*(\boldsymbol{w},\tau)$. We impose the following additional assumption to establish the asymptotic behavior of $\widehat{\theta}_\tau$.

Assumption S.8. *(i)* $\widetilde{\xi}_n > 0$. *(ii)* $(\widetilde{\xi}_n \log n)^{-1} = o(1)$.

Assumption S.8 imposes the restriction on the convergence rate of the minimum risk $\widetilde{\xi}_n$, implying that when the total weights on the misspecified models have a positive lower bound, this minimum risk has an asymptotic behavior similar to ξ_n^*. The similar assumption can be found in Zhang and Liu (2023) and Tu and Wang (2025).

A.5 Notations and Assumptions for Theorem 5

We impose the following additional assumption for Theorem 5.

Assumption S.9. *For each* $i = 1, \ldots, n$, $|f_{\eta^*}(\cdot|\boldsymbol{X}_i)| \leq c_1$ *and* $|f'_{\eta^*}(\cdot|\boldsymbol{X}_i)| \leq c_1$ *, where* c_1 *is a positive constant.*

Assumption S.9 requires that the conditional density function of η_i^* and its first derivative are bounded. A similar assumption can be found in Lu and Su (2015), Wang et al. (2021) and Tu and Wang (2025).

B Detailed proofs of the theorems

In this section, we provide the proofs of Theorems 1–6. Before presenting the formal proofs, we first introduce some notation and lemmas that will be used in the proofs.

In the following proof, we will use Knight's (1998) identity repeatedly,

$$\rho_\tau(u+v) - \rho_\tau(u) = v\psi_\tau(u) + \int_0^{-v} [\mathbf{1}\{u \leq s\} - \mathbf{1}\{u \leq 0\}]\, ds. \tag{S.2}$$

Since we decompose the conditional quantile treatment effect into the condition quantile of the treatment effect under Assumption S.1, we have

$$\begin{aligned} P(\eta_i^* \leq 0|\boldsymbol{X}_i) &= P(Y_i(1) - Y_i(0) - Q_1(\tau;\boldsymbol{X}_i) + Q_0(\tau;\boldsymbol{X}_i) \leq 0|\boldsymbol{X}_i) \\ &= P(Y_i(1) - Y_i(0) \leq Q_1(\tau;\boldsymbol{X}_i) - Q_0(\tau;\boldsymbol{X}_i)|\boldsymbol{X}_i) = \tau, \end{aligned} \tag{S.3}$$

which are also repeatedly used in the following proof. Additionally, the following lemmas are essential for the study of the asymptotic properties of our model averaging procedure. We use C to denote a generic constant. For simplicity, we suppress the dependence of $\varepsilon_i(d,\tau)$ on τ, $i = 1, \ldots, n$.

Lemma 1. *(Theorem 3.1 of Lu and Su (2015)) Let n_1 and n_0 denote the number of treatment observations and control observations, respectively. Suppose Assumptions 3, S.3, S.4 hold, then*

$$\|\widehat{\boldsymbol{\beta}}_d^{(m)}(\tau) - \boldsymbol{\beta}_d^{*(m)}(\tau)\| = O_p\Big(\sqrt{p_m/n_d}\Big),\ d = 0, 1. \tag{S.4}$$

Assumption 3 indicates that n_1, n_0, and n are of the same order in probability. Therefore, (S.4) is equivalent to

$$\|\widehat{\boldsymbol{\beta}}_d^{(m)}(\tau) - \boldsymbol{\beta}_d^{*(m)}(\tau)\| = O_p\Big(\sqrt{p_m/n}\Big),\ d = 0, 1. \tag{S.5}$$

Because we assume that the number of candidate models, M, and the number of the fold of the cross-validation process, K, are fixed, we further have

$$\max_{1\le m\le M}\|\widehat{\boldsymbol{\beta}}_d^{(m)}(\tau) - \boldsymbol{\beta}_d^{*(m)}(\tau)\| = O_p\Big(\sqrt{\overline{p}/n}\Big), \quad d = 0, 1, \tag{S.6}$$

and

$$\max_{1\le k\le K}\max_{1\le m\le M}\|\widehat{\boldsymbol{\beta}}_{d,-k}^{(m)}(\tau) - \boldsymbol{\beta}_d^{*(m)}(\tau)\| = O_p\Big(\sqrt{\overline{p}/n}\Big), \quad d = 0, 1, \tag{S.7}$$

where $\widehat{\boldsymbol{\beta}}_{d,-k}^{(m)}(\tau)$ denotes the quantile regression coefficient estimator obtained without using the observations in fold k for the mth model, where $m = 1, \ldots, M$ and $k = 1, \ldots, K$.

Lemma 2. *(Theorem 1 of Abadie and Imbens (2006)) Suppose Assumptions 2, 3, S.2 hold, then*

$$\frac{1}{n}\sum_{i=1}^{n} b_i(\tau) = O_p(n^{-1/p}), \tag{S.8}$$

where $b_i(\tau) = D_i\left\{Q_0(\tau; \boldsymbol{X}_i) - Q_0(\tau; \boldsymbol{X}_{i^*})\right\} + (1 - D_i)\left\{Q_1(\tau; \boldsymbol{X}_{i^*}) - Q_1(\tau; \boldsymbol{X}_i)\right\}$.

Lemma 3. *(Lemma 3 of Abadie and Imbens (2006)) Let i^* denote the unit whose covariate value is closest to that of unit i among units with treatment status opposite to that of unit i. Let $Z_1(i)$ denote the number of times unit i is used as a match when one match per unit is selected, that is, $Z_1(i) = \sum_{j=1}^{n} \mathbf{1}\{i = j^*\}$. Suppose Assumptions 2, 3, S.2 hold, then*

(i) $Z_1(i) = O_p(1)$.

(ii) $\mathsf{E}\{[Z_1(i)]^q\}$ *is bounded uniformly in* n *for any* $q > 0$.

Lemma 4. *Suppose Assumptions 2, 3, S.2, S.3 hold, then*

$$\frac{1}{n}\sum_{i=1}^{n}(\eta_i^* - \widetilde{\eta}_i) = O_p(n^{-1/2}), \tag{S.9}$$

where $\widetilde{\eta}_i = D_i\{\varepsilon_i(1) - \varepsilon_{i^*}(0)\} + (1 - D_i)\{\varepsilon_{i^*}(1) - \varepsilon_i(0)\}$.

Proof of Lemma 4. We first note that $\eta_i^* - \widetilde{\eta}_i = D_i(\varepsilon_{i^*}(0) - \varepsilon_i(0)) + (1 - D_i)(\varepsilon_i(1) - \varepsilon_{i^*}(1))$. In addition, Assumption 2 implies that $(\varepsilon_i(1), \varepsilon_i(0)) \perp D_i|\boldsymbol{X}_i$. Therefore,

$$\begin{aligned}
\mathsf{E}[\eta_i^* - \widetilde{\eta}_i] &= \mathsf{E}[D_i(\varepsilon_{i^*}(0) - \varepsilon_i(0))] + \mathsf{E}[(1 - D_i)(\varepsilon_i(1) - \varepsilon_{i^*}(1))] \\
&= \mathsf{E}\{\mathsf{E}[D_i|\boldsymbol{X}_i]\mathsf{E}[(\varepsilon_{i^*}(0) - \varepsilon_i(0))|\boldsymbol{X}_i, \boldsymbol{X}_{i^*}]\} \\
&\quad + \mathsf{E}\{\mathsf{E}[(1 - D_i)|\boldsymbol{X}_i]\mathsf{E}[(\varepsilon_i(1) - \varepsilon_{i^*}(1))|\boldsymbol{X}_i, \boldsymbol{X}_{i^*}]\} \\
&= 0,
\end{aligned}$$

which indicates that

$$\mathsf{E}\Big[\frac{1}{n}\sum_{i=1}^{n}(\eta_i^* - \widetilde{\eta}_i)\Big] = 0. \tag{S.10}$$

In addition, under Assumption S.3 (ii), we have

$$\begin{aligned}
\mathsf{E}[\eta_i^* - \widetilde{\eta}_i]^2 &= \mathsf{E}[D_i(\varepsilon_{i^*}(0) - \varepsilon_i(0)) + (1 - D_i)(\varepsilon_i(1) - \varepsilon_{i^*}(1))]^2 \\
&\leq C\{\mathsf{E}[\varepsilon_{i^*}(0) - \varepsilon_i(0)]^2 + \mathsf{E}[\varepsilon_i(1) - \varepsilon_{i^*}(1)]^2\} \\
&\leq C\{\mathsf{E}[\varepsilon_i^2(0)] + \mathsf{E}[\varepsilon_i^2(1)]\} \\
&= C\{\mathsf{E}[\mathsf{E}(\varepsilon_i^2(0)|\boldsymbol{X}_i)] + \mathsf{E}[\mathsf{E}(\varepsilon_i^2(1)|\boldsymbol{X}_i)]\} \\
&\leq C.
\end{aligned} \tag{S.11}$$

Hence, by Cauchy-Schwarz inequality, we have

$$\mathsf{E}\Big|(\eta_i^* - \widetilde{\eta}_i)(\eta_j^* - \widetilde{\eta}_j)\Big| \leq \sqrt{\mathsf{E}[(\eta_i^* - \widetilde{\eta}_i)^2]\mathsf{E}[(\eta_j^* - \widetilde{\eta}_j)^2]} \leq C. \tag{S.12}$$

Similarly, under Assumption S.3 (ii), we have

$$\mathsf{E}\left[\left|(\eta_i^* - \widetilde{\eta}_i)(\eta_j^* - \widetilde{\eta}_j)\right| \middle| \mathcal{X}_n\right] \leq \sqrt{\mathsf{E}[(\eta_i^* - \widetilde{\eta}_i)^2|\mathcal{X}_n]\mathsf{E}[(\eta_j^* - \widetilde{\eta}_j)^2|\mathcal{X}_n]} \leq C, \tag{S.13}$$

where $\mathcal{X}_n = \{\boldsymbol{X}_1, \ldots, \boldsymbol{X}_n\}$. It is seen from (S.10) that

$$\begin{aligned} Var\Big[\frac{1}{n}\sum_{i=1}^{n}(\eta_i^* - \widetilde{\eta}_i)\Big] &= \mathsf{E}\left\{\Big[\frac{1}{n}\sum_{i=1}^{n}(\eta_i^* - \widetilde{\eta}_i)\Big]^2\right\} \\ &= \frac{1}{n^2}\sum_{i=1}^{n}\mathsf{E}[(\eta_i^* - \widetilde{\eta}_i)^2] + \frac{2}{n^2}\sum_{1\leq i<j\leq n}\mathsf{E}[(\eta_i^* - \widetilde{\eta}_i)(\eta_j^* - \widetilde{\eta}_j)]. \end{aligned} \tag{S.14}$$

Let $V_n = \sum_{1\leq i<j\leq n}\mathsf{E}[(\eta_i^* - \widetilde{\eta}_i)(\eta_j^* - \widetilde{\eta}_j)]$. By $\mathsf{E}[\eta_i^* - \widetilde{\eta}_i] = 0$, we have $\mathsf{E}[(\eta_i^* - \widetilde{\eta}_i)(\eta_j^* - \widetilde{\eta}_j)] = Cov(\eta_i^* - \widetilde{\eta}_i, \eta_j^* - \widetilde{\eta}_j)$. Under the matching framework in this paper, it follows that

$$Cov(\eta_i^* - \widetilde{\eta}_i, \eta_j^* - \widetilde{\eta}_j) = \begin{cases} 0, & \text{if } D_i \neq D_j, \quad i^* \neq j, \quad \text{and } j^* \neq i, \\ \mathsf{E}[(\eta_i^* - \widetilde{\eta}_i)(\eta_j^* - \widetilde{\eta}_j)], & \text{if } D_i \neq D_j, \quad \text{and } i^* = j \text{ or } j^* = i, \\ 0, & \text{if } D_i = D_j, \quad \text{and } i^* \neq j^*, \\ \mathsf{E}[(\eta_i^* - \widetilde{\eta}_i)(\eta_j^* - \widetilde{\eta}_j)], & \text{if } D_i = D_j, \quad \text{and } i^* = j^*. \end{cases}$$

Denote $\mathcal{V}_1 = \{(i,j) : 1 \leq i < j \leq n, D_i = D_j, i^* = j^*\}$. Recall that $Z_1(i)$ denotes the number of times unit i is used as a match, which implies that these $Z_1(i)$ units share the same matching object and are therefore pairwise correlated. Thus $|\mathcal{V}_1| = \sum_{i=1}^{n}\binom{Z_1(i)}{2}$, where $|\mathcal{V}_1|$ denotes the cardinality of the set $\mathcal{V}_1$. Similarly, let $\mathcal{V}_2 = \{(i,j) : 1 \leq i < j \leq n, D_i \neq D_j, \text{ and } i^* = j \text{ or } j^* = i\}$. Since each unit i has at most one match i^*, it follows that $|\mathcal{V}_2| \leq n$. Therefore, by Lemma 3 (ii) and (S.13), we have

$$\begin{aligned} V_n &= \mathsf{E}\left\{\sum_{1\leq i<j\leq n}(\eta_i^* - \widetilde{\eta}_i)(\eta_j^* - \widetilde{\eta}_j)\right\} \\ &= \mathsf{E}\left\{\sum_{1\leq i<j\leq n}\mathsf{E}\Big[(\eta_i^* - \widetilde{\eta}_i)(\eta_j^* - \widetilde{\eta}_j)\Big| D_1, \ldots, D_n, \boldsymbol{X}_1, \ldots, \boldsymbol{X}_n\Big]\right\} \\ &= \mathsf{E}\left(\sum_{(i,j)\in\mathcal{V}_1}\mathsf{E}\Big[(\eta_i^* - \widetilde{\eta}_i)(\eta_j^* - \widetilde{\eta}_j)\Big| D_1, \ldots, D_n, \boldsymbol{X}_1, \ldots, \boldsymbol{X}_n\Big]\right. \end{aligned}$$

$$
\begin{aligned}
&\quad + \sum_{(i,j)\in\mathcal{V}_2} \mathsf{E}\Big[(\eta_i^* - \widetilde{\eta}_i)(\eta_j^* - \widetilde{\eta}_j)\Big| D_1, \ldots, D_n, \boldsymbol{X}_1, \ldots, \boldsymbol{X}_n\Big]\Bigg) \\
&\le \mathsf{E}\Bigg(\sum_{(i,j)\in\mathcal{V}_1} \mathsf{E}\Big[|(\eta_i^* - \widetilde{\eta}_i)(\eta_j^* - \widetilde{\eta}_j)|\Big| D_1, \ldots, D_n, \boldsymbol{X}_1, \ldots, \boldsymbol{X}_n\Big] \\
&\quad + \sum_{(i,j)\in\mathcal{V}_2} \mathsf{E}\Big[|(\eta_i^* - \widetilde{\eta}_i)(\eta_j^* - \widetilde{\eta}_j)|\Big| D_1, \ldots, D_n, \boldsymbol{X}_1, \ldots, \boldsymbol{X}_n\Big]\Bigg) \\
&\le \mathsf{E}\left\{ C \cdot \sum_{i=1}^{n} \binom{Z_1(i)}{2} + Cn \right\} \\
&\le C \sum_{i=1}^{n} \mathsf{E}\{[Z_1(i)]^2\} + Cn \\
&\le Cn. \qquad \text{(S.15)}
\end{aligned}
$$

Therefore, by (S.11), (S.14), and (S.15), we have

$$
Var\Big[\frac{1}{n}\sum_{i=1}^{n}(\eta_i^* - \widetilde{\eta}_i)\Big] \le \frac{1}{n^2}\cdot Cn + \frac{2}{n^2}\cdot Cn = O\Big(\frac{1}{n}\Big), \qquad \text{(S.16)}
$$

which, by Markov inequality, indicates that

$$
P\Big(\frac{1}{n}\sum_{i=1}^{n}(\eta_i^* - \widetilde{\eta}_i) > \kappa/\sqrt{n}\Big) \le n\kappa^{-2}\mathsf{E}\left\{\Big[\frac{1}{n}\sum_{i=1}^{n}(\eta_i^* - \widetilde{\eta}_i)\Big]^2\right\} \le C\kappa^{-2}. \qquad \text{(S.17)}
$$

Hence, we have shown that $\frac{1}{n}\sum_{i=1}^{n}(\eta_i^* - \widetilde{\eta}_i) = O_p(n^{-1/2})$. This completes the proof of Lemma 4. □

B.1 Proof of Theorem 1

Proof of Theorem 1. Following the proof of Theorem 2.1 in Li (1987), Theorem 1 is valid if we can show that

$$
\sup_{\boldsymbol{w}\in\mathcal{W}} \left| \frac{\mathrm{CV}(\boldsymbol{w},\tau) - \mathrm{FPE}_n(\boldsymbol{w},\tau)}{\mathrm{FPE}_n(\boldsymbol{w},\tau)} \right| = o_p(1). \qquad \text{(S.18)}
$$

By triangle inequality, we can decompose the above equality as follows

$$\begin{aligned}
&\sup_{\boldsymbol{w}\in\mathcal{W}}\Big|\frac{\mathrm{CV}(\boldsymbol{w},\tau)-\mathrm{FPE}_n(\boldsymbol{w},\tau)}{\mathrm{FPE}_n(\boldsymbol{w},\tau)}\Big| \\
\le &\sup_{\boldsymbol{w}\in\mathcal{W}}\Big|\frac{\mathrm{CV}(\boldsymbol{w},\tau)-\mathrm{FPE}_n^*(\boldsymbol{w},\tau)}{\mathrm{FPE}_n(\boldsymbol{w},\tau)}\Big|+\sup_{\boldsymbol{w}\in\mathcal{W}}\Big|\frac{\mathrm{FPE}_n^*(\boldsymbol{w},\tau)-\mathrm{FPE}_n(\boldsymbol{w},\tau)}{\mathrm{FPE}_n(\boldsymbol{w},\tau)}\Big|,
\end{aligned}$$

where $\mathrm{FPE}_n^*(\boldsymbol{w},\tau)=\mathsf{E}\Big[\rho_\tau\left\{\alpha_{n+1}^*-\Delta^*(\tau;\boldsymbol{X}_{n+1},\boldsymbol{w})\right\}\Big]$. Therefore, in order to prove (S.18), it is sufficient to show that

$$\sup_{\boldsymbol{w}\in\mathcal{W}}\Big|\frac{\mathrm{CV}(\boldsymbol{w},\tau)-\mathrm{FPE}_n^*(\boldsymbol{w},\tau)}{\mathrm{FPE}_n(\boldsymbol{w},\tau)}\Big|=o_p(1), \tag{S.19}$$

and

$$\sup_{\boldsymbol{w}\in\mathcal{W}}\Big|\frac{\mathrm{FPE}_n^*(\boldsymbol{w},\tau)-\mathrm{FPE}_n(\boldsymbol{w},\tau)}{\mathrm{FPE}_n(\boldsymbol{w},\tau)}\Big|=o_p(1). \tag{S.20}$$

Let $\widetilde{\Delta}_{-k}^{(m)}(\tau;\boldsymbol{X}_i)=\boldsymbol{X}_i^{(m)\top}\widehat{\boldsymbol{\beta}}_{1,-k}^{(m)}(\tau)-\boldsymbol{X}_i^{(m)\top}\widehat{\boldsymbol{\beta}}_{0,-k}^{(m)}(\tau)$ and $\widetilde{\Delta}^{(-k)}(\tau;\boldsymbol{X}_i,\boldsymbol{w})=\sum_{m=1}^{M}w_m\widetilde{\Delta}_{-k}^{(m)}(\tau;\boldsymbol{X}_i)$ for convenience, where $i\in\mathcal{O}_k,k=1,\ldots,K$. Let $\mathsf{E}_{\boldsymbol{X}_i}$ denote the expectation with respect to $\boldsymbol{X}_i$. By (S.2) and (S.3), we obtain

$$\begin{aligned}
&\mathrm{CV}(\boldsymbol{w},\tau)-\mathrm{FPE}_n^*(\boldsymbol{w},\tau) \\
=&\frac{1}{n}\sum_{k=1}^{K}\sum_{i\in\mathcal{O}_k}\rho_\tau\left\{\widetilde{\alpha}_i-\widetilde{\Delta}^{(-k)}(\tau;\boldsymbol{X}_i,\boldsymbol{w})\right\}-\mathsf{E}\Big[\rho_\tau\left\{\alpha_{n+1}^*-\Delta^*(\tau;\boldsymbol{X}_{n+1},\boldsymbol{w})\right\}\Big] \\
=&\frac{1}{n}\sum_{k=1}^{K}\sum_{i\in\mathcal{O}_k}\Big[\rho_\tau\left\{\widetilde{\alpha}_i-\widetilde{\Delta}^{(-k)}(\tau;\boldsymbol{X}_i,\boldsymbol{w})\right\}-\rho_\tau(\eta_i^*)\Big] \\
&-\Big\{\mathsf{E}\Big[\rho_\tau\left\{\alpha_{n+1}^*-\Delta^*(\tau;\boldsymbol{X}_{n+1},\boldsymbol{w})\right\}\Big]-\mathsf{E}\Big[\rho_\tau(\eta_{n+1}^*)\Big]\Big\} \\
&+\frac{1}{n}\sum_{k=1}^{K}\sum_{i\in\mathcal{O}_k}\Big(\rho_\tau(\eta_i^*)-\mathsf{E}\Big[\rho_\tau(\eta_{n+1}^*)\Big]\Big) \\
=&\frac{1}{n}\sum_{k=1}^{K}\sum_{i\in\mathcal{O}_k}\Big\{\Big[\Delta(\tau;\boldsymbol{X}_i)-\widetilde{\Delta}^{(-k)}(\tau;\boldsymbol{X}_i,\boldsymbol{w})+b_i(\tau)+\widetilde{\eta}_i-\eta_i^*\Big]\psi_\tau(\eta_i^*)\Big\} \\
&+\frac{1}{n}\sum_{k=1}^{K}\sum_{i\in\mathcal{O}_k}\int_0^{\widetilde{\Delta}^{(-k)}(\tau;\boldsymbol{X}_i,\boldsymbol{w})-\Delta(\tau;\boldsymbol{X}_i)-b_i(\tau)+\eta_i^*-\widetilde{\eta}_i}[\mathbf{1}\left\{\eta_i^*\le s\right\}-\mathbf{1}\left\{\eta_i^*\le 0\right\}]ds
\end{aligned}$$

$$
\begin{aligned}
&- \mathsf{E}_{\boldsymbol{X}_i}\left\{\int_0^{\Delta^*(\tau;\boldsymbol{X}_i,\boldsymbol{w})-\Delta(\tau;\boldsymbol{X}_i)}[F_{\eta^*}(s|\boldsymbol{X}_i)-F_{\eta^*}(0|\boldsymbol{X}_i)]ds\right\}\\
&+\frac{1}{n}\sum_{k=1}^{K}\sum_{i\in\mathcal{O}_k}\rho_\tau(\eta_i^*)-\mathsf{E}\Big[\rho_\tau(\eta_{n+1}^*)\Big]\\
&=\sum_{l=1}^{7}\mathrm{CV}_{ln}(\boldsymbol{w},\tau),
\end{aligned}
$$

where

$$
\mathrm{CV}_{1n}(\boldsymbol{w},\tau)=\frac{1}{n}\sum_{k=1}^{K}\sum_{i\in\mathcal{O}_k}\Big[\Delta(\tau;\boldsymbol{X}_i)-\widetilde{\Delta}^{(-k)}(\tau;\boldsymbol{X}_i,\boldsymbol{w})\Big]\psi_\tau(\eta_i^*),
$$

$$
\begin{aligned}
&\mathrm{CV}_{2n}(\boldsymbol{w},\tau)\\
&=\frac{1}{n}\sum_{k=1}^{K}\sum_{i\in\mathcal{O}_k}\int_0^{\widetilde{\Delta}^{(-k)}(\tau;\boldsymbol{X}_i,\boldsymbol{w})-\Delta(\tau;\boldsymbol{X}_i)}[\mathbf{1}\{\eta_i^*\le s\}-\mathbf{1}\{\eta_i^*\le 0\}-F_{\eta^*}(s|\boldsymbol{X}_i)+F_{\eta^*}(0|\boldsymbol{X}_i)]ds,
\end{aligned}
$$

$$
\begin{aligned}
\mathrm{CV}_{3n}(\boldsymbol{w},\tau)=\frac{1}{n}\sum_{k=1}^{K}\sum_{i\in\mathcal{O}_k}\Bigg[&\int_0^{\widetilde{\Delta}^{(-k)}(\tau;\boldsymbol{X}_i,\boldsymbol{w})-\Delta(\tau;\boldsymbol{X}_i)}[F_{\eta^*}(s|\boldsymbol{X}_i)-F_{\eta^*}(0|\boldsymbol{X}_i)]ds\\
&-\mathsf{E}_{\boldsymbol{X}_i}\left\{\int_0^{\widetilde{\Delta}^{(-k)}(\tau;\boldsymbol{X}_i,\boldsymbol{w})-\Delta(\tau;\boldsymbol{X}_i)}[F_{\eta^*}(s|\boldsymbol{X}_i)-F_{\eta^*}(0|\boldsymbol{X}_i)]ds\right\}\Bigg],
\end{aligned}
$$

$$
\begin{aligned}
\mathrm{CV}_{4n}(\boldsymbol{w},\tau)=\frac{1}{n}\sum_{k=1}^{K}\sum_{i\in\mathcal{O}_k}\mathsf{E}_{\boldsymbol{X}_i}\Bigg[&\int_0^{\widetilde{\Delta}^{(-k)}(\tau;\boldsymbol{X}_i,\boldsymbol{w})-\Delta(\tau;\boldsymbol{X}_i)}[F_{\eta^*}(s|\boldsymbol{X}_i)-F_{\eta^*}(0|\boldsymbol{X}_i)]ds\\
&-\int_0^{\Delta^*(\tau;\boldsymbol{X}_i,\boldsymbol{w})-\Delta(\tau;\boldsymbol{X}_i)}[F_{\eta^*}(s|\boldsymbol{X}_i)-F_{\eta^*}(0|\boldsymbol{X}_i)]ds\Bigg],
\end{aligned}
$$

$$
\mathrm{CV}_{5n}(\boldsymbol{w},\tau)=\frac{1}{n}\sum_{k=1}^{K}\sum_{i\in\mathcal{O}_k}\Big[b_i(\tau)\psi_\tau(\eta_i^*)+\int_{\widetilde{\Delta}^{(-k)}(\tau;\boldsymbol{X}_i,\boldsymbol{w})-\Delta(\tau;\boldsymbol{X}_i)}^{\widetilde{\Delta}^{(-k)}(\tau;\boldsymbol{X}_i,\boldsymbol{w})-\Delta(\tau;\boldsymbol{X}_i)-b_i(\tau)}[\mathbf{1}\{\eta_i^*\le s\}-\mathbf{1}\{\eta_i^*\le 0\}]ds\Big],
$$

$$\mathrm{CV}_{6n}(\boldsymbol{w},\tau) = \frac{1}{n}\sum_{k=1}^{K}\sum_{i\in\mathcal{O}_k}\Big[(\widetilde{\eta}_i-\eta_i^*)\psi_\tau(\eta_i^*) + \int_{\widetilde{\Delta}^{(-k)}(\tau;\boldsymbol{X}_i,\boldsymbol{w})-\Delta(\tau;\boldsymbol{X}_i)-b_i(\tau)}^{\widetilde{\Delta}^{(-k)}(\tau;\boldsymbol{X}_i,\boldsymbol{w})-\Delta(\tau;\boldsymbol{X}_i)-b_i(\tau)+\eta_i^*-\widetilde{\eta}_i} [\mathbf{1}\{\eta_i^*\le s\}-\mathbf{1}\{\eta_i^*\le 0\}]ds\Big],$$

and

$$\mathrm{CV}_{7n}(\boldsymbol{w},\tau) = \frac{1}{n}\sum_{k=1}^{K}\sum_{i\in\mathcal{O}_k}\rho_\tau(\eta_i^*) - \mathsf{E}\Big[\rho_\tau(\eta_{n+1}^*)\Big].$$

Thus,

$$\begin{aligned}&\sup_{\boldsymbol{w}\in\mathcal{W}}\Big|\frac{\mathrm{CV}(\boldsymbol{w},\tau)-\mathrm{FPE}_n^*(\boldsymbol{w},\tau)}{\mathrm{FPE}_n(\boldsymbol{w},\tau)}\Big|\\ &\le\Big[\sum_{l=1}^{7}\sup_{\boldsymbol{w}\in\mathcal{W}}|\mathrm{CV}_{ln}(\boldsymbol{w},\tau)|\Big]\Big/\inf_{\boldsymbol{w}\in\mathcal{W}}|\mathrm{FPE}_n(\boldsymbol{w},\tau)|.\end{aligned}$$

Consequently, to prove (S.19), it is enough to show that

$$\text{(i)}\ \inf_{\boldsymbol{w}\in\mathcal{W}}\mathrm{FPE}_n(\boldsymbol{w},\tau)\ge\mathsf{E}[\rho_\tau(\eta_{n+1}^*)]-o_p(1),$$

$$\text{(ii)}\ \sup_{\boldsymbol{w}\in\mathcal{W}}|\mathrm{CV}_{1n}(\boldsymbol{w},\tau)|=o_p(1),$$

$$\text{(iii)}\ \sup_{\boldsymbol{w}\in\mathcal{W}}|\mathrm{CV}_{2n}(\boldsymbol{w},\tau)|=o_p(1),$$

$$\text{(iv)}\ \sup_{\boldsymbol{w}\in\mathcal{W}}|\mathrm{CV}_{3n}(\boldsymbol{w},\tau)|=o_p(1),$$

$$\text{(v)}\ \sup_{\boldsymbol{w}\in\mathcal{W}}|\mathrm{CV}_{4n}(\boldsymbol{w},\tau)|=o_p(1),$$

$$\text{(vi)} \sup_{\boldsymbol{w}\in\mathcal{W}} |\mathrm{CV}_{5n}(\boldsymbol{w},\tau)| = o_p(1),$$

$$\text{(vii)} \sup_{\boldsymbol{w}\in\mathcal{W}} |\mathrm{CV}_{6n}(\boldsymbol{w},\tau)| = o_p(1),$$

and

$$\text{(viii)} \sup_{\boldsymbol{w}\in\mathcal{W}} |\mathrm{CV}_{7n}(\boldsymbol{w},\tau)| = o_p(1).$$

It should be noted that $\mathrm{CV}_{7n}(\boldsymbol{w},\tau) = \frac{1}{n}\sum_{i=1}^{n} \rho_\tau(\eta_i^*) - \mathsf{E}[\rho_\tau(\eta_{n+1}^*)]$. Hence, (vi) holds under the weak law of large numbers. Additionally, note that $\mathsf{E}[\mathrm{CV}_{7n}(\boldsymbol{w},\tau)] = 0$. Under Assumption S.3 (ii), we have

$$\begin{aligned}
Var\{\mathrm{CV}_{7n}(\boldsymbol{w},\tau)\} &= \mathsf{E}\left\{\frac{1}{n}\sum_{i=1}^{n}[\rho_\tau(\eta_i^*) - E(\rho_\tau(\eta_i^*))]\right\}^2 \\
&\le \frac{1}{n^2}\sum_{i=1}^{n}\mathsf{E}\{\rho_\tau(\eta_i^*)\}^2 \\
&\le \frac{1}{n^2}\sum_{i=1}^{n}\mathsf{E}[\eta_i^*]^2 \\
&= \frac{1}{n^2}\sum_{i=1}^{n}\mathsf{E}\{\varepsilon_i(1) - \varepsilon_i(0)\}^2 \\
&\le \frac{C}{n^2}\sum_{i=1}^{n}\left\{\mathsf{E}[\varepsilon_i(1)]^2 + +\mathsf{E}[\varepsilon_i(0)]^2\right\} \\
&= O\Big(\frac{1}{n}\Big).
\end{aligned}$$

Therefore, by Chebyshev's inequality,

$$P\Big(\mathrm{CV}_{7n}(\boldsymbol{w},\tau) > \kappa/\sqrt{n}\Big) \le \kappa^{-2} n Var\{\mathrm{CV}_{7n}(\boldsymbol{w},\tau)\} \le C\kappa^{-2},$$

which implies that $\sup_{\boldsymbol{w}\in\mathcal{W}} |\mathrm{CV}_{7n}(\boldsymbol{w},\tau)| = O_p(n^{-1/2})$. Hence, it remains to verify (i)-(vii) below.

We first prove (i). Let $u_{n+1}(\boldsymbol{w}) = \Delta(\tau;\boldsymbol{X}_{n+1}) - \Delta^*(\tau;\boldsymbol{X}_{n+1},\boldsymbol{w})$. By (S.2),

$$\mathrm{FPE}_n(\boldsymbol{w},\tau) - \mathsf{E}[\rho_\tau\left\{\eta_{n+1}^* + u_{n+1}(\boldsymbol{w})\right\}]$$

$$
\begin{aligned}
&= \mathsf{E}\Big[\rho_\tau\left\{\alpha^*_{n+1} - \widehat{\Delta}(\tau; \boldsymbol{X}_{n+1}, \boldsymbol{w})\right\} - \rho_\tau\left\{\eta^*_{n+1} + u_{n+1}(\boldsymbol{w})\right\}|\mathcal{S}_n\Big] \\
&= \mathsf{E}\Big[\rho_\tau\left\{\eta^*_{n+1} + u_{n+1}(\boldsymbol{w}) + \Delta^*(\tau; \boldsymbol{X}_{n+1}, \boldsymbol{w}) - \widehat{\Delta}(\tau; \boldsymbol{X}_{n+1}, \boldsymbol{w})\right\} - \rho_\tau\left\{\eta^*_{n+1} + u_{n+1}(\boldsymbol{w})\right\}|\mathcal{S}_n\Big] \\
&= \mathsf{E}\Big[\left\{\Delta^*(\tau; \boldsymbol{X}_{n+1}, \boldsymbol{w}) - \widehat{\Delta}(\tau; \boldsymbol{X}_{n+1}, \boldsymbol{w})\right\}\psi_\tau\left\{\eta^*_{n+1} + u_{n+1}(\boldsymbol{w})\right\}|\mathcal{S}_n\Big] \\
&\quad + \mathsf{E}\left\{\int_0^{\widehat{\Delta}(\tau;\boldsymbol{X}_{n+1},\boldsymbol{w}) - \Delta^*(\tau;\boldsymbol{X}_{n+1},\boldsymbol{w})} \Big[\mathbf{1}\left\{\eta^*_{n+1} + u_{n+1}(\boldsymbol{w}) \le s\right\} - \mathbf{1}\left\{\eta^*_{n+1} + u_{n+1}(\boldsymbol{w}) \le 0\right\}\Big] ds |\mathcal{S}_n\right\} \\
&= \mathsf{E}\Big[\Lambda_1(\boldsymbol{w}, \tau)|\mathcal{S}_n\Big] + \mathsf{E}\Big[\Lambda_2(\boldsymbol{w}, \tau)|\mathcal{S}_n\Big],
\end{aligned}
$$

where

$$
\Lambda_1(\boldsymbol{w}, \tau) = \Big[\Delta^*(\tau; \boldsymbol{X}_{n+1}, \boldsymbol{w}) - \widehat{\Delta}(\tau; \boldsymbol{X}_{n+1}, \boldsymbol{w})\Big]\psi_\tau\left\{\eta^*_{n+1} + u_{n+1}(\boldsymbol{w})\right\},
$$

and

$$
\Lambda_2(\boldsymbol{w}, \tau) = \int_0^{\widehat{\Delta}(\tau;\boldsymbol{X}_{n+1},\boldsymbol{w}) - \Delta^*(\tau;\boldsymbol{X}_{n+1},\boldsymbol{w})} \Big[\mathbf{1}\left\{\eta^*_{n+1} + u_{n+1}(\boldsymbol{w}) \le s\right\} - \mathbf{1}\left\{\eta^*_{n+1} + u_{n+1}(\boldsymbol{w}) \le 0\right\}\Big] ds.
$$

Consequently,

$$
\begin{aligned}
&\sup_{\boldsymbol{w}\in\mathcal{W}} \Big|\mathrm{FPE}_n(\boldsymbol{w}, \tau) - \mathsf{E}[\rho_\tau\left\{\eta^*_{n+1} + u_{n+1}(\boldsymbol{w})\right\}]\Big| \\
&\le \sup_{\boldsymbol{w}\in\mathcal{W}} \mathsf{E}\Big[|\Lambda_1(\boldsymbol{w}, \tau)|\Big|\mathcal{S}_n\Big] + \sup_{\boldsymbol{w}\in\mathcal{W}} \mathsf{E}\Big[|\Lambda_2(\boldsymbol{w}, \tau)|\Big|\mathcal{S}_n\Big]. \qquad \text{(S.21)}
\end{aligned}
$$

By Cauchy-Schwarz inequality, triangle inequality, $|\psi_\tau(u)| = |\tau - \mathbf{1}\{u \le 0\}| \le 1$, Assumption S.3 (iii), Assumption S.5 (i) and (S.6), we have

$$
\begin{aligned}
&\sup_{\boldsymbol{w}\in\mathcal{W}} \mathsf{E}\Big[|\Lambda_1(\boldsymbol{w}, \tau)|\Big|\mathcal{S}_n\Big] \\
&= \sup_{\boldsymbol{w}\in\mathcal{W}} \mathsf{E}\left\{\left|\Big[\Delta^*(\tau; \boldsymbol{X}_{n+1}, \boldsymbol{w}) - \widehat{\Delta}(\tau; \boldsymbol{X}_{n+1}, \boldsymbol{w})\Big]\psi_\tau\left\{\eta^*_{n+1} + u_{n+1}(\boldsymbol{w})\right\}\right|\Big|\mathcal{S}_n\right\} \\
&\le \sup_{\boldsymbol{w}\in\mathcal{W}} \mathsf{E}\left\{\left|\Delta^*(\tau; \boldsymbol{X}_{n+1}, \boldsymbol{w}) - \widehat{\Delta}(\tau; \boldsymbol{X}_{n+1}, \boldsymbol{w})\right|\Big|\mathcal{S}_n\right\} \\
&\le \sup_{\boldsymbol{w}\in\mathcal{W}} \mathsf{E}\left[\sum_{m=1}^M w_m\left|\boldsymbol{X}^{(m)\top}_{n+1}\left\{\Big[\boldsymbol{\beta}^{*(m)}_1(\tau) - \boldsymbol{\beta}^{*(m)}_0(\tau)\Big] - \Big[\widehat{\boldsymbol{\beta}}^{(m)}_1(\tau) - \widehat{\boldsymbol{\beta}}^{(m)}_0(\tau)\Big]\right\}\right|\Big|\mathcal{S}_n\right] \\
&\le \max_{1\le m\le M}\left\|\Big[\boldsymbol{\beta}^{*(m)}_1(\tau) - \boldsymbol{\beta}^{*(m)}_0(\tau)\Big] - \Big[\widehat{\boldsymbol{\beta}}^{(m)}_1(\tau) - \widehat{\boldsymbol{\beta}}^{(m)}_0(\tau)\Big]\right\| \max_{1\le m\le M} \mathsf{E}\|\boldsymbol{X}^{(m)}_{n+1}\|
\end{aligned}
$$

$$
\begin{aligned}
&\leq \max_{1\leq m\leq M}\left(\mathsf{E}\|\boldsymbol{X}_{n+1}^{(m)}\|^2\right)^{1/2}\left\{\max_{1\leq m\leq M}\|\widehat{\boldsymbol{\beta}}_1^{(m)}(\tau)-\boldsymbol{\beta}_1^{*(m)}(\tau)\|+\max_{1\leq m\leq M}\|\widehat{\boldsymbol{\beta}}_0^{(m)}(\tau)-\boldsymbol{\beta}_0^{*(m)}(\tau)\|\right\}\\
&= O_p(\overline{p}n^{-1/2})\\
&= o_p(1), \qquad \text{(S.22)}
\end{aligned}
$$

and

$$
\begin{aligned}
&\sup_{\boldsymbol{w}\in\mathcal{W}}\mathsf{E}\Big[|\Lambda_2(\boldsymbol{w},\tau)|\Big|\mathcal{S}_n\Big]\\
&=\sup_{\boldsymbol{w}\in\mathcal{W}}\mathsf{E}\left\{\Big|\int_0^{\widehat{\Delta}(\tau;\boldsymbol{X}_{n+1},\boldsymbol{w})-\Delta^*(\tau;\boldsymbol{X}_{n+1},\boldsymbol{w})}\Big[\mathbf{1}\left\{\eta_{n+1}^*+u_{n+1}(\boldsymbol{w})\leq s\right\}-\mathbf{1}\left\{\eta_{n+1}^*+u(\boldsymbol{w})\leq 0\right\}\Big]ds\Big|\Big|\mathcal{S}_n\right\}\\
&\leq\sup_{\boldsymbol{w}\in\mathcal{W}}\mathsf{E}\left\{\Big|\widehat{\Delta}(\tau;\boldsymbol{X}_{n+1},\boldsymbol{w})-\Delta^*(\tau;\boldsymbol{X}_{n+1},\boldsymbol{w})\Big|\Big|\mathcal{S}_n\right\}\\
&=o_p(1). \qquad \text{(S.23)}
\end{aligned}
$$

Combining (S.21), (S.22) and (S.23), we have

$$
\sup_{\boldsymbol{w}\in\mathcal{W}}\Big|\mathrm{FPE}_n(\boldsymbol{w},\tau)-\mathsf{E}[\rho_\tau\left\{\eta_{n+1}^*+u_{n+1}(\boldsymbol{w})\right\}]\Big|=o_p(1). \qquad \text{(S.24)}
$$

Now let $D(t)=\mathsf{E}\Big[\rho_\tau(\eta_{n+1}^*+t)-\rho_\tau(\eta_{n+1}^*)|\boldsymbol{X}_{n+1}\Big]$, where $t\in\mathbb{R}$. It is seen that $D(t)$ has a global minimum at $t=0$, which implies that

$$
\min_{\boldsymbol{w}\in\mathcal{W}}\mathsf{E}\Big[\rho_\tau\left\{\eta_{n+1}^*+u_{n+1}(\boldsymbol{w})\right\}|\boldsymbol{X}_{n+1}\Big]\geq\mathsf{E}[\rho_\tau(\eta_{n+1}^*)|\boldsymbol{X}_{n+1}]. \qquad \text{(S.25)}
$$

Additionally, note that

$$
\begin{aligned}
&\mathsf{E}\left\{\min_{\boldsymbol{w}\in\mathcal{W}}\mathsf{E}\Big[\rho_\tau\left\{\eta_{n+1}^*+u_{n+1}(\boldsymbol{w})\right\}|\boldsymbol{X}_{n+1}\Big]\right\}\\
\leq&\min_{\boldsymbol{w}\in\mathcal{W}}\mathsf{E}\left\{\mathsf{E}\Big[\rho_\tau\left\{\eta_{n+1}^*+u_{n+1}(\boldsymbol{w})\right\}|\boldsymbol{X}_{n+1}\Big]\right\}\\
=&\min_{\boldsymbol{w}\in\mathcal{W}}\mathsf{E}\Big[\rho_\tau\left\{\eta_{n+1}^*+u_{n+1}(\boldsymbol{w})\right\}\Big].
\end{aligned}
$$

Therefore, taking expectation on both side of (S.25) yields

$$
\min_{\boldsymbol{w}\in\mathcal{W}}\mathsf{E}\Big[\rho_\tau\left\{\eta_{n+1}^*+u_{n+1}(\boldsymbol{w})\right\}\Big]\geq\mathsf{E}[\rho_\tau(\eta_{n+1}^*)]. \qquad \text{(S.26)}
$$

Combining (S.24) and (S.26), we have

$$\begin{aligned}\min_{\boldsymbol{w}\in\mathcal{W}} \mathrm{FPE}_n(\boldsymbol{w},\tau) &\geq \min_{\boldsymbol{w}\in\mathcal{W}} \mathsf{E}\Big[\rho_\tau\left\{\eta^*_{n+1}+u_{n+1}(\boldsymbol{w})\right\}\Big] - o_p(1)\\ &\geq \mathsf{E}[\rho_\tau(\eta^*_{n+1})] - o_p(1),\end{aligned} \tag{S.27}$$

which establishes (i).

Second, we prove (ii). We decompose $\mathrm{CV}_{1n}(\boldsymbol{w},\tau)$ as

$$\begin{aligned}\mathrm{CV}_{1n}(\boldsymbol{w},\tau) &= \frac{1}{n}\sum_{k=1}^{K}\sum_{i\in\mathcal{O}_k}\left\{\Big[\Delta(\tau;\boldsymbol{X}_i) - \widetilde{\Delta}^{(-k)}(\tau;\boldsymbol{X}_i,\boldsymbol{w})\Big]\psi_\tau(\eta^*_i)\right\}\\ &= \mathrm{CV}_{1n,1}(\boldsymbol{w},\tau) - \mathrm{CV}_{1n,2}(\boldsymbol{w},\tau),\end{aligned}$$

where

$$\mathrm{CV}_{1n,1}(\boldsymbol{w},\tau) = \frac{1}{n}\sum_{i=1}^{n}\left\{\Big[\Delta(\tau;\boldsymbol{X}_i) - \Delta^*(\tau;\boldsymbol{X}_i,\boldsymbol{w})\Big]\psi_\tau(\eta^*_i)\right\},$$

and

$$\mathrm{CV}_{1n,2}(\boldsymbol{w},\tau) = \frac{1}{n}\sum_{k=1}^{K}\sum_{i\in\mathcal{O}_k}\left\{\Big[\widetilde{\Delta}^{(-k)}(\tau;\boldsymbol{X}_i,\boldsymbol{w}) - \Delta^*(\tau;\boldsymbol{X}_i,\boldsymbol{w})\Big]\psi_\tau(\eta^*_i)\right\}.$$

Therefore, in order to prove (ii), it is enough to show that $\sup_{\boldsymbol{w}\in\mathcal{W}}|\mathrm{CV}_{1n,1}(\boldsymbol{w},\tau)| = o_p(1)$ and $\sup_{\boldsymbol{w}\in\mathcal{W}}|\mathrm{CV}_{1n,2}(\boldsymbol{w},\tau)| = o_p(1)$. Similar to the proof of Theorem 3.3 in Lu and Su (2015), we let $h_n = 1/(\overline{p}\log n)$ and create grids using regions of the form $\mathcal{W}_l = \{\boldsymbol{w}_l : |\boldsymbol{w}-\boldsymbol{w}_l|_1 \leq h_n\}$, where $\boldsymbol{w}_l = (w_{l1}, w_{l2}, \ldots, w_{lM})^\top$ and $|\boldsymbol{w}-\boldsymbol{w}_l|_1 = \sum_{m=1}^{M}|w_m - w_{lm}|$. By selecting $\boldsymbol{w}_l$ to lay on a grid, $\mathcal{W}$ can be covered by $N = O(1/h_n^{M-1})$ regions $\mathcal{W}_l$, $l = 1,2,\ldots,N$. By triangle inequality, Jensen inequality, and Assumption S.3 (iii), we have

$$\max_{1\leq m\leq M}\|\boldsymbol{\beta}_1^{*(m)}(\tau) - \boldsymbol{\beta}_0^{*(m)}(\tau)\| \leq \max_{1\leq m\leq M}\|\boldsymbol{\beta}_1^{*(m)}(\tau)\| + \max_{1\leq m\leq M}\|\boldsymbol{\beta}_0^{*(m)}(\tau)\| = O_p(\overline{p}^{1/2}), \tag{S.28}$$

and

$$\mathsf{E}\Big[\frac{1}{n}\sum_{i=1}^{n}\|\boldsymbol{X}_i^{(m)}\|\Big]^2 \leq \frac{1}{n}\sum_{i=1}^{n}\mathsf{E}\|\boldsymbol{X}_i^{(m)}\|^2 = O_p(p_m). \tag{S.29}$$

Combining $|\psi_\tau(u)|\leq 1$, (S.28), and (S.29), we know that the following result holds uniformly over l,

$$\begin{aligned}
&\sup_{\boldsymbol{w}\in\mathcal{W}_l}|\mathrm{CV}_{1n,1}(\boldsymbol{w},\tau)-\mathrm{CV}_{1n,1}(\boldsymbol{w}_l,\tau)| \\
=&\sup_{\boldsymbol{w}\in\mathcal{W}_l}\Big|\sum_{m=1}^{M}(w_m-w_{lm})\frac{1}{n}\sum_{i=1}^{n}\boldsymbol{X}_i^{(m)\top}\left\{\boldsymbol{\beta}_1^{*(m)}(\tau)-\boldsymbol{\beta}_0^{*(m)}(\tau)\right\}\psi_\tau(\eta_i^*)\Big| \\
\leq&\max_{1\leq m\leq M}\|\boldsymbol{\beta}_1^{*(m)}(\tau)-\boldsymbol{\beta}_0^{*(m)}(\tau)\|\max_{1\leq m\leq M}\frac{1}{n}\sum_{i=1}^{n}\|\boldsymbol{X}_i^{(m)}\|\sup_{\boldsymbol{w}\in\mathcal{W}_l}\sum_{m=1}^{M}|w_m-w_{lm}| \\
=&O_p(\overline{p}h_n) \\
=&o_p(1). \qquad \text{(S.30)}
\end{aligned}$$

Therefore, by (S.30), we obtain

$$\begin{aligned}
&\sup_{\boldsymbol{w}\in\mathcal{W}}|\mathrm{CV}_{1n,1}(\boldsymbol{w},\tau)| \\
\leq&\max_{1\leq l\leq N}\sup_{\boldsymbol{w}\in\mathcal{W}_l}|\mathrm{CV}_{1n,1}(\boldsymbol{w},\tau)| \\
\leq&\max_{1\leq l\leq N}|\mathrm{CV}_{1n,1}(\boldsymbol{w}_l,\tau)|+\max_{1\leq l\leq N}\sup_{\boldsymbol{w}\in\mathcal{W}_l}|\mathrm{CV}_{1n,1}(\boldsymbol{w},\tau)-\mathrm{CV}_{1n,1}(\boldsymbol{w}_l,\tau)| \\
=&\max_{1\leq l\leq N}|\mathrm{CV}_{1n,1}(\boldsymbol{w}_l,\tau)|+o_p(1). \qquad \text{(S.31)}
\end{aligned}$$

Hence, to prove $\sup_{\boldsymbol{w}\in\mathcal{W}}|\mathrm{CV}_{1n,1}(\boldsymbol{w},\tau)|=o_p(1)$, we only need to show that $\max_{1\leq l\leq N}|\mathrm{CV}_{1n,1}(\boldsymbol{w}_l,\tau)|=o_p(1)$. Let $u_i(\boldsymbol{w}_l)=\Delta(\tau;\boldsymbol{X}_i)-\Delta^*(\tau;\boldsymbol{X}_i,\boldsymbol{w}_l)$ and $t_n=(n^{1+\delta}\overline{p}^2)^{1/2}$. Note that

$$\begin{aligned}
|u_i(\boldsymbol{w}_l)|&\leq\sum_{m=1}^{M}w_{lm}|\Delta(\tau;\boldsymbol{X}_i)-\Delta^{*(m)}(\tau;\boldsymbol{X}_i)| \\
&\leq\max_{1\leq m\leq M}|\Delta(\tau;\boldsymbol{X}_i)-\Delta^{*(m)}(\tau;\boldsymbol{X}_i)| \\
&\equiv U_i.
\end{aligned}$$

Thus, for any $\kappa>0$,

$$P\Big(\max_{1\leq l\leq N}|\mathrm{CV}_{1n,1}(\boldsymbol{w}_l,\tau)|>2\kappa(\log n)^{-1}\Big)$$

$$
\begin{aligned}
&= P\Big(\max_{1\le l\le N} \Big| \frac{1}{n}\sum_{i=1}^{n} u_i(\boldsymbol{w}_l)\psi_\tau(\eta_i^*) \Big| > 2\kappa(\log n)^{-1} \Big) \\
&\le P\Big(\max_{1\le l\le N} \Big| \frac{1}{n}\sum_{i=1}^{n} u_i(\boldsymbol{w}_l)\psi_\tau(\eta_i^*)\mathbf{1}\left\{U_i \le t_n\right\} \Big| > \kappa(\log n)^{-1} \Big) \\
&\quad + P\Big(\max_{1\le l\le N} \Big| \frac{1}{n}\sum_{i=1}^{n} u_i(\boldsymbol{w}_l)\psi_\tau(\eta_i^*)\mathbf{1}\left\{U_i > t_n\right\} \Big| > \kappa(\log n)^{-1} \Big) \\
&\equiv T_{11} + T_{12}.
\end{aligned}
$$

Hence, to establish $\max_{1\le l\le N}|\mathrm{CV}_{1n,1}(\boldsymbol{w}_l,\tau)| = O_p(\frac{1}{\log n})$, it is suffices to show that $T_{11} = o(1)$ and $T_{12} = o(1)$. By the law of iterated expectations and (S.3), we obtain

$$
\begin{aligned}
&\mathsf{E}\Big[u_i(\boldsymbol{w}_l)\psi_\tau(\eta_i^*)\mathbf{1}\left\{U_i \le t_n\right\} \Big] \\
&= \mathsf{E}\Big\{ \mathsf{E}\Big[u_i(\boldsymbol{w}_l)\psi_\tau(\eta_i^*)\mathbf{1}\left\{U_i \le t_n\right\} | \boldsymbol{X}_i \Big] \Big\} \\
&= \mathsf{E}\left\{ u_i(\boldsymbol{w}_l)\mathbf{1}\left\{U_i \le t_n\right\} \mathsf{E}[\psi_\tau(\eta_i^*)|\boldsymbol{X}_i] \right\} \\
&= 0.
\end{aligned}
$$

We notice that

$$
\Big| u_i(\boldsymbol{w}_l)\psi_\tau(\eta_i^*)\mathbf{1}\left\{U_i \le t_n\right\} \Big| \le \Big| u_i(\boldsymbol{w}_l) \Big| \le t_n.
$$

By Jensen inequality, triangle inequality, and Assumption S.3 (iii), we obtain

$$
\begin{aligned}
&Var\Big[u_i(\boldsymbol{w}_l)\psi_\tau(\eta_i^*)\mathbf{1}\left\{U_i \le t_n\right\} \Big] \\
&= \mathsf{E}\Big[u_i(\boldsymbol{w}_l)\psi_\tau(\eta_i^*)\mathbf{1}\left\{U_i \le t_n\right\} \Big]^2 \\
&\le \mathsf{E}\Big[\sum_{m=1}^{M} w_{lm}\left\{ \Delta(\tau;\boldsymbol{X}_i) - \Delta^{*(m)}(\tau;\boldsymbol{X}_i) \right\} \Big]^2 \\
&\le \sum_{m=1}^{M} w_{lm}\mathsf{E}\Big[\left\{ \Delta(\tau;\boldsymbol{X}_i) - \Delta^{*(m)}(\tau;\boldsymbol{X}_i) \right\}^2 \Big] \\
&\le C \max_{1\le m\le M} \mathsf{E}\Big[\left\{ \Delta(\tau;\boldsymbol{X}_i) \right\}^2 + \left\{ \Delta^{*(m)}(\tau;\boldsymbol{X}_i) \right\}^2 \Big] \\
&\le C \max_{1\le m\le M} \mathsf{E}\Big[\left\{ Q_1(\tau;\boldsymbol{X}_i) \right\}^2 + \left\{ Q_0(\tau;\boldsymbol{X}_i) \right\}^2 + \Big\{ \boldsymbol{X}_i^{(m)\top}\Big(\boldsymbol{\beta}_1^{*(m)}(\tau) - \boldsymbol{\beta}_0^{*(m)}(\tau) \Big) \Big\}^2 \Big]
\end{aligned}
$$

$$\leq C \max_{1\leq m\leq M} \|\boldsymbol{\beta}_1^{*(m)}(\tau) - \boldsymbol{\beta}_0^{*(m)}(\tau)\|^2 \max_{1\leq m\leq M} \mathsf{E}\|\boldsymbol{X}_i^{(m)}\|^2 + C$$
$$\leq C\overline{p}^2. \tag{S.32}$$

Therefore, by Boole inequality, Bernstein inequality (Serfling, 1980), and Assumption S.5, we have

$$\begin{aligned}
T_{11} &= P\Big(\max_{1\leq l\leq N} \Big| \frac{1}{n}\sum_{i=1}^{n} u_i(\boldsymbol{w}_l)\psi_\tau(\eta_i^*)\mathbf{1}\left\{U_i \leq t_n\right\} \Big| > \kappa(\log n)^{-1} \Big) \\
&\leq N \max_{1\leq l\leq N} P\Big(\Big| \frac{1}{n}\sum_{i=1}^{n} u_i(\boldsymbol{w}_l)\psi_\tau(\eta_i^*)\mathbf{1}\left\{U_i \leq t_n\right\} \Big| > \kappa(\log n)^{-1} \Big) \\
&\leq 2N \exp\left\{ -\frac{n^2\kappa^2(\log n)^{-2}}{2nC\overline{p}^2 + 2n\kappa(\log n)^{-1}t_n/3} \right\} \\
&= 2\exp\left\{ \log N - \frac{n\kappa^2(\log n)^{-2}}{2C\overline{p}^2 + 2\kappa(\log n)^{-1}t_n/3} \right\} \\
&= o(1).
\end{aligned} \tag{S.33}$$

We notice that

$$\left\{ \max_{1\leq l\leq N} \Big| \frac{1}{n}\sum_{i=1}^{n} u_i(\boldsymbol{w}_l)\psi_\tau(\eta_i^*)\mathbf{1}\left\{U_i > t_n\right\} \Big| > \kappa(\log n)^{-1} \right\} \subset \left\{ \max_{1\leq i\leq n} U_i > t_n \right\}.$$

Therefore, by Boole inequality, Markov inequality, triangle inequality, Assumption S.3 (iii) and (S.32), we have

$$\begin{aligned}
T_{12} &= P\Big(\max_{1\leq l\leq N} \Big| \frac{1}{n}\sum_{i=1}^{n} u_i(\boldsymbol{w}_l)\psi_\tau(\eta_i^*)\mathbf{1}\left\{U_i > t_n\right\} \Big| > \kappa(\log n)^{-1} \Big) \\
&\leq P\Big(\max_{1\leq i\leq n} U_i > t_n \Big) \\
&= P\Big(\max_{1\leq i\leq n} \max_{1\leq m\leq M} |\Delta(\tau;\boldsymbol{X}_i) - \Delta^{*(m)}(\tau;\boldsymbol{X}_i)| > t_n \Big) \\
&\leq \sum_{i=1}^{n}\sum_{m=1}^{M} P\Big(|\Delta(\tau;\boldsymbol{X}_i) - \Delta^{*(m)}(\tau;\boldsymbol{X}_i)| > t_n \Big) \\
&\leq \sum_{i=1}^{n}\sum_{m=1}^{M} \frac{1}{t_n^2}\mathsf{E}\Big[|\Delta(\tau;\boldsymbol{X}_i) - \Delta^{*(m)}(\tau;\boldsymbol{X}_i)| \Big]^2 \\
&= O(n\overline{p}^2 t_n^{-2})
\end{aligned}$$

$$= o(1). \tag{S.34}$$

Combining (S.33) and (S.34), we obtain

$$\max_{1\le l\le N}|\mathrm{CV}_{1n,1}(\boldsymbol{w}_l,\tau)|= O_p(\frac{1}{\log n}),$$

and thus combining (S.30), and (S.31), we obtain

$$\sup_{\boldsymbol{w}\in\mathcal{W}}|\mathrm{CV}_{1n,1}(\boldsymbol{w},\tau)|= O_p(\frac{1}{\log n}) = o_p(1). \tag{S.35}$$

Further, by triangle inequality, $|\psi_\tau(u)|\le 1$, Assumption S.5 (i), (S.7), and (S.29), we have

$$\begin{aligned}
&\sup_{\boldsymbol{w}\in\mathcal{W}}|\mathrm{CV}_{1n,2}(\boldsymbol{w},\tau)| \\
&= \sup_{\boldsymbol{w}\in\mathcal{W}}\Big|\frac{1}{n}\sum_{k=1}^{K}\sum_{i\in\mathcal{O}_k}\Big\{\Big[\widetilde{\Delta}^{(-k)}(\tau;\boldsymbol{X}_i,\boldsymbol{w})-\Delta^*(\tau;\boldsymbol{X}_i,\boldsymbol{w})\Big]\psi_\tau(\eta_i^*)\Big\}\Big| \\
&\le \sup_{\boldsymbol{w}\in\mathcal{W}}\frac{1}{n}\sum_{k=1}^{K}\sum_{i\in\mathcal{O}_k}\Big|\Big[\sum_{m=1}^{M}w_m\boldsymbol{X}_i^{(m)\top}\Big\{\Big(\widehat{\boldsymbol{\beta}}_{1,-k}^{(m)}(\tau)-\widehat{\boldsymbol{\beta}}_{0,-k}^{(m)}(\tau)\Big)-\Big(\boldsymbol{\beta}_1^{*(m)}(\tau)-\boldsymbol{\beta}_0^{*(m)}(\tau)\Big)\Big\}\Big]\Big| \\
&\le \max_{1\le k\le K}\max_{1\le m\le M}\Big\|\Big(\widehat{\boldsymbol{\beta}}_{1,-k}^{(m)}(\tau)-\widehat{\boldsymbol{\beta}}_{0,-k}^{(m)}(\tau)\Big)-\Big(\boldsymbol{\beta}_1^{*(m)}(\tau)-\boldsymbol{\beta}_0^{*(m)}(\tau)\Big)\Big\|\max_{1\le m\le M}\frac{1}{n}\sum_{i=1}^{n}\|\boldsymbol{X}_i^{(m)}\| \\
&= O_p(\overline{p}n^{-1/2}) \\
&= o_p(1).
\end{aligned} \tag{S.36}$$

Third, we establish (iii). Similar to $\mathrm{CV}_{1n}(\boldsymbol{w},\tau)$, we have the following decomposition:

$$\begin{aligned}
&\mathrm{CV}_{2n}(\boldsymbol{w},\tau) \\
&= \frac{1}{n}\sum_{k=1}^{K}\sum_{i\in\mathcal{O}_k}\int_0^{\widetilde{\Delta}^{(-k)}(\tau;\boldsymbol{X}_i,\boldsymbol{w})-\Delta(\tau;\boldsymbol{X}_i)}[\mathbf{1}\{\eta_i^*\le s\}-\mathbf{1}\{\eta_i^*\le 0\}-F_{\eta^*}(s|\boldsymbol{X}_i)+F_{\eta^*}(0|\boldsymbol{X}_i)]ds \\
&= \mathrm{CV}_{2n,1}(\boldsymbol{w},\tau)+\mathrm{CV}_{2n,2}(\boldsymbol{w},\tau),
\end{aligned} \tag{S.37}$$

where

$$\mathrm{CV}_{2n,1}(\boldsymbol{w},\tau)$$

$$= \frac{1}{n}\sum_{i=1}^{n}\int_{0}^{\Delta^*(\tau;\boldsymbol{X}_i,\boldsymbol{w})-\Delta(\tau;\boldsymbol{X}_i)} [\mathbf{1}\{\eta_i^* \le s\} - \mathbf{1}\{\eta_i^* \le 0\} - F_{\eta^*}(s|\boldsymbol{X}_i) + F_{\eta^*}(0|\boldsymbol{X}_i)]ds,$$

and

$$\begin{aligned}
&\mathrm{CV}_{2n,2}(\boldsymbol{w},\tau)\\
&= \frac{1}{n}\sum_{k=1}^{K}\sum_{i\in\mathcal{O}_k}\int_{\Delta^*(\tau;\boldsymbol{X}_i,\boldsymbol{w})-\Delta(\tau;\boldsymbol{X}_i)}^{\widetilde{\Delta}^{(-k)}(\tau;\boldsymbol{X}_i,\boldsymbol{w})-\Delta(\tau;\boldsymbol{X}_i)} [\mathbf{1}\{\eta_i^* \le s\} - \mathbf{1}\{\eta_i^* \le 0\} - F_{\eta^*}(s|\boldsymbol{X}_i) + F_{\eta^*}(0|\boldsymbol{X}_i)]ds.
\end{aligned}$$

Hence, we prove (iii) by showing that $\sup_{\boldsymbol{w}\in\mathcal{W}}|\mathrm{CV}_{2n,1}(\boldsymbol{w},\tau)| = o_p(1)$ and $\sup_{\boldsymbol{w}\in\mathcal{W}}|\mathrm{CV}_{2n,2}(\boldsymbol{w},\tau)| = o_p(1)$. For the former, the proof is analogous to that of $\sup_{\boldsymbol{w}\in\mathcal{W}}|\mathrm{CV}_{1n,1}(\boldsymbol{w},\tau)| = o_p(1)$, and thus we omit it for saving space. For the latter, noting that $|\mathbf{1}\{\eta_i^* \le s\} - \mathbf{1}\{\eta_i^* \le 0\} - F_{\eta^*}(s|\boldsymbol{X}_i) + F_{\eta^*}(0|\boldsymbol{X}_i)| \le 2$ and similar to (S.36), it is seen that

$$\begin{aligned}
&\sup_{\boldsymbol{w}\in\mathcal{W}}|\mathrm{CV}_{2n,2}(\boldsymbol{w},\tau)|\\
&\le \sup_{\boldsymbol{w}\in\mathcal{W}}\frac{2}{n}\sum_{k=1}^{K}\sum_{i\in\mathcal{O}_k}\Big|\sum_{m=1}^{M} w_m \boldsymbol{X}_i^{(m)\top}\left\{\left(\widehat{\boldsymbol{\beta}}_{1,-k}^{(m)}(\tau) - \widehat{\boldsymbol{\beta}}_{0,-k}^{(m)}(\tau)\right) - \left(\boldsymbol{\beta}_1^{*(m)}(\tau) - \boldsymbol{\beta}_0^{*(m)}(\tau)\right)\right\}\Big|\\
&= o_p(1). \qquad \text{(S.38)}
\end{aligned}$$

Fourth, we establish (iv). Decompose $\mathrm{CV}_{3n}(\boldsymbol{w},\tau)$ as

$$\begin{aligned}
\mathrm{CV}_{3n}(\boldsymbol{w},\tau) &= \frac{1}{n}\sum_{k=1}^{K}\sum_{i\in\mathcal{O}_k}\left[\int_{0}^{\widetilde{\Delta}^{(-k)}(\tau;\boldsymbol{X}_i,\boldsymbol{w})-\Delta(\tau;\boldsymbol{X}_i)} [F_{\eta^*}(s|\boldsymbol{X}_i) - F_{\eta^*}(0|\boldsymbol{X}_i)]ds\right.\\
&\qquad \left. - \mathsf{E}_{\boldsymbol{X}_i}\left\{\int_{0}^{\widetilde{\Delta}^{(-k)}(\tau;\boldsymbol{X}_i,\boldsymbol{w})-\Delta(\tau;\boldsymbol{X}_i)} [F_{\eta^*}(s|\boldsymbol{X}_i) - F_{\eta^*}(0|\boldsymbol{X}_i)]ds\right\}\right]\\
&= \mathrm{CV}_{3n,1}(\boldsymbol{w},\tau) + \mathrm{CV}_{3n,2}(\boldsymbol{w},\tau), \qquad \text{(S.39)}
\end{aligned}$$

where

$$\begin{aligned}
\mathrm{CV}_{3n,1}(\boldsymbol{w},\tau) &= \frac{1}{n}\sum_{i=1}^{n}\left[\int_{0}^{\Delta^*(\tau;\boldsymbol{X}_i,\boldsymbol{w})-\Delta(\tau;\boldsymbol{X}_i)} [F_{\eta^*}(s|\boldsymbol{X}_i) - F_{\eta^*}(0|\boldsymbol{X}_i)]ds\right.\\
&\qquad \left. - \mathsf{E}_{\boldsymbol{X}_i}\left\{\int_{0}^{\Delta^*(\tau;\boldsymbol{X}_i,\boldsymbol{w})-\Delta(\tau;\boldsymbol{X}_i)} [F_{\eta^*}(s|\boldsymbol{X}_i) - F_{\eta^*}(0|\boldsymbol{X}_i)]ds\right\}\right],
\end{aligned}$$

and

$$\mathrm{CV}_{3n,2}(\boldsymbol{w},\tau)=\frac{1}{n}\sum_{k=1}^{K}\sum_{i\in\mathcal{O}_k}\Bigg[\int_{\Delta^*(\tau;\boldsymbol{X}_i,\boldsymbol{w})-\Delta(\tau;\boldsymbol{X}_i)}^{\widetilde{\Delta}^{(-k)}(\tau;\boldsymbol{X}_i,\boldsymbol{w})-\Delta(\tau;\boldsymbol{X}_i)}[F_{\eta^*}(s|\boldsymbol{X}_i)-F_{\eta^*}(0|\boldsymbol{X}_i)]ds$$
$$-\mathsf{E}_{\boldsymbol{X}_i}\left\{\int_{\Delta^*(\tau;\boldsymbol{X}_i,\boldsymbol{w})-\Delta(\tau;\boldsymbol{X}_i)}^{\widetilde{\Delta}^{(-k)}(\tau;\boldsymbol{X}_i,\boldsymbol{w})-\Delta(\tau;\boldsymbol{X}_i)}[F_{\eta^*}(s|\boldsymbol{X}_i)-F_{\eta^*}(0|\boldsymbol{X}_i)]ds\right\}\Bigg].$$

Therefore, to prove $\sup_{\boldsymbol{w}\in\mathcal{W}}|\mathrm{CV}_{3n}(\boldsymbol{w},\tau)|=o_p(1)$, we only need to show that $\sup_{\boldsymbol{w}\in\mathcal{W}}|\mathrm{CV}_{3n,1}(\boldsymbol{w},\tau)|=o_p(1)$ and $\sup_{\boldsymbol{w}\in\mathcal{W}}|\mathrm{CV}_{3n,2}(\boldsymbol{w},\tau)|=o_p(1)$. Similarly, the proof of $\sup_{\boldsymbol{w}\in\mathcal{W}}|\mathrm{CV}_{3n,1}(\boldsymbol{w},\tau)|=o_p(1)$ is analogous to that of $\sup_{\boldsymbol{w}\in\mathcal{W}}|\mathrm{CV}_{1n,1}(\boldsymbol{w},\tau)|=o_p(1)$ and thus omitted. For $\sup_{\boldsymbol{w}\in\mathcal{W}}|\mathrm{CV}_{3n,2}(\boldsymbol{w},\tau)|=o_p(1)$, in view of that $|F_{\eta^*}(s|\boldsymbol{X}_i)-F_{\eta^*}(0|\boldsymbol{X}_i)|\leq 1$, we have

$$\begin{aligned}|\mathrm{CV}_{3n,2}(\boldsymbol{w},\tau)|&\leq\frac{1}{n}\sum_{k=1}^{K}\sum_{i\in\mathcal{O}_k}\left|\widetilde{\Delta}^{(-k)}(\tau;\boldsymbol{X}_i,\boldsymbol{w})-\Delta^*(\tau;\boldsymbol{X}_i,\boldsymbol{w})\right|\\&\quad+\frac{1}{n}\sum_{k=1}^{K}\sum_{i\in\mathcal{O}_k}\mathsf{E}_{\boldsymbol{X}_i}\left|\widetilde{\Delta}^{(-k)}(\tau;\boldsymbol{X}_i,\boldsymbol{w})-\Delta^*(\tau;\boldsymbol{X}_i,\boldsymbol{w})\right|\\&\equiv\mathrm{CV}_{3n,21}(\boldsymbol{w},\tau)+\mathrm{CV}_{3n,22}(\boldsymbol{w},\tau),\end{aligned}\tag{S.40}$$

and hence we prove $\sup_{\boldsymbol{w}\in\mathcal{W}}|\mathrm{CV}_{3n,2}(\boldsymbol{w},\tau)|=o_p(1)$ by showing that $\sup_{\boldsymbol{w}\in\mathcal{W}}\mathrm{CV}_{3n,21}(\boldsymbol{w},\tau)=o_p(1)$ and $\sup_{\boldsymbol{w}\in\mathcal{W}}\mathrm{CV}_{3n,22}(\boldsymbol{w},\tau)=o_p(1)$. Analogous to (S.36), we obtain

$$\begin{aligned}&\sup_{\boldsymbol{w}\in\mathcal{W}}\mathrm{CV}_{3n,21}(\boldsymbol{w},\tau)\\&=\sup_{\boldsymbol{w}\in\mathcal{W}}\frac{1}{n}\sum_{k=1}^{K}\sum_{i\in\mathcal{O}_k}\left|\left[\sum_{m=1}^{M}w_m\boldsymbol{X}_i^{(m)\top}\left\{\left(\widehat{\boldsymbol{\beta}}_{1,-k}^{(m)}(\tau)-\widehat{\boldsymbol{\beta}}_{0,-k}^{(m)}(\tau)\right)-\left(\boldsymbol{\beta}_1^{*(m)}(\tau)-\boldsymbol{\beta}_0^{*(m)}(\tau)\right)\right\}\right]\right|\\&=o_p(1).\end{aligned}\tag{S.41}$$

Further, by Cauchy-Schwarz inequality, triangle inequality, Assumption S.4 (ii), Assumption S.5 (i), and (S.7), we have

$$\sup_{\boldsymbol{w}\in\mathcal{W}}\mathrm{CV}_{3n,22}(\boldsymbol{w},\tau)$$

$$
\begin{aligned}
&= \sup_{\boldsymbol{w}\in\mathcal{W}} \frac{1}{n}\sum_{k=1}^{K}\sum_{i\in\mathcal{O}_k} \mathsf{E}_{\boldsymbol{X}_i}\Big|\sum_{m=1}^{M} w_m \boldsymbol{X}_i^{(m)\top}\Big\{\Big(\widehat{\boldsymbol{\beta}}_{1,-k}^{(m)}(\tau)-\widehat{\boldsymbol{\beta}}_{0,-k}^{(m)}(\tau)\Big)-\Big(\boldsymbol{\beta}_1^{*(m)}(\tau)-\boldsymbol{\beta}_0^{*(m)}(\tau)\Big)\Big\}\Big| \\
&\le \sup_{\boldsymbol{w}\in\mathcal{W}} \frac{1}{n}\sum_{k=1}^{K}\sum_{i\in\mathcal{O}_k}\sum_{m=1}^{M} w_m \mathsf{E}_{\boldsymbol{X}_i}\Big|\boldsymbol{X}_i^{(m)\top}\Big\{\Big(\widehat{\boldsymbol{\beta}}_{1,-k}^{(m)}(\tau)-\widehat{\boldsymbol{\beta}}_{0,-k}^{(m)}(\tau)\Big)-\Big(\boldsymbol{\beta}_1^{*(m)}(\tau)-\boldsymbol{\beta}_0^{*(m)}(\tau)\Big)\Big\}\Big| \\
&= \sup_{\boldsymbol{w}\in\mathcal{W}} \frac{1}{n}\sum_{k=1}^{K}\sum_{i\in\mathcal{O}_k}\sum_{m=1}^{M} w_m \Bigg[\Big\{\Big(\widehat{\boldsymbol{\beta}}_{1,-k}^{(m)}(\tau)-\widehat{\boldsymbol{\beta}}_{0,-k}^{(m)}(\tau)\Big)-\Big(\boldsymbol{\beta}_1^{*(m)}(\tau)-\boldsymbol{\beta}_0^{*(m)}(\tau)\Big)\Big\}^{\top} \\
&\qquad \cdot \mathsf{E}\Big[\boldsymbol{X}_i^{(m)}\boldsymbol{X}_i^{(m)\top}\Big]\Big\{\Big(\widehat{\boldsymbol{\beta}}_{1,-k}^{(m)}(\tau)-\widehat{\boldsymbol{\beta}}_{0,-k}^{(m)}(\tau)\Big)-\Big(\boldsymbol{\beta}_1^{*(m)}(\tau)-\boldsymbol{\beta}_0^{*(m)}(\tau)\Big)\Big\}\Bigg]^{1/2} \\
&\le \max_{1\le k\le K}\max_{1\le m\le M}\Big\|\Big\{\Big(\widehat{\boldsymbol{\beta}}_{1,-k}^{(m)}(\tau)-\widehat{\boldsymbol{\beta}}_{0,-k}^{(m)}(\tau)\Big)-\Big(\boldsymbol{\beta}_1^{*(m)}(\tau)-\boldsymbol{\beta}_0^{*(m)}(\tau)\Big)\Big\}\Big\| \\
&\qquad \cdot \max_{1\le m\le M}\lambda_{\max}\Big\{\mathsf{E}\Big[\boldsymbol{X}_i^{(m)}\boldsymbol{X}_i^{(m)\top}\Big]\Big\}^{1/2} \\
&= O_p(\overline{p}^{1/2}\overline{c}_A^{1/2}n^{-1/2}) \\
&= o_p(1). \qquad \text{(S.42)}
\end{aligned}
$$

Consequently, combining (S.41) and (S.42), we obtain $\sup_{\boldsymbol{w}\in\mathcal{W}}|\mathrm{CV}_{3n,2}(\boldsymbol{w},\tau)|= o_p(1)$ and thus $\sup_{\boldsymbol{w}\in\mathcal{W}}|\mathrm{CV}_{3n}(\boldsymbol{w},\tau)|= o_p(1)$ holds.

Fifth, we prove (v). Similar to (S.42), we obtain

$$
\begin{aligned}
&\sup_{\boldsymbol{w}\in\mathcal{W}}|\mathrm{CV}_{4n}(\boldsymbol{w},\tau)| \\
&= \sup_{\boldsymbol{w}\in\mathcal{W}}\Bigg|\frac{1}{n}\sum_{k=1}^{K}\sum_{i\in\mathcal{O}_k}\mathsf{E}_{\boldsymbol{X}_i}\Bigg[\int_0^{\widetilde{\Delta}^{(-k)}(\tau;\boldsymbol{X}_i,\boldsymbol{w})-\Delta(\tau;\boldsymbol{X}_i)}[F_{\eta^*}(s|\boldsymbol{X}_i)-F_{\eta^*}(0|\boldsymbol{X}_i)]ds \\
&\quad -\int_0^{\Delta^*(\tau;\boldsymbol{X}_i,\boldsymbol{w})-\Delta(\tau;\boldsymbol{X}_i)}[F_{\eta^*}(s|\boldsymbol{X}_i)-F_{\eta^*}(0|\boldsymbol{X}_i)]ds\Bigg]\Bigg| \\
&\le \sup_{\boldsymbol{w}\in\mathcal{W}}\frac{1}{n}\sum_{k=1}^{K}\sum_{i\in\mathcal{O}_k}\mathsf{E}_{\boldsymbol{X}_i}\Bigg|\int_{\Delta^*(\tau;\boldsymbol{X}_i,\boldsymbol{w})-\Delta(\tau;\boldsymbol{X}_i)}^{\widetilde{\Delta}^{(-k)}(\tau;\boldsymbol{X}_i,\boldsymbol{w})-\Delta(\tau;\boldsymbol{X}_i)}[F_{\eta^*}(s|\boldsymbol{X}_i)-F_{\eta^*}(0|\boldsymbol{X}_i)]ds\Bigg| \\
&\le \sup_{\boldsymbol{w}\in\mathcal{W}}\frac{1}{n}\sum_{k=1}^{K}\sum_{i\in\mathcal{O}_k}\mathsf{E}_{\boldsymbol{X}_i}\Big|\widetilde{\Delta}^{(-k)}(\tau;\boldsymbol{X}_i,\boldsymbol{w})-\Delta^*(\tau;\boldsymbol{X}_i,\boldsymbol{w})\Big| \\
&= o_p(1). \qquad \text{(S.43)}
\end{aligned}
$$

Next, we prove (vi). We first notice that,

$$
\begin{aligned}
&\sup_{\boldsymbol{w}\in\mathcal{W}}|\mathrm{CV}_{5n}(\boldsymbol{w},\tau)| \\
&= \sup_{\boldsymbol{w}\in\mathcal{W}}\Big|\frac{1}{n}\sum_{k=1}^{K}\sum_{i\in\mathcal{O}_k}\Big[b_i(\tau)\psi_\tau(\eta_i^*)+\int_{\widetilde{\Delta}^{(-k)}(\tau;\boldsymbol{X}_i,\boldsymbol{w})-\Delta(\tau;\boldsymbol{X}_i)}^{\widetilde{\Delta}^{(-k)}(\tau;\boldsymbol{X}_i,\boldsymbol{w})-\Delta(\tau;\boldsymbol{X}_i)-b_i(\tau)}[\mathbf{1}\{\eta_i^*\le s\}-\mathbf{1}\{\eta_i^*\le 0\}]ds\Big]\Big| \\
&\le \frac{2}{n}\sum_{i=1}^{n}|b_i(\tau)|, \qquad \text{(S.44)}
\end{aligned}
$$

and hence by Lemma 2, Cauchy-Schwarz inequality, and Assumption S.2 (iii), we obtain

$$
\begin{aligned}
&\mathsf{E}\Big[\frac{2}{n}\sum_{i=1}^{n}|b_i(\tau)|\Big]^2 \\
&\le \frac{4}{n}\mathsf{E}[\sum_{i=1}^{n}b_i^2(\tau)] \\
&= \frac{4}{n}\mathsf{E}\Bigg[n_0^{-2/p}\sum_{D_i=1}\mathsf{E}\left\{n_0^{2/p}b_i^2(\tau)|D_1,D_2,\ldots,D_n,\boldsymbol{X}_i\right\} \\
&\qquad + n_1^{-2/p}\sum_{D_i=0}\mathsf{E}\left\{n_1^{2/p}b_i^2(\tau)|D_1,D_2,\ldots,D_n,\boldsymbol{X}_i\right\}\Bigg] \\
&\le \frac{C}{n}\mathsf{E}\Bigg[n_0^{-2/p}\sum_{D_i=1}\mathsf{E}\left\{n_0^{2/p}||\boldsymbol{X}_i-\boldsymbol{X}_{i^*}||^2|D_1,D_2,\ldots,D_n,\boldsymbol{X}_i\right\} \\
&\qquad + n_1^{-2/p}\sum_{D_i=0}\mathsf{E}\left\{n_1^{2/p}||\boldsymbol{X}_i-\boldsymbol{X}_{i^*}||^2|D_1,D_2,\ldots,D_n,\boldsymbol{X}_i\right\}\Bigg] \\
&= Cn^{-2/p}\mathsf{E}\left\{\left(\frac{n}{n_0}\right)^{2/p}\left(\frac{n_1}{n}\right)+\left(\frac{n}{n_1}\right)^{2/p}\left(\frac{n_0}{n}\right)\right\} \\
&= O(n^{-2/p}), \qquad \text{(S.45)}
\end{aligned}
$$

where the last equality holds by the fact that under Assumption 3, any moment of n/n_1 or n/n_0 is uniformly bounded (see Lemma A.2 in Li et al. (2025)). Therefore, by Markov inequality, we have

$$
\sup_{\boldsymbol{w}\in\mathcal{W}}|\mathrm{CV}_{5n}(\boldsymbol{w},\tau)|= O_p(n^{-1/p})=o_p(1). \qquad \text{(S.46)}
$$

Then, we show (vii). We observe that

$$
\begin{aligned}
&\sup_{\boldsymbol{w}\in\mathcal{W}}|\mathrm{CV}_{6n}(\boldsymbol{w},\tau)| \\
&= \sup_{\boldsymbol{w}\in\mathcal{W}}\Big|\frac{1}{n}\sum_{k=1}^{K}\sum_{i\in\mathcal{O}_k}\Big[(\widetilde{\eta}_i-\eta_i^*)\psi_\tau(\eta_i^*) \\
&\qquad + \int_{\widetilde{\Delta}^{(-k)}(\tau;\boldsymbol{X}_i,\boldsymbol{w})-\Delta(\tau;\boldsymbol{X}_i)-b_i(\tau)}^{\widetilde{\Delta}^{(-k)}(\tau;\boldsymbol{X}_i,\boldsymbol{w})-\Delta(\tau;\boldsymbol{X}_i)-b_i(\tau)+\eta_i^*-\widetilde{\eta}_i}[\mathbf{1}\{\eta_i^*\le s\}-\mathbf{1}\{\eta_i^*\le 0\}]ds\Big| \\
&\le \Big|\frac{\tau}{n}\sum_{i=1}^{n}(\widetilde{\eta}_i-\eta_i^*)\Big| + \sup_{\boldsymbol{w}\in\mathcal{W}}\Big|\frac{1}{n}\sum_{k=1}^{K}\sum_{i\in\mathcal{O}_k}\int_{\widetilde{\Delta}^{(-k)}(\tau;\boldsymbol{X}_i,\boldsymbol{w})-\Delta(\tau;\boldsymbol{X}_i)-b_i(\tau)}^{\widetilde{\Delta}^{(-k)}(\tau;\boldsymbol{X}_i,\boldsymbol{w})-\Delta(\tau;\boldsymbol{X}_i)-b_i(\tau)+\eta_i^*-\widetilde{\eta}_i}\mathbf{1}\{\eta_i^*\le s\}\,ds\Big| \\
&\equiv |\mathrm{CV}_{6n,1}(\boldsymbol{w},\tau)| + \sup_{\boldsymbol{w}\in\mathcal{W}}|\mathrm{CV}_{6n,2}(\boldsymbol{w},\tau)|. \qquad \text{(S.47)}
\end{aligned}
$$

Therefore, to prove (vii), we only need to prove that $|\mathrm{CV}_{6n,1}(\boldsymbol{w},\tau)|= o_p(1)$ and $\sup_{\boldsymbol{w}\in\mathcal{W}}|\mathrm{CV}_{6n,2}(\boldsymbol{w},\tau)|= o_p(1)$. For the former, by Lemma 4, we know that $|\mathrm{CV}_{6n,1}(\boldsymbol{w},\tau)|= O_p(\sqrt{1/n}) = o_p(1)$. For the latter, let $A_{i(k)}^{(\boldsymbol{w})} = \widetilde{\Delta}^{(-k)}(\tau;\boldsymbol{X}_i,\boldsymbol{w}) - \Delta(\tau;\boldsymbol{X}_i) - b_i(\tau)$ and $B_{i(k)}^{(\boldsymbol{w})} = \widetilde{\Delta}^{(-k)}(\tau;\boldsymbol{X}_i,\boldsymbol{w}) - \Delta(\tau;\boldsymbol{X}_i) - b_i(\tau) + \eta_i^* - \widetilde{\eta}_i$. Note that when $\eta_i^* - \widetilde{\eta}_i \neq 0$, we have

$$
\int_{A_{i(k)}^{(\boldsymbol{w})}}^{B_{i(k)}^{(\boldsymbol{w})}}[\mathbf{1}\{\eta_i^*\le s\}]ds = \begin{cases} \eta_i^*-\widetilde{\eta}_i, & \text{if } \eta_i^* < \min\{A_{i(k)}^{(\boldsymbol{w})}, B_{i(k)}^{(\boldsymbol{w})}\}, \\ [\eta_i^*-\widetilde{\eta}_i]\frac{\eta_i^*-A_{i(k)}^{(\boldsymbol{w})}}{\eta_i^*-\widetilde{\eta}_i}, & \text{if } B_{i(k)}^{(\boldsymbol{w})} \le \eta_i^* \le A_{i(k)}^{(\boldsymbol{w})}, \\ [\eta_i^*-\widetilde{\eta}_i]\frac{B_{i(k)}^{(\boldsymbol{w})}-\eta_i^*}{\eta_i^*-\widetilde{\eta}_i}, & \text{if } A_{i(k)}^{(\boldsymbol{w})} \le \eta_i^* \le B_{i(k)}^{(\boldsymbol{w})}, \\ 0, & \text{if } \eta_i^* > \max\{A_{i(k)}^{(\boldsymbol{w})}, B_{i(k)}^{(\boldsymbol{w})}\}. \end{cases}
$$

Denote $\nu_i = \mathbf{1}\{\eta_i^*-\widetilde{\eta}_i=0\} + \mathbf{1}\{\eta_i^* < \min\{A_{i(k)}^{(\boldsymbol{w})}, B_{i(k)}^{(\boldsymbol{w})}\}\} + \frac{\eta_i^*-A_{i(k)}^{(\boldsymbol{w})}}{\eta_i^*-\widetilde{\eta}_i}\cdot\mathbf{1}\{B_{i(k)}^{(\boldsymbol{w})} \le \eta_i^* \le A_{i(k)}^{(\boldsymbol{w})}\} + \frac{B_{i(k)}^{(\boldsymbol{w})}-\eta_i^*}{\eta_i^*-\widetilde{\eta}_i}\mathbf{1}\{A_{i(k)}^{(\boldsymbol{w})} \le \eta_i^* \le B_{i(k)}^{(\boldsymbol{w})}\}$, $i = 1, 2\ldots, n$. Hence

$$
\mathrm{CV}_{6n,2}(\boldsymbol{w},\tau) = \frac{1}{n}\sum_{i=1}^{n}\int_{A_{i(k)}^{(\boldsymbol{w})}}^{B_{i(k)}^{(\boldsymbol{w})}}[\mathbf{1}\{\eta_i^*\le s\}]ds = \frac{1}{n}\sum_{i=1}^{n}\nu_i[\eta_i^*-\widetilde{\eta}_i].
$$

It is obvious that $0 \le \nu_i \le 1, i = 1, 2, \ldots n$, and therefore

$$
\mathsf{E}\{[\sup_{\boldsymbol{w}\in\mathcal{W}}|\mathrm{CV}_{6n,2}(\boldsymbol{w},\tau)|]^2\}
$$

$$
\begin{aligned}
&= \mathsf{E}\{\sup_{\boldsymbol{w}\in\mathcal{W}}[\frac{1}{n}\sum_{i=1}^{n}\nu_i(\eta_i^*-\widetilde{\eta}_i)]^2\} \\
&= \mathsf{E}\left\{\sup_{\boldsymbol{w}\in\mathcal{W}}\left[\frac{1}{n}\sum_{i=1}^{n}\nu_i^2(\eta_i^*-\widetilde{\eta}_i)^2+\frac{2}{n}\sum_{1\le i<j\le n}\nu_i\nu_j(\eta_i^*-\widetilde{\eta}_i)(\eta_j^*-\widetilde{\eta}_j)\right]\right\} \\
&\le \mathsf{E}\left\{\sup_{\boldsymbol{w}\in\mathcal{W}}\left[\frac{1}{n^2}\sum_{i=1}^{n}\nu_i^2(\eta_i^*-\widetilde{\eta}_i)^2\right]\right\}+\mathsf{E}\left\{\sup_{\boldsymbol{w}\in\mathcal{W}}\left[\frac{2}{n^2}\sum_{1\le i<j\le n}\nu_i\nu_j(\eta_i^*-\widetilde{\eta}_i)(\eta_j^*-\widetilde{\eta}_j)\right]\right\} \\
&\le \frac{1}{n^2}\sum_{i=1}^{n}\mathsf{E}\{\sup_{\boldsymbol{w}\in\mathcal{W}}\nu_i^2(\eta_i^*-\widetilde{\eta}_i)^2\}+\frac{2}{n^2}\mathsf{E}\left\{\sum_{1\le i<j\le n}[\sup_{\boldsymbol{w}\in\mathcal{W}}\nu_i\nu_j(\eta_i^*-\widetilde{\eta}_i)(\eta_i^*-\widetilde{\eta}_i)]\right\} \\
&\le \frac{1}{n^2}\sum_{i=1}^{n}\mathsf{E}\{(\eta_i^*-\widetilde{\eta}_i)^2\}+\frac{2}{n^2}\mathsf{E}\left\{\sum_{1\le i<j\le n}[\sup_{\boldsymbol{w}\in\mathcal{W}}\nu_i\nu_j(\eta_i^*-\widetilde{\eta}_i)(\eta_i^*-\widetilde{\eta}_i)]\right\} \\
&\equiv \frac{1}{n^2}\sum_{i=1}^{n}\mathsf{E}\{(\eta_i^*-\widetilde{\eta}_i)^2\}+\mathrm{CV}_{6n,21}(\boldsymbol{w},\tau). \qquad \text{(S.48)}
\end{aligned}
$$

Similar to (S.15), we have

$$
\begin{aligned}
\mathrm{CV}_{6n,21}(\boldsymbol{w},\tau) &= \frac{2}{n^2}\mathsf{E}\left\{\sum_{1\le i<j\le n}[\sup_{\boldsymbol{w}\in\mathcal{W}}\nu_i\nu_j(\eta_i^*-\widetilde{\eta}_i)(\eta_j^*-\widetilde{\eta}_j)]\right\} \\
&= \frac{2}{n^2}\mathsf{E}\left\{\sum_{1\le i<j\le n}\mathsf{E}[\sup_{\boldsymbol{w}\in\mathcal{W}}\nu_i\nu_j(\eta_i^*-\widetilde{\eta}_i)(\eta_j^*-\widetilde{\eta}_j)|D_1,\ldots,D_n,\boldsymbol{X}_1,\ldots,\boldsymbol{X}_n]\right\} \\
&\le \frac{2}{n^2}\mathsf{E}\left\{\sum_{1\le i<j\le n}\left|\mathsf{E}[\sup_{\boldsymbol{w}\in\mathcal{W}}\nu_i\nu_j(\eta_i^*-\widetilde{\eta}_i)(\eta_j^*-\widetilde{\eta}_j)|D_1,\ldots,D_n,\boldsymbol{X}_1,\ldots,\boldsymbol{X}_n]\right|\right\} \\
&\le \frac{2}{n^2}\mathsf{E}\left\{\sum_{1\le i<j\le n}\left|\mathsf{E}[(\eta_i^*-\widetilde{\eta}_i)(\eta_j^*-\widetilde{\eta}_j)|D_1,\ldots,D_n,\boldsymbol{X}_1,\ldots,\boldsymbol{X}_n]\right|\right\} \\
&\le Cn^{-1}. \qquad \text{(S.49)}
\end{aligned}
$$

Combining (S.11), (S.48), and (S.49), we obtain

$$
\mathsf{E}\{[\sup_{\boldsymbol{w}\in\mathcal{W}}|\mathrm{CV}_{6n,2}(\boldsymbol{w},\tau)|]^2\}=O(\frac{1}{n}).
$$

Therefore, by Markov inequality, we show that

$$
\sup_{\boldsymbol{w}\in\mathcal{W}}|\mathrm{CV}_{6n,2}(\boldsymbol{w},\tau)|=O_p(n^{-1/2})=o_p(1). \qquad \text{(S.50)}
$$

Therefore, we have proved (vii).

We now consider (S.20). By (S.2), we have

$$
\begin{aligned}
&\mathrm{FPE}_n^*(\boldsymbol{w},\tau)-\mathrm{FPE}_n(\boldsymbol{w},\tau)\\
&=\mathsf{E}\Big[\rho_\tau\left\{\alpha_{n+1}^*-\Delta^*(\tau;\boldsymbol{X}_{n+1},\boldsymbol{w})\right\}-\rho_\tau\left\{\alpha_{n+1}^*-\widehat{\Delta}(\tau;\boldsymbol{X}_{n+1},\boldsymbol{w})\right\}\Big|\mathcal{S}_n\Big]\\
&=\mathsf{E}\Big[\Big(\rho_\tau\left\{\alpha_{n+1}^*-\Delta^*(\tau;\boldsymbol{X}_{n+1},\boldsymbol{w})\right\}-\rho_\tau(\eta_{n+1}^*)\Big)-\Big(\rho_\tau\left\{\alpha_{n+1}^*-\widehat{\Delta}(\tau;\boldsymbol{X}_{n+1},\boldsymbol{w})\right\}-\rho_\tau(\eta_{n+1}^*)\Big)\Big|\mathcal{S}_n\Big]\\
&=\mathsf{E}\Big[\left\{\widehat{\Delta}(\tau;\boldsymbol{X}_{n+1},\boldsymbol{w})-\Delta^*(\tau;\boldsymbol{X}_{n+1},\boldsymbol{w})\right\}\psi_\tau(\eta_{n+1}^*)\Big|\mathcal{S}_n\Big]\\
&\quad+\mathsf{E}\left[\int_{\widehat{\Delta}(\tau;\boldsymbol{X}_{n+1},\boldsymbol{w})-\Delta(\tau;\boldsymbol{X}_{n+1})}^{\Delta^*(\tau;\boldsymbol{X}_{n+1},\boldsymbol{w})-\Delta(\tau;\boldsymbol{X}_{n+1})}[\mathbf{1}\left\{\eta_{n+1}^*\le s\right\}-\mathbf{1}\left\{\eta_{n+1}^*\le 0\right\}]ds\Big|\mathcal{S}_n\right]\\
&\equiv\Gamma_1(\boldsymbol{w},\tau)+\Gamma_2(\boldsymbol{w},\tau).
\end{aligned}
\tag{S.51}
$$

Hence, to prove $\sup_{\boldsymbol{w}\in\mathcal{W}}|\mathrm{FPE}_n^*(\boldsymbol{w},\tau)-\mathrm{FPE}_n(\boldsymbol{w},\tau)|=o_p(1)$, it suffices to show that $\sup_{\boldsymbol{w}\in\mathcal{W}}|\Gamma_1(\boldsymbol{w},\tau)|=o_p(1)$ and $\sup_{\boldsymbol{w}\in\mathcal{W}}|\Gamma_2(\boldsymbol{w},\tau)|=o_p(1)$. Analogous to (S.22), we have

$$
\begin{aligned}
&\sup_{\boldsymbol{w}\in\mathcal{W}}|\Gamma_1(\boldsymbol{w},\tau)|\\
&=\sup_{\boldsymbol{w}\in\mathcal{W}}\Big|\mathsf{E}\Big[\left\{\widehat{\Delta}(\tau;\boldsymbol{X}_{n+1},\boldsymbol{w})-\Delta^*(\tau;\boldsymbol{X}_{n+1},\boldsymbol{w})\right\}\psi_\tau(\eta_{n+1}^*)\Big|\mathcal{S}_n\Big]\Big|\\
&\le\sup_{\boldsymbol{w}\in\mathcal{W}}\mathsf{E}\Big|\Big[\Big(\Delta^*(\tau;\boldsymbol{X}_{n+1},\boldsymbol{w})-\widehat{\Delta}(\tau;\boldsymbol{X}_{n+1},\boldsymbol{w})\Big)\Big|\mathcal{S}_n\Big]\Big|\\
&=o_p(1),
\end{aligned}
\tag{S.52}
$$

and

$$
\begin{aligned}
&\sup_{\boldsymbol{w}\in\mathcal{W}}|\Gamma_2(\boldsymbol{w},\tau)|\\
&=\sup_{\boldsymbol{w}\in\mathcal{W}}\left|\mathsf{E}\left[\int_{\widehat{\Delta}(\tau;\boldsymbol{X}_{n+1},\boldsymbol{w})-\Delta(\tau;\boldsymbol{X}_{n+1})}^{\Delta^*(\tau;\boldsymbol{X}_{n+1},\boldsymbol{w})-\Delta(\tau;\boldsymbol{X}_{n+1})}[\mathbf{1}\left\{\eta_{n+1}^*\le s\right\}-\mathbf{1}\left\{\eta_{n+1}^*\le 0\right\}]ds\Big|\mathcal{S}_n\right]\right|\\
&\le\sup_{\boldsymbol{w}\in\mathcal{W}}\mathsf{E}\Big|\Big[\Big(\Delta^*(\tau;\boldsymbol{X}_{n+1},\boldsymbol{w})-\widehat{\Delta}(\tau;\boldsymbol{X}_{n+1},\boldsymbol{w})\Big)\Big|\mathcal{S}_n\Big]\Big|\\
&=o_p(1).
\end{aligned}
\tag{S.53}
$$

This completes the proof of Theorem 1. □

B.2 Proof of Theorem 2

Proof of Theorem 2. Because $\mathsf{E}[\rho_\tau(\eta^*_{n+1})]$ is independent of $\boldsymbol{w}$, therefore

$$\widehat{\boldsymbol{w}}_\tau = \arg\min_{\boldsymbol{w}\in\mathcal{W}} \mathrm{CV}(\boldsymbol{w},\tau) = \arg\min_{\boldsymbol{w}\in\mathcal{W}} \{\mathrm{CV}(\boldsymbol{w},\tau) - \mathsf{E}[\rho_\tau(\eta^*_{n+1})]\}.$$

Denote $\mathrm{CV}^*(\boldsymbol{w},\tau) = \mathrm{CV}(\boldsymbol{w},\tau) - \mathsf{E}[\rho_\tau(\eta^*_{n+1})]$. Based on the proof of Theorem 1, Theorem 2 is valid if we can show that

$$\sup_{\boldsymbol{w}\in\mathcal{W}} \left| \frac{\mathrm{CV}^*(\boldsymbol{w},\tau) - R_n(\boldsymbol{w},\tau)}{R_n(\boldsymbol{w},\tau)} \right| = o_p(1). \tag{S.54}$$

To prove (S.54), it is sufficient to show that

$$\sup_{\boldsymbol{w}\in\mathcal{W}} \left| \frac{\mathrm{CV}^*(\boldsymbol{w},\tau) - R^*_n(\boldsymbol{w},\tau)}{R^*_n(\boldsymbol{w},\tau)} \right| = o_p(1), \tag{S.55}$$

and

$$\sup_{\boldsymbol{w}\in\mathcal{W}} \left| \frac{R^*_n(\boldsymbol{w},\tau) - R_n(\boldsymbol{w},\tau)}{R^*_n(\boldsymbol{w},\tau)} \right| = o_p(1). \tag{S.56}$$

It should be noted that

$$\mathrm{CV}^*(\boldsymbol{w},\tau) - R^*_n(\boldsymbol{w},\tau) = \mathrm{CV}(\boldsymbol{w},\tau) - \mathrm{FPE}^*_n(\boldsymbol{w},\tau), \tag{S.57}$$

and

$$R^*_n(\boldsymbol{w},\tau) - R_n(\boldsymbol{w},\tau) = \mathrm{FPE}^*_n(\boldsymbol{w},\tau) - \mathrm{FPE}_n(\boldsymbol{w},\tau). \tag{S.58}$$

In addition, from the proof of (S.19) and (S.20) in Theorem 1, we have

$$\begin{aligned} &\sup_{\boldsymbol{w}\in\mathcal{W}} \left| \mathrm{CV}(\boldsymbol{w},\tau) - \mathrm{FPE}^*_n(\boldsymbol{w},\tau) \right| \\ &= O_p(\frac{1}{\log n}) + O_p(\overline{p} n^{-1/2}) + O_p(\overline{p}^{1/2}\overline{c}_A^{1/2} n^{-1/2}) + O_p(n^{-1/p}) + O_p(n^{-1/2}), \end{aligned} \tag{S.59}$$

and

$$\sup_{\boldsymbol{w}\in\mathcal{W}}\left|\mathrm{FPE}_n^*(\boldsymbol{w},\tau)-\mathrm{FPE}_n(\boldsymbol{w},\tau)\right|=O_p(\overline{p}n^{-1/2}). \tag{S.60}$$

Consequently, by Assumption S.5 (i), Assumption S.6, (S.57), (S.58), (S.59), and (S.60), we have

$$\begin{aligned}
&\sup_{\boldsymbol{w}\in\mathcal{W}}\left|\frac{\mathrm{CV}^*(\boldsymbol{w},\tau)-R_n^*(\boldsymbol{w},\tau)}{R_n^*(\boldsymbol{w},\tau)}\right| \\
&\le \frac{1}{\xi_n^*}\sup_{\boldsymbol{w}\in\mathcal{W}}\left|\mathrm{CV}(\boldsymbol{w},\tau)-\mathrm{FPE}_n^*(\boldsymbol{w},\tau)\right| \\
&= O_p(\frac{1}{\xi_n^*\log n})+O_p(\frac{\overline{p}}{\xi_n^* n^{1/2}})+O_p(\frac{\overline{p}^{1/2}\overline{c}_A^{1/2}}{\xi_n^* n^{1/2}})+O_p(\frac{1}{\xi_n^* n^{1/p}})+O_p(\frac{1}{\xi_n^* n^{1/2}}) \\
&= o_p(1),
\end{aligned} \tag{S.61}$$

and

$$\begin{aligned}
&\sup_{\boldsymbol{w}\in\mathcal{W}}\left|\frac{R_n^*(\boldsymbol{w},\tau)-R_n(\boldsymbol{w},\tau)}{R_n^*(\boldsymbol{w},\tau)}\right| \\
&\le \frac{1}{\xi_n^*}\sup_{\boldsymbol{w}\in\mathcal{W}}\left|\mathrm{FPE}_n^*(\boldsymbol{w},\tau)-\mathrm{FPE}_n(\boldsymbol{w},\tau)\right| \\
&= O_p(\xi_n^{*-1}\overline{p}n^{-1/2}) \\
&= o_p(1),
\end{aligned} \tag{S.62}$$

where the last equalities in (S.61) and (S.62) hold by the fact that

$$\frac{\overline{p}}{\xi_n^* n^{1/2}}+\frac{\overline{p}^{1/2}\overline{c}_A^{1/2}}{\xi_n^* n^{1/2}}=O(\frac{1}{\xi_n^*\log n}\cdot\frac{\overline{p}\,\overline{c}_A\log n}{n^{1/2}})=o(1),$$

and

$$\frac{1}{\xi_n^* n^{1/p}}+\frac{1}{\xi_n^* n^{1/2}}=\frac{1}{\xi_n^*\log n}\cdot(\frac{\log n}{n^{1/2}}+\frac{\log n}{n^{1/p}})=o(1).$$

Therefore, we have proved (S.55) and (S.56), from which (S.54) follows. This completes the proof of Theorem 2. □

B.3 Proof of Theorem 3

Proof of Theorem 3. Denote $e_n = n^{-\lambda}$. Similar to the proof of Theorem 2 in Liao and Zou (2020), it is sufficient to show that there exists a constant C_0 such that, for the $M \times 1$ vector $\boldsymbol{v} = (v_1, \ldots, v_M)^\top$,

$$\lim_{n\to\infty} P\left(\inf_{\|\boldsymbol{v}\|=C_0, \boldsymbol{w}_\tau^0 + e_n\boldsymbol{v}\in\mathcal{W}} \mathrm{CV}(\boldsymbol{w}_\tau^0 + e_n\boldsymbol{v}, \tau) > \mathrm{CV}(\boldsymbol{w}_\tau^0, \tau) \right) = 1. \tag{S.63}$$

We first note that, by the proof of Theorem 2, we have

$$\sup_{\boldsymbol{w}\in\mathcal{W}} |\mathrm{CV}(\boldsymbol{w}, \tau) - \mathrm{FPE}_n^*(\boldsymbol{w}, \tau)| = o_p(\xi_n^*).$$

It should be noted that

$$\begin{aligned}
&\mathrm{CV}(\boldsymbol{w}_\tau^0 + e_n\boldsymbol{v}, \tau) - \mathrm{CV}(\boldsymbol{w}_\tau^0, \tau) \\
&= \Big[\mathrm{CV}(\boldsymbol{w}_\tau^0 + e_n\boldsymbol{v}, \tau) - R_n^*(\boldsymbol{w}_\tau^0 + e_n\boldsymbol{v}, \tau) - \mathsf{E}[\rho_\tau(\eta_{n+1}^*)] \\
&\quad - \mathrm{CV}(\boldsymbol{w}_\tau^0, \tau) + R_n^*(\boldsymbol{w}_\tau^0, \tau) + \mathsf{E}[\rho_\tau(\eta_{n+1}^*)] + R_n^*(\boldsymbol{w}_\tau^0 + e_n\boldsymbol{v}, \tau) - R_n^*(\boldsymbol{w}_\tau^0, \tau) \Big] \\
&= \Big[\mathrm{CV}(\boldsymbol{w}_\tau^0 + e_n\boldsymbol{v}, \tau) - \mathrm{FPE}_n^*(\boldsymbol{w}_\tau^0 + e_n\boldsymbol{v}, \tau) - \mathrm{CV}(\boldsymbol{w}_\tau^0, \tau) \\
&\quad + \mathrm{FPE}_n^*(\boldsymbol{w}_\tau^0, \tau) + R_n^*(\boldsymbol{w}_\tau^0 + e_n\boldsymbol{v}, \tau) - R_n^*(\boldsymbol{w}_\tau^0, \tau) \Big] \\
&= O_p(\xi_n^*) + R_n^*(\boldsymbol{w}_\tau^0 + e_n\boldsymbol{v}, \tau) - R_n^*(\boldsymbol{w}_\tau^0, \tau).
\end{aligned} \tag{S.64}$$

Consequently,

$$\begin{aligned}
&\inf_{\|\boldsymbol{v}\|=C_0, \boldsymbol{w}_\tau^0 + e_n\boldsymbol{v}\in\mathcal{W}} \mathrm{CV}(\boldsymbol{w}_\tau^0 + e_n\boldsymbol{v}, \tau) - \mathrm{CV}(\boldsymbol{w}_\tau^0, \tau) \\
&= O_p(\xi_n^*) + \inf_{\|\boldsymbol{v}\|=C_0, \boldsymbol{w}_\tau^0 + e_n\boldsymbol{v}\in\mathcal{W}} R_n^*(\boldsymbol{w}_\tau^0 + e_n\boldsymbol{v}, \tau) - R_n^*(\boldsymbol{w}_\tau^0, \tau).
\end{aligned} \tag{S.65}$$

Recalling that $\boldsymbol{w}_\tau^0 = \arg\min_{\boldsymbol{w}\in\mathcal{W}} R_n^*(\boldsymbol{w}, \tau)$ and $\xi_n^* = \inf_{\boldsymbol{w}\in\mathcal{W}} R_n^*(\boldsymbol{w}, \tau)$, by Assumption S.7, we have

$$\xi_n^* / [R_n^*(\boldsymbol{w}_\tau^0 + e_n\boldsymbol{v}, \tau) - R_n^*(\boldsymbol{w}_\tau^0, \tau)] = o(1). \tag{S.66}$$

In addition, we have

$$R_n^*(\boldsymbol{w}_\tau^0 + e_n \boldsymbol{v}, \tau) - R_n^*(\boldsymbol{w}_\tau^0, \tau) > 0. \tag{S.67}$$

Combining (S.65), (S.66), (S.67), we conclude that (S.65) is positive with probability approaching 1. This completes the proof of Theorem 3. □

B.4 Proof of Theorem 4

Proof of Theorem 4. By the proof of Theorem 1 and Assumption S.8, we have

$$\sup_{\boldsymbol{w} \in \mathcal{W}} |\mathrm{CV}(\boldsymbol{w}, \tau) - \mathrm{FPE}_n^*(\boldsymbol{w}, \tau)| = o_p(\widetilde{\xi}_n). \tag{S.68}$$

Recall that $\mathrm{CV}^*(\boldsymbol{w}, \tau) = \mathrm{CV}(\boldsymbol{w}, \tau) - \mathsf{E}[\rho_\tau(\eta_{n+1}^*)]$. It is seen that

$$\mathrm{CV}(\boldsymbol{w}, \tau) - \mathrm{FPE}_n^*(\boldsymbol{w}, \tau) = \mathrm{CV}^*(\boldsymbol{w}, \tau) - R_n^*(\boldsymbol{w}, \tau). \tag{S.69}$$

For any correctly specified model $m \in \mathcal{M}_c$, we note that

$$P\Big(\alpha_{n+1}^* \le \Delta^{*(m)}(\tau; \boldsymbol{X}_{n+1}) | \boldsymbol{X}_{n+1}\Big) = \tau, \tag{S.70}$$

which implies that $\Delta^{*(m)}(\tau; \boldsymbol{X}_{n+1}) = \Delta(\tau; \boldsymbol{X}_{n+1})$. Therefore, we have

$$\begin{aligned}
&R_n^*(\boldsymbol{w}, \tau) \\
&= \mathsf{E}[\rho_\tau\{\alpha_{n+1}^* - \Delta^*(\tau; \boldsymbol{X}_{n+1}, \boldsymbol{w})\} - \rho_\tau\{\alpha_{n+1}^* - \Delta(\tau; \boldsymbol{X}_{n+1})\}] \\
&= \mathsf{E}\{\psi_\tau\{\widetilde{\varepsilon}_{n+1}(\boldsymbol{w})\}[\Delta(\tau; \boldsymbol{X}_{n+1}) - \Delta^*(\tau; \boldsymbol{X}_{n+1}, \boldsymbol{w})]\} \\
&= \mathsf{E}\left\{\sum_{m=1}^{M} w_m \psi_\tau\{\widetilde{\varepsilon}_{n+1}(\boldsymbol{w})\}[\Delta(\tau; \boldsymbol{X}_{n+1}) - \Delta^{*(m)}(\tau; \boldsymbol{X}_{n+1})]\right\} \\
&= \mathsf{E}\left\{\sum_{m \in \mathcal{M}_c} w_m \psi_\tau\{\widetilde{\varepsilon}_{n+1}(\boldsymbol{w})\}[\Delta(\tau; \boldsymbol{X}_{n+1}) - \Delta^{*(m)}(\tau; \boldsymbol{X}_{n+1})]\right\} \\
&\quad + \mathsf{E}\left\{\sum_{m \notin \mathcal{M}_c} w_m \psi_\tau\{\widetilde{\varepsilon}_{n+1}(\boldsymbol{w})\}[\Delta(\tau; \boldsymbol{X}_{n+1}) - \Delta^{*(m)}(\tau; \boldsymbol{X}_{n+1})]\right\} \\
&= \mathsf{E}\left\{\sum_{m \notin \mathcal{M}_c} w_m \psi_\tau\{\widetilde{\varepsilon}_{n+1}(\boldsymbol{w})\}[\Delta(\tau; \boldsymbol{X}_{n+1}) - \Delta^{*(m)}(\tau; \boldsymbol{X}_{n+1})]\right\}
\end{aligned}$$

$$\equiv R^*_{n,1}(\boldsymbol{w},\tau), \tag{S.71}$$

where $\widetilde{\varepsilon}_{n+1}(\boldsymbol{w})$ is between $\alpha^*_{n+1}-\Delta^*(\tau;\boldsymbol{X}_{n+1},\boldsymbol{w})$ and $\alpha^*_{n+1}-\Delta(\tau;\boldsymbol{X}_{n+1})$. Hence, combining (S.68), (S.69), and (S.71), we obtain

$$\mathrm{CV}^*(\boldsymbol{w},\tau)=R^*_{n,1}(\boldsymbol{w},\tau)+o_p(\widetilde{\xi}_n). \tag{S.72}$$

If $\widehat{\theta}_\tau=1$, then the conclusion of Theorem 4 is obviously true. In the case $\widehat{\theta}_\tau<1$, we state that $\widehat{\theta}_\tau\to 1$ in probability when $n\to\infty$. Otherwise, there exists a positive constant $c_\tau<1$, such that $\widehat{\theta}_\tau\le c_\tau$ as $n\to\infty$. In this case, for a sufficiently large n, we have

$$\begin{aligned} R^*_{n,1}(\widehat{\boldsymbol{w}}_\tau,\tau) &= (1-\widehat{\theta}_\tau)(1-\widehat{\theta}_\tau)^{-1}R^*_{n,1}(\widehat{\boldsymbol{w}}_\tau,\tau)\\ &\ge (1-\widehat{\theta}_\tau)\inf_{\boldsymbol{w}\in\mathcal{W}_S}(1-\theta_\tau)^{-1}R^*_{n,1}(\boldsymbol{w},\tau)\\ &= (1-\widehat{\theta}_\tau)\zeta_n, \end{aligned} \tag{S.73}$$

where $\zeta_n=\inf_{\boldsymbol{w}\in\mathcal{W}_S}(1-\theta_\tau)^{-1}R^*_{n,1}(\boldsymbol{w},\tau)$ with $\mathcal{W}_S=\{\boldsymbol{w}\in\mathcal{W}:\theta_\tau\le c_\tau\}$. By Assumption S.8 (i), we have

$$\zeta_n=\inf_{\boldsymbol{w}\in\mathcal{W}_S}(1-\theta_\tau)^{-1}R^*_{n,1}(\boldsymbol{w},\tau)\ge\inf_{\boldsymbol{w}\in\mathcal{W}_S}R^*_n(\boldsymbol{w},\tau)=\widetilde{\xi}_n>0. \tag{S.74}$$

Let $\boldsymbol{w}^m\in\mathcal{W}$ be a vector with the mth element taking a value 1 and all others taking a value 0, where $m\in\mathcal{M}_c$. We note that

$$\begin{aligned} &R^*_n(\boldsymbol{w}^m,\tau)\\ &=\mathsf{E}\Big[\rho_\tau\left\{\alpha^*_{n+1}-\Delta^*(\tau;\boldsymbol{X}_{n+1},\boldsymbol{w}^m)\right\}-\rho_\tau(\eta^*_{n+1})\Big]\\ &=\mathsf{E}\Big[\rho_\tau\left\{\Delta(\tau;\boldsymbol{X}_{n+1})-\Delta^{*(m)}(\tau;\boldsymbol{X}_{n+1})+\eta^*_{n+1}\right\}-\rho_\tau(\eta^*_{n+1})\Big]\\ &=0. \end{aligned} \tag{S.75}$$

By the definition of $\mathrm{CV}^*(\boldsymbol{w},\tau)$, we have $\mathrm{CV}^*(\widehat{\boldsymbol{w}}_\tau,\tau)\le\mathrm{CV}^*(\boldsymbol{w}^m,\tau)$. Combining (S.72), (S.73), (S.74), and (S.75), we have

$$(1-\widehat{\theta}_\tau)\zeta_n\le R^*_{n,1}(\widehat{\boldsymbol{w}}_\tau,\tau)\le o_p(\widetilde{\xi}_n). \tag{S.76}$$

Hence, $\widehat{\theta}_\tau \to 1$ in probability. This contradicts the initial assumption that $\widehat{\theta}_\tau \le c_\tau$ as $n \to \infty$, thereby establishing the desired result. This completes the proof of Theorem 4. □

B.5 Proof of Theorem 5

Proof of Theorem 5. Define $\overline{\boldsymbol{w}}_\tau = (0, \ldots, 0, \sum_{m=1}^{M_0+1} \widehat{w}_{\tau,m}, \widehat{w}_{\tau,M_0+2}, \ldots, \widehat{w}_{\tau,M})^\top$. Because $\widehat{\boldsymbol{w}}_\tau = \arg\min_{\boldsymbol{w}\in\mathcal{W}} \mathrm{CV}(\boldsymbol{w}, \tau)$, we have

$$\mathrm{CV}(\widehat{\boldsymbol{w}}_\tau, \tau) - \mathrm{CV}(\overline{\boldsymbol{w}}_\tau, \tau) \le 0. \tag{S.77}$$

Let $R_n^0(\boldsymbol{w}, \tau) = \frac{1}{n}\sum_{k=1}^{K}\sum_{i\in\mathcal{O}_k}\int_0^{\widetilde{\Delta}^{(-k)}(\tau;\boldsymbol{X}_i,\boldsymbol{w})-\Delta(\tau;\boldsymbol{X}_i)}[F_{\eta^*}(s|\boldsymbol{X}_i) - F_{\eta^*}(0|\boldsymbol{X}_i)]ds$. By (S.2), it follows that

$$\begin{aligned}
&\mathrm{CV}(\widehat{\boldsymbol{w}}_\tau, \tau) - \mathrm{CV}(\overline{\boldsymbol{w}}_\tau, \tau) \\
&= \mathrm{CV}(\widehat{\boldsymbol{w}}_\tau, \tau) - \frac{1}{n}\sum_{i=1}^{n}\rho_\tau(\eta_i^*) - R_n^0(\widehat{\boldsymbol{w}}_\tau, \tau) \\
&\quad - \mathrm{CV}(\overline{\boldsymbol{w}}_\tau, \tau) + \frac{1}{n}\sum_{i=1}^{n}\rho_\tau(\eta_i^*) + R_n^0(\overline{\boldsymbol{w}}_\tau, \tau) + R_n^0(\widehat{\boldsymbol{w}}_\tau, \tau) - R_n^0(\overline{\boldsymbol{w}}_\tau, \tau) \\
&= D_1(\widehat{\boldsymbol{w}}_\tau, \overline{\boldsymbol{w}}_\tau) + D_2(\widehat{\boldsymbol{w}}_\tau, \overline{\boldsymbol{w}}_\tau) + D_3(\widehat{\boldsymbol{w}}_\tau, \overline{\boldsymbol{w}}_\tau), + D_4(\widehat{\boldsymbol{w}}_\tau, \overline{\boldsymbol{w}}_\tau), + D_5(\widehat{\boldsymbol{w}}_\tau, \overline{\boldsymbol{w}}_\tau),
\end{aligned} \tag{S.78}$$

where

$$D_1(\widehat{\boldsymbol{w}}_\tau, \overline{\boldsymbol{w}}_\tau) = \frac{1}{n}\sum_{k=1}^{K}\sum_{i\in\mathcal{O}_k}[\widetilde{\Delta}^{(-k)}(\tau; \boldsymbol{X}_i, \overline{\boldsymbol{w}}_\tau) - \widetilde{\Delta}^{(-k)}(\tau; \boldsymbol{X}_i, \widehat{\boldsymbol{w}}_\tau)]\psi_\tau(\eta_i^*),$$

$$\begin{aligned}
&D_2(\widehat{\boldsymbol{w}}_\tau, \overline{\boldsymbol{w}}_\tau) \\
&= \frac{1}{n}\sum_{k=1}^{K}\sum_{i\in\mathcal{O}_k}\int_{\widetilde{\Delta}^{(-k)}(\tau;\boldsymbol{X}_i,\overline{\boldsymbol{w}}_\tau)-\Delta(\tau;\boldsymbol{X}_i)}^{\widetilde{\Delta}^{(-k)}(\tau;\boldsymbol{X}_i,\widehat{\boldsymbol{w}}_\tau)-\Delta(\tau;\boldsymbol{X}_i)}[\mathbf{1}\{\eta_i^* \le s\} - \mathbf{1}\{\eta_i^* \le 0\} - F_{\eta^*}(s|\boldsymbol{X}_i) + F_{\eta^*}(0|\boldsymbol{X}_i)]ds,
\end{aligned}$$

$$D_3(\widehat{\boldsymbol{w}}_\tau, \overline{\boldsymbol{w}}_\tau)$$

$$= \frac{1}{n}\sum_{k=1}^{K}\sum_{i\in\mathcal{O}_k}\left(\int_{\widetilde{\Delta}^{(-k)}(\tau;\boldsymbol{X}_i,\widehat{\boldsymbol{w}}_\tau)-\Delta(\tau;\boldsymbol{X}_i)}^{\widetilde{\Delta}^{(-k)}(\tau;\boldsymbol{X}_i,\widehat{\boldsymbol{w}}_\tau)-\Delta(\tau;\boldsymbol{X}_i)-b_i(\tau)} [\mathbf{1}\{\eta_i^* \le s\} - \mathbf{1}\{\eta_i^* \le 0\}]ds \right.$$
$$\left. - \int_{\widetilde{\Delta}^{(-k)}(\tau;\boldsymbol{X}_i,\overline{\boldsymbol{w}}_\tau)-\Delta(\tau;\boldsymbol{X}_i)}^{\widetilde{\Delta}^{(-k)}(\tau;\boldsymbol{X}_i,\overline{\boldsymbol{w}}_\tau)-\Delta(\tau;\boldsymbol{X}_i)-b_i(\tau)} [\mathbf{1}\{\eta_i^* \le s\} - \mathbf{1}\{\eta_i^* \le 0\}]ds \right),$$

$$D_4(\widehat{\boldsymbol{w}}_\tau, \overline{\boldsymbol{w}}_\tau)$$
$$= \frac{1}{n}\sum_{k=1}^{K}\sum_{i\in\mathcal{O}_k}\left(\int_{\widetilde{\Delta}^{(-k)}(\tau;\boldsymbol{X}_i,\widehat{\boldsymbol{w}}_\tau)-\Delta(\tau;\boldsymbol{X}_i)-b_i(\tau)}^{\widetilde{\Delta}^{(-k)}(\tau;\boldsymbol{X}_i,\widehat{\boldsymbol{w}}_\tau)-\Delta(\tau;\boldsymbol{X}_i)-b_i(\tau)+\eta_i^*-\widetilde{\eta}_i} \mathbf{1}\{\eta_i^* \le s\}\, ds \right.$$
$$\left. - \int_{\widetilde{\Delta}^{(-k)}(\tau;\boldsymbol{X}_i,\overline{\boldsymbol{w}}_\tau)-\Delta(\tau;\boldsymbol{X}_i)-b_i(\tau)}^{\widetilde{\Delta}^{(-k)}(\tau;\boldsymbol{X}_i,\overline{\boldsymbol{w}}_\tau)-\Delta(\tau;\boldsymbol{X}_i)-b_i(\tau)+\eta_i^*-\widetilde{\eta}_i} \mathbf{1}\{\eta_i^* \le s\}\, ds \right),$$

and

$$D_5(\widehat{\boldsymbol{w}}_\tau, \overline{\boldsymbol{w}}_\tau) = R_n^0(\widehat{\boldsymbol{w}}_\tau, \tau) - R_n^0(\overline{\boldsymbol{w}}_\tau, \tau).$$

By the definition of $\widehat{\boldsymbol{w}}_\tau$ and $\overline{\boldsymbol{w}}_\tau$, we have

$$\widetilde{\Delta}^{(-k)}(\tau;\boldsymbol{X}_i,\overline{\boldsymbol{w}}_\tau) - \widetilde{\Delta}^{(-k)}(\tau;\boldsymbol{X}_i,\widehat{\boldsymbol{w}}_\tau) = \sum_{m=1}^{M_0}\widehat{w}_{\tau,m}\left\{\widetilde{\Delta}_{-k}^{(M_0+1)}(\tau;\boldsymbol{X}_i) - \widetilde{\Delta}_{-k}^{(m)}(\tau;\boldsymbol{X}_i)\right\},$$

where $\widetilde{\Delta}_{-k}^{(m)}(\tau;\boldsymbol{X}_i) = \boldsymbol{X}_i^{(m)\top}\left\{\widehat{\boldsymbol{\beta}}_{1,-k}^{(m)}(\tau) - \widehat{\boldsymbol{\beta}}_{0,-k}^{(m)}(\tau)\right\}$. Therefore, we have

$$D_1(\widehat{\boldsymbol{w}}_\tau, \overline{\boldsymbol{w}}_\tau) = \frac{1}{n}\sum_{k=1}^{K}\sum_{i\in\mathcal{O}_k}\left[\sum_{m=1}^{M_0}\widehat{w}_{\tau,m}\left\{\widetilde{\Delta}_{-k}^{(M_0+1)}(\tau;\boldsymbol{X}_i) - \widetilde{\Delta}_{-k}^{(m)}(\tau;\boldsymbol{X}_i)\right\}\right]\psi_\tau(\eta_i^*).$$

For fixed p, Lemma 1 implies that, with probability approaching 1, $\widehat{\boldsymbol{\beta}}_{d,-k}^{(m)}(\tau)$ lies in the ball centered at $\boldsymbol{\beta}_d^{*(m)}(\tau)$ with radius $C/\sqrt{n}$, that is,

$$\left\{\boldsymbol{\beta}_d^{*(m)}(\tau) + \frac{1}{\sqrt{n}}l_d^{(m)}(\tau) : \|l_d^{(m)}(\tau)\| = C\right\}$$

for $m = 1,\ldots,M$, $k = 1,\ldots,K$, and $d = 0, 1$. Let $l_d(\tau) = (l_d^{(1)}(\tau), l_d^{(2)}(\tau), \ldots, l_d^{(M)}(\tau))^\top$. Then, for each $\boldsymbol{w} \in \mathcal{W}$, let $\boldsymbol{w}^U = (w_1,\ldots,w_{M_0})^\top$ and $\overline{\boldsymbol{w}} = (0,\ldots,0,\sum_{m=1}^{M_0+1}w_m, w_{M_0+2},\ldots,w_M)^\top$.

Define

$$D(\boldsymbol{w}, l_1(\tau), l_0(\tau)) = \frac{1}{n}\sum_{i=1}^{n}\psi_\tau(\eta_i^*)\Bigg(\sum_{m=1}^{M_0} w_m\Big[g(\boldsymbol{X}_i, l_1^{(M_0+1)}(\tau), l_0^{(M_0+1)}(\tau)) - g(\boldsymbol{X}_i, l_1^{(m)}(\tau), l_0^{(m)}(\tau))\Big]\Bigg),$$

where

$$\begin{aligned} g(\boldsymbol{X}_i, l_1^{(m)}(\tau), l_0^{(m)}(\tau)) &= \boldsymbol{X}_i^{(m)\top}\left\{\boldsymbol{\beta}_1^{*(m)}(\tau) + \frac{1}{\sqrt{n}}l_1^{(m)} - \boldsymbol{\beta}_0^{*(m)}(\tau) - \frac{1}{\sqrt{n}}l_0^{(m)}\right\} \\ &= \boldsymbol{X}_i^{(m)\top}\left\{\boldsymbol{\beta}_1^{*(m)}(\tau) - \boldsymbol{\beta}_0^{*(m)}(\tau) + \frac{1}{\sqrt{n}}\Big(l_1^{(m)} - l_0^{(m)}\Big)\right\}. \end{aligned}$$

Note that $\mathsf{E}[D(\boldsymbol{w}, l_1(\tau), l_0(\tau))] = 0$ by (S.3). Hence,

$$\begin{aligned} &Var[D(\boldsymbol{w}, l_1(\tau), l_0(\tau))] \\ &= \mathsf{E}\left[\frac{1}{n}\sum_{i=1}^{n}\psi_\tau(\eta_i^*)\left\{\sum_{m=1}^{M_0} w_m[g(\boldsymbol{X}_i, l_1^{(M_0+1)}(\tau), l_0^{(M_0+1)}(\tau)) - g(\boldsymbol{X}_i, l_1^{(m)}(\tau), l_0^{(m)}(\tau))]\right\}\right]^2 \\ &\le 2\mathsf{E}\left[\frac{1}{n}\sum_{i=1}^{n}\psi_\tau(\eta_i^*)\left\{\sum_{m=1}^{M_0} w_m[\Delta^{*(M_0+1)}(\tau;\boldsymbol{X}_i) - \Delta^{*(m)}(\tau;\boldsymbol{X}_i)]\right\}\right]^2 \\ &\quad + 2\mathsf{E}\left[\frac{1}{n}\sum_{i=1}^{n}\psi_\tau(\eta_i^*)\left\{\sum_{m=1}^{M_0} w_m[\Phi^{(M_0+1)}(\tau;\boldsymbol{X}_i) - \Phi^{(m)}(\tau;\boldsymbol{X}_i)]\right\}\right]^2, \end{aligned} \tag{S.79}$$

where $\Phi^{(m)}(\tau;\boldsymbol{X}_i) = \frac{1}{\sqrt{n}}\boldsymbol{X}_i^{(m)\top}(l_1^{(m)}(\tau) - l_0^{(m)}(\tau))$. By the definition of $\psi_\tau(u)$, we have $|\psi_\tau(u)| \le 1$. Hence, by Cauchy-Schwarz inequality, Assumption S.3(iii) and for fixed p, we obtain

$$\begin{aligned} &\mathsf{E}\left[\frac{1}{n}\sum_{i=1}^{n}\psi_\tau(\eta_i^*)\left\{\sum_{m=1}^{M_0} w_m[\Delta^{*(M_0+1)}(\tau;\boldsymbol{X}_i) - \Delta^{*(m)}(\tau;\boldsymbol{X}_i)]\right\}\right]^2 \\ &\le \frac{1}{n^2}\sum_{i=1}^{n}\mathsf{E}\left[\left\{\sum_{m=1}^{M_0} w_m[\Delta^{*(M_0+1)}(\tau;\boldsymbol{X}_i) - \Delta^{*(m)}(\tau;\boldsymbol{X}_i)]\right\}^2\right] \\ &\le \frac{1}{n^2}\sum_{i=1}^{n}\Big(\sum_{m=1}^{M_0} w_m^2\Big)\sum_{m=1}^{M_0}\mathsf{E}[\Delta^{*(M_0+1)}(\tau;\boldsymbol{X}_i) - \Delta^{*(m)}(\tau;\boldsymbol{X}_i)]^2 \\ &= O\Big(\frac{1}{n}\Big)\|\boldsymbol{w}^U\|^2, \end{aligned} \tag{S.80}$$

and

$$\begin{aligned}
&\mathsf{E}\left[\frac{1}{n}\sum_{i=1}^{n}\psi_\tau(\eta_i^*)\left\{\sum_{m=1}^{M_0}w_m[\Phi^{(M_0+1)}(\tau;\boldsymbol{X}_i)-\Phi^{(m)}(\tau;\boldsymbol{X}_i)]\right\}\right]^2\\
&\le \frac{1}{n^2}\sum_{i=1}^{n}\Big(\sum_{m=1}^{M_0}w_m^2\Big)\sum_{m=1}^{M_0}\mathsf{E}[\Phi^{(M_0+1)}(\tau;\boldsymbol{X}_i)-\Phi^{(m)}(\tau;\boldsymbol{X}_i)]^2\\
&= O\Big(\frac{1}{n^2}\Big)\|\boldsymbol{w}^U\|^2. \qquad \text{(S.81)}
\end{aligned}$$

Therefore, combining (S.79), (S.80), (S.81), and by Chebyshev inequality, we have

$$|D(\boldsymbol{w},l_1(\tau),l_0(\tau))|= O_p\Big(\frac{1}{\sqrt{n}}\Big)\|\boldsymbol{w}^U\|. \qquad \text{(S.82)}$$

It follows that

$$|D_1(\widehat{\boldsymbol{w}}_\tau,\overline{\boldsymbol{w}}_\tau)|= O_p(\frac{1}{\sqrt{n}})\|\widehat{\boldsymbol{w}}_\tau^U\|. \qquad \text{(S.83)}$$

In a similar process, we can prove that

$$|D_2(\widehat{\boldsymbol{w}}_\tau,\overline{\boldsymbol{w}}_\tau)|= O_p(\frac{1}{\sqrt{n}})\|\widehat{\boldsymbol{w}}_\tau^U\|. \qquad \text{(S.84)}$$

Then, we consider $D_3(\widehat{\boldsymbol{w}}_\tau,\overline{\boldsymbol{w}}_\tau)$. Note that

$$|D_3(\widehat{\boldsymbol{w}}_\tau,\overline{\boldsymbol{w}}_\tau)|\le \frac{2}{n}\sum_{i=1}^{n}|b_i(\tau)|,$$

and hence, by the argument of (S.46) in the proof of Theorem 1, we can prove that

$$|D_3(\widehat{\boldsymbol{w}}_\tau,\overline{\boldsymbol{w}}_\tau)|= O_p(n^{-1/p}). \qquad \text{(S.85)}$$

Next, we deal with $D_4(\widehat{\boldsymbol{w}}_\tau,\overline{\boldsymbol{w}}_\tau)$. We notice that

$$\begin{aligned}
|D_4(\widehat{\boldsymbol{w}}_\tau,\overline{\boldsymbol{w}}_\tau)| \le &\left|\frac{1}{n}\sum_{k=1}^{K}\sum_{i\in\mathcal{O}_k}\int_{\widetilde{\Delta}^{(-k)}(\tau;\boldsymbol{X}_i,\widehat{\boldsymbol{w}}_\tau)-\Delta(\tau;\boldsymbol{X}_i)-b_i(\tau)}^{\widetilde{\Delta}^{(-k)}(\tau;\boldsymbol{X}_i,\widehat{\boldsymbol{w}}_\tau)-\Delta(\tau;\boldsymbol{X}_i)-b_i(\tau)+\eta_i^*-\widetilde{\eta}_i}\mathbf{1}\{\eta_i^*\le s\}\,ds\right|\\
&+\left|\frac{1}{n}\sum_{k=1}^{K}\sum_{i\in\mathcal{O}_k}\int_{\widetilde{\Delta}^{(-k)}(\tau;\boldsymbol{X}_i,\overline{\boldsymbol{w}}_\tau)-\Delta(\tau;\boldsymbol{X}_i)-b_i(\tau)}^{\widetilde{\Delta}^{(-k)}(\tau;\boldsymbol{X}_i,\overline{\boldsymbol{w}}_\tau)-\Delta(\tau;\boldsymbol{X}_i)-b_i(\tau)+\eta_i^*-\widetilde{\eta}_i}\mathbf{1}\{\eta_i^*\le s\}\,ds\right|
\end{aligned}$$

$$\equiv |D_{41}(\widehat{\boldsymbol{w}}_\tau)|+|D_{42}(\overline{\boldsymbol{w}}_\tau)|.$$

By the proof of (S.50), we have

$$|D_{41}(\widehat{\boldsymbol{w}}_\tau)|= O_p(\frac{1}{\sqrt{n}}),$$

and

$$|D_{42}(\overline{\boldsymbol{w}}_\tau)|= O_p(\frac{1}{\sqrt{n}}),$$

which indicate that

$$|D_4(\widehat{\boldsymbol{w}}_\tau, \overline{\boldsymbol{w}}_\tau)|= O_p(\frac{1}{\sqrt{n}}). \tag{S.86}$$

Finally, we focus on $D_5(\widehat{\boldsymbol{w}}_\tau, \overline{\boldsymbol{w}}_\tau)$. Note that

$$\begin{aligned}
&D_5(\widehat{\boldsymbol{w}}_\tau, \overline{\boldsymbol{w}}_\tau) \\
&= R_n^0(\widehat{\boldsymbol{w}}_\tau, \tau) - R_n^0(\overline{\boldsymbol{w}}_\tau, \tau) \\
&= \frac{1}{n}\sum_{k=1}^{K}\sum_{i\in\mathcal{O}_k}\int_{\widetilde{\Delta}^{(-k)}(\tau;\boldsymbol{X}_i,\overline{\boldsymbol{w}}_\tau)-\Delta(\tau;\boldsymbol{X}_i)}^{\widetilde{\Delta}^{(-k)}(\tau;\boldsymbol{X}_i,\widehat{\boldsymbol{w}}_\tau)-\Delta(\tau;\boldsymbol{X}_i)} [F_{\eta^*}(s|\boldsymbol{X}_i) - F_{\eta^*}(0|\boldsymbol{X}_i)]ds \\
&= D_{51}(\widehat{\boldsymbol{w}}_\tau, \overline{\boldsymbol{w}}_\tau) + D_{52}(\widehat{\boldsymbol{w}}_\tau, \overline{\boldsymbol{w}}_\tau) + D_{53}(\widehat{\boldsymbol{w}}_\tau, \overline{\boldsymbol{w}}_\tau),
\end{aligned} \tag{S.87}$$

where

$$D_{51}(\widehat{\boldsymbol{w}}_\tau, \overline{\boldsymbol{w}}_\tau) = \frac{1}{n}\sum_{k=1}^{K}\sum_{i\in\mathcal{O}_k}\frac{1}{2}f_{\eta^*}(0|\boldsymbol{X}_i)\left\{\widetilde{\Delta}^{(-k)}(\tau;\boldsymbol{X}_i,\widehat{\boldsymbol{w}}_\tau) - \widetilde{\Delta}^{(-k)}(\tau;\boldsymbol{X}_i,\overline{\boldsymbol{w}}_\tau)\right\}^2,$$

$$\begin{aligned}
D_{52}(\widehat{\boldsymbol{w}}_\tau, \overline{\boldsymbol{w}}_\tau) = \frac{1}{n}\sum_{k=1}^{K}\sum_{i\in\mathcal{O}_k} f_{\eta^*}(0|\boldsymbol{X}_i)&\left\{\widetilde{\Delta}^{(-k)}(\tau;\boldsymbol{X}_i,\widehat{\boldsymbol{w}}_\tau) - \widetilde{\Delta}^{(-k)}(\tau;\boldsymbol{X}_i,\overline{\boldsymbol{w}}_\tau)\right\} \\
&\cdot\left\{\widetilde{\Delta}^{(-k)}(\tau;\boldsymbol{X}_i,\overline{\boldsymbol{w}}_\tau) - \Delta(\tau;\boldsymbol{X}_i)\right\},
\end{aligned}$$

and

$$D_{53}(\widehat{\boldsymbol{w}}_\tau, \overline{\boldsymbol{w}}_\tau) = \frac{1}{n}\sum_{k=1}^{K}\sum_{i\in\mathcal{O}_k}\int_{\widetilde{\Delta}^{(-k)}(\tau;\boldsymbol{X}_i,\overline{\boldsymbol{w}}_\tau)-\Delta(\tau;\boldsymbol{X}_i)}^{\widetilde{\Delta}^{(-k)}(\tau;\boldsymbol{X}_i,\widehat{\boldsymbol{w}}_\tau)-\Delta(\tau;\boldsymbol{X}_i)} \frac{1}{2} f'_{\eta^*}(\overline{s}|\boldsymbol{X}_i)s^2 ds.$$

Here $\overline{s}$ is between $\widetilde{\Delta}^{(-k)}(\tau;\boldsymbol{X}_i,\overline{\boldsymbol{w}}_\tau)-\Delta(\tau;\boldsymbol{X}_i)$ and $\widetilde{\Delta}^{(-k)}(\tau;\boldsymbol{X}_i,\widehat{\boldsymbol{w}}_\tau)-\Delta(\tau;\boldsymbol{X}_i)$.

By Cauchy-Schwarz inequality, Assumption S.3 (iii) for fixed p, and Assumption S.9, we have

$$\begin{aligned}
&|D_{51}(\widehat{\boldsymbol{w}}_\tau, \overline{\boldsymbol{w}}_\tau)| \\
&\le \frac{C}{n}\sum_{i=1}^{n}\Big|\Delta^*(\tau;\boldsymbol{X}_i,\widehat{\boldsymbol{w}}_\tau)-\Delta^*(\tau;\boldsymbol{X}_i,\overline{\boldsymbol{w}}_\tau)+\Phi(\tau;\boldsymbol{X}_i,\widehat{\boldsymbol{w}}_\tau)-\Phi(\tau;\boldsymbol{X}_i,\overline{\boldsymbol{w}}_\tau)\Big|^2 \\
&\le C||\widehat{\boldsymbol{w}}_\tau^U||^2\Big(\sum_{m=1}^{M_0}\frac{1}{n}\sum_{i=1}^{n}[\Delta^{*(m)}(\tau;\boldsymbol{X}_i)-\Delta^{*(M_0+1)}(\tau;\boldsymbol{X}_i)]^2\Big) \\
&\quad + C||\widehat{\boldsymbol{w}}_\tau^U||^2\Big(\sum_{m=1}^{M_0}\frac{1}{n}\sum_{i=1}^{n}[\Phi^{*(m)}(\tau;\boldsymbol{X}_i)-\Phi^{*(M_0+1)}(\tau;\boldsymbol{X}_i)]^2\Big) \\
&\le C||\widehat{\boldsymbol{w}}_\tau^U||^2\Big(\max_{1\le m\le M}\frac{1}{n}\sum_{i=1}^{n}[\Delta^{*(m)}(\tau;\boldsymbol{X}_i)-\Delta^{*(M_0+1)}(\tau;\boldsymbol{X}_i)]^2\Big) \\
&\quad + C||\widehat{\boldsymbol{w}}_\tau^U||^2\Big(\max_{1\le m\le M}\frac{1}{n}\sum_{i=1}^{n}[\Phi^{*(m)}(\tau;\boldsymbol{X}_i)-\Phi^{*(M_0+1)}(\tau;\boldsymbol{X}_i)]^2\Big) \\
&\le C\|\widehat{\boldsymbol{w}}_\tau^U\|^2, \qquad \text{(S.88)}
\end{aligned}$$

where $\Phi(\tau;\boldsymbol{X}_i,\boldsymbol{w}) = \frac{1}{\sqrt{n}}\sum_{m=1}^{M} w_m\Phi^{(m)}(\tau;\boldsymbol{X}_i)$.

Then, we consider $D_{52}(\widehat{\boldsymbol{w}}_\tau, \overline{\boldsymbol{w}}_\tau)$. We notice that for any correct models $m \in \{M_0+1,\ldots,M\}$,

$$\begin{aligned}
&g(\boldsymbol{X}_i, l_1^{(m)}(\tau), l_0^{(m)}(\tau)) - \Delta(\tau;\boldsymbol{X}_i) \\
&= \boldsymbol{X}_i^{(m)\top}\left\{\boldsymbol{\beta}_1^{*(m)}(\tau)-\boldsymbol{\beta}_0^{*(m)}(\tau)+\frac{1}{\sqrt{n}}\Big(l_1^{(m)}-l_0^{(m)}\Big)\right\}-\Delta^{*(m)}(\tau;\boldsymbol{X}_i) \\
&= O_p(\frac{1}{\sqrt{n}}).
\end{aligned}$$

Therefore, by Assumption S.3 (iii) for fixed p and Assumption S.9, we obtain

$$
\begin{aligned}
&|D_{52}(\widehat{\boldsymbol{w}}_\tau, \overline{\boldsymbol{w}}_\tau)| \\
&= \left| \frac{1}{n} \sum_{i=1}^{n} f_{\eta^*}(0|\boldsymbol{X}_i) \left\{ \sum_{m=1}^{M_0} \widehat{w}_{\tau,m} [g(\boldsymbol{X}_i, l_1^{(m)}(\tau), l_0^{(m)}(\tau)) - g(\boldsymbol{X}_i, l_1^{(M_0+1)}(\tau), l_0^{(M_0+1)}(\tau))] \right\} \right. \\
&\qquad \left. \cdot \left\{ \sum_{m=M_0+1}^{M} \overline{w}_{\tau,m} [g(\boldsymbol{X}_i, l_1^{(m)}(\tau), l_0^{(m)}(\tau)) - \Delta(\tau; \boldsymbol{X}_i)] \right\} \right| \\
&\leq \frac{C}{\sqrt{n}} (\sum_{m=1}^{M_0} \widehat{w}_{\tau,m}) \max_{1 \leq m \leq M} \frac{1}{n} \sum_{i=1}^{n} |g(\boldsymbol{X}_i, l_1^{(m)}(\tau), l_0^{(m)}(\tau)) - g(\boldsymbol{X}_i, l_1^{(M_0+1)}(\tau), l_0^{(M_0+1)}(\tau))| \\
&= O_p(\frac{1}{\sqrt{n}}) ||\widehat{\boldsymbol{w}}_\tau^U||. \qquad \text{(S.89)}
\end{aligned}
$$

Next, we focus on $D_{53}(\widehat{\boldsymbol{w}}_\tau, \overline{\boldsymbol{w}}_\tau)$. By Assumption S.9, we have

$$
\begin{aligned}
&|D_{53}(\widehat{\boldsymbol{w}}_\tau, \overline{\boldsymbol{w}}_\tau)| \\
&\leq \frac{C}{n} \sum_{k=1}^{K} \sum_{i \in \mathcal{O}_k} \left| [\widetilde{\Delta}^{(-k)}(\tau; \boldsymbol{X}_i, \widehat{\boldsymbol{w}}_\tau) - \Delta(\tau; \boldsymbol{X}_i)]^3 - [\widetilde{\Delta}^{(-k)}(\tau; \boldsymbol{X}_i, \overline{\boldsymbol{w}}_\tau) - \Delta(\tau; \boldsymbol{X}_i)]^3 \right| \\
&\leq \frac{C}{n} \sum_{i=1}^{n} \left| \Delta^*(\tau; \boldsymbol{X}_i, \widehat{\boldsymbol{w}}_\tau) - \Delta(\tau; \boldsymbol{X}_i) + \Phi(\tau; \boldsymbol{X}_i, \widehat{\boldsymbol{w}}_\tau) \right|^3 \\
&\quad + \frac{C}{n} \sum_{i=1}^{n} \left| \Delta^*(\tau; \boldsymbol{X}_i, \overline{\boldsymbol{w}}_\tau) - \Delta(\tau; \boldsymbol{X}_i) + \Phi(\tau; \boldsymbol{X}_i, \overline{\boldsymbol{w}}_\tau) \right|^3 \\
&\equiv D_{53,1}(\widehat{\boldsymbol{w}}_\tau, \overline{\boldsymbol{w}}_\tau) + D_{53,2}(\widehat{\boldsymbol{w}}_\tau, \overline{\boldsymbol{w}}_\tau). \qquad \text{(S.90)}
\end{aligned}
$$

It is easy to see that

$$
\begin{aligned}
&D_{53,1}(\widehat{\boldsymbol{w}}_\tau, \overline{\boldsymbol{w}}_\tau) \\
&\leq \frac{C}{n} \sum_{i=1}^{n} \left\{ \left| \sum_{m=1}^{M_0} \widehat{w}_{\tau,m} [\Delta^{*(m)}(\tau; \boldsymbol{X}_i) - \Delta(\tau; \boldsymbol{X}_i)] \right|^3 + \left| \frac{1}{\sqrt{n}} \sum_{m=1}^{M} \widehat{w}_{\tau,m} \boldsymbol{X}_i^{(m)\top} \{l_1^{(m)}(\tau) - l_0^{(m)}(\tau)\} \right|^3 \right\} \\
&\leq C ||\widehat{\boldsymbol{w}}_\tau^U||^3 \left\{ \frac{1}{n} \sum_{i=1}^{n} \Big(\sum_{m=1}^{M_0} [\Delta^{*(m)}(\tau; \boldsymbol{X}_i) - \Delta(\tau; \boldsymbol{X}_i)]^2 \Big)^{3/2} \right\} \\
&\quad + C(\frac{1}{\sqrt{n}})^3 \max_{1 \leq m \leq M} \frac{1}{n} \sum_{i=1}^{n} \left| \boldsymbol{X}_i^{(m)\top} \{l_1^{(m)}(\tau) - l_0^{(m)}(\tau)\} \right|^3. \qquad \text{(S.91)}
\end{aligned}
$$

By Cauchy-Schwarz inequality, triangle inequality, and Assumption S.3 (iii) for fixed p, we

obtain

$$\begin{aligned}
&\mathsf{E}\left|\frac{1}{n}\sum_{i=1}^{n}\left|\boldsymbol{X}_i^{(m)\top}\{l_1^{(m)}(\tau)-l_0^{(m)}(\tau)\}\right|^3\right| \\
&\leq C\mathsf{E}||\boldsymbol{X}_i^{(m)}||^3 \\
&\leq C\Big(\mathsf{E}||\boldsymbol{X}_i^{(m)}||^4\Big)^{1/2}\Big(\mathsf{E}||\boldsymbol{X}_i^{(m)}||^2\Big)^{1/2} \\
&\leq C, \qquad \text{(S.92)}
\end{aligned}$$

and

$$\begin{aligned}
&\mathsf{E}\left|\frac{1}{n}\sum_{i=1}^{n}\Big(\sum_{m=1}^{M_0}[\Delta^{*(m)}(\tau;\boldsymbol{X}_i)-\Delta(\tau;\boldsymbol{X}_i)]^2\Big)^{3/2}\right| \\
&=\mathsf{E}\Big[\Big(\sum_{m=1}^{M_0}[\Delta^{*(m)}(\tau;\boldsymbol{X}_i)-\Delta(\tau;\boldsymbol{X}_i)]^2\Big)^{3/2}\Big] \\
&\leq\left(\mathsf{E}\left\{\Big[\sum_{m=1}^{M_0}[\Delta^{*(m)}(\tau;\boldsymbol{X}_i)-\Delta(\tau;\boldsymbol{X}_i)]^2\Big]^2\right\}\right)^{1/2}\left(\mathsf{E}\left\{\Big[\sum_{m=1}^{M_0}[\Delta^{*(m)}(\tau;\boldsymbol{X}_i)-\Delta(\tau;\boldsymbol{X}_i)]^2\Big]\right\}\right)^{1/2} \\
&\leq\left(M_0\sum_{m=1}^{M_0}\mathsf{E}\left\{[\Delta^{*(m)}(\tau;\boldsymbol{X}_i)-\Delta(\tau;\boldsymbol{X}_i)]^4\right\}\right)^{1/2}\left(\sum_{m=1}^{M_0}\mathsf{E}\left\{[\Delta^{*(m)}(\tau;\boldsymbol{X}_i)-\Delta(\tau;\boldsymbol{X}_i)]^2\right\}\right)^{1/2} \\
&\leq C. \qquad \text{(S.93)}
\end{aligned}$$

Combining (S.91), (S.92), and (S.93), we have

$$D_{53,1}(\widehat{\boldsymbol{w}}_\tau,\overline{\boldsymbol{w}}_\tau)\leq C||\widehat{\boldsymbol{w}}_\tau^U||^3+O_p(n^{-3/2}). \qquad \text{(S.94)}$$

Similarly, we obtain

$$D_{53,2}(\widehat{w}_\tau,\overline{\boldsymbol{w}}_\tau)=O_p(n^{-3/2}). \qquad \text{(S.95)}$$

Thus, by the triangle inequality and (S.87)–(S.90), (S.94), and (S.95), we have

$$|D_5(\widehat{\boldsymbol{w}}_\tau,\overline{\boldsymbol{w}}_\tau)|\leq C||\widehat{\boldsymbol{w}}_\tau^U||^2+O_p(\frac{1}{\sqrt{n}})||\widehat{\boldsymbol{w}}_\tau^U||+O_p(n^{-3/2}), \qquad \text{(S.96)}$$

where $||\widehat{\boldsymbol{w}}_\tau^U||\to 0$ as $n\to\infty$ under Assumptions 1–3, S.1–S.5, and S.8–S.9; see Theorem 4.

Finally, combining (S.77), (S.78), (S.83), (S.84), (S.85), (S.86), and (S.96), we have

$$C\|\widehat{\boldsymbol{w}}_\tau^U\|^2 + \|\widehat{\boldsymbol{w}}_\tau^U\| O_p(n^{-1/2}) + O_p(n^{-1/p}) + O_p(n^{-1/2}) + O_p(n^{-3/2}) \leq 0.$$

Equivalently,

$$(\|\widehat{\boldsymbol{w}}_\tau^U\| + O_p(n^{-1/2}))^2 - O_p(n^{-1}) + O_p(n^{-1/p}) + O_p(n^{-1/2}) + O_p(n^{-3/2}) \leq 0,$$

which implies

$$\|\widehat{\boldsymbol{w}}_\tau^U\| = O_p(n^{-1/(2p)} + n^{-1/4}). \tag{S.97}$$

This completes the proof of Theorem 5. □

B.6 Proof of Theorem 6

Proof of Theorem 6. Note that

$$\begin{aligned}
&|\widehat{\Delta}(\tau; \boldsymbol{X}_{n+1}, \widehat{\boldsymbol{w}}_\tau) - \Delta(\tau; \boldsymbol{X}_{n+1})|^2 \\
&\leq 2\Big(\Big| \sum_{m=1}^{M_0} \widehat{w}_{\tau,m} \{\widehat{\Delta}^{(m)}(\tau; \boldsymbol{X}_{n+1}) - \Delta(\tau; \boldsymbol{X}_{n+1})\} \Big|^2 \\
&\qquad + \Big| \sum_{m=M_0+1}^{M} \widehat{w}_{\tau,m} \{\widehat{\Delta}^{(m)}(\tau; \boldsymbol{X}_{n+1}) - \Delta(\tau; \boldsymbol{X}_{n+1})\} \Big|^2 \Big) \\
&\equiv 2(\Psi_1 + \Psi_2).
\end{aligned} \tag{S.98}$$

We first consider Ψ_2. It should be noted that for the correctly specified models $m \in \{M_0 + 1, \ldots, M\}$,

$$\Delta(\tau; \boldsymbol{X}_{n+1}) = \Delta^{*(m)}(\tau; \boldsymbol{X}_{n+1}). \tag{S.99}$$

Hence, for fixed p, by triangle inequality, Assumption S.3 (iii), (S.6), and (S.99), we have

$$\Psi_2 = \Big| \sum_{m=M_0+1}^{M} \widehat{w}_{\tau,m} \{\widehat{\Delta}^{(m)}(\tau; \boldsymbol{X}_{n+1}) - \Delta(\tau; \boldsymbol{X}_{n+1})\} \Big|^2$$

$$
\begin{aligned}
&= \Big| \sum_{m=M_0+1}^{M} \widehat{w}_{\tau,m} \{\widehat{\Delta}^{(m)}(\tau; \boldsymbol{X}_{n+1}) - \Delta^{*(m)}(\tau; \boldsymbol{X}_{n+1})\} \Big|^2 \\
&= \Big| \sum_{m=M_0+1}^{M} \widehat{w}_{\tau,m} \boldsymbol{X}_{n+1}^{(m)\top} \{\widehat{\boldsymbol{\beta}}_1^{(m)}(\tau) - \widehat{\boldsymbol{\beta}}_0^{(m)}(\tau) - \boldsymbol{\beta}_1^{*(m)}(\tau) + \boldsymbol{\beta}_1^{*(m)}(\tau)\} \Big|^2 \\
&\leq (M - M_0)^2 \max_{M_0+1 \leq m \leq M} ||\boldsymbol{X}_{n+1}^{(m)}||^2 \max_{M_0+1 \leq m \leq M} \|\widehat{\boldsymbol{\beta}}_1^{(m)}(\tau) - \widehat{\boldsymbol{\beta}}_0^{(m)}(\tau) - \boldsymbol{\beta}_1^{*(m)}(\tau) + \boldsymbol{\beta}_1^{*(m)}(\tau)\}\|^2 \\
&\leq C \max_{1 \leq m \leq M} ||\boldsymbol{X}_{n+1}^{(m)}||^2 \Big(\max_{1 \leq m \leq M} \|\widehat{\boldsymbol{\beta}}_1^{(m)}(\tau)) - \boldsymbol{\beta}_1^{*(m)}(\tau)\}\|^2 + \max_{1 \leq m \leq M} \|\widehat{\boldsymbol{\beta}}_0^{(m)}(\tau)) - \boldsymbol{\beta}_0^{*(m)}(\tau)\}\|^2 \Big) \\
&= O_p(\frac{1}{n}).
\end{aligned} \tag{S.100}
$$

Then, we focus on Ψ_1. Note that

$$
\begin{aligned}
\Psi_1 &= \Big| \sum_{m=1}^{M_0} \widehat{w}_{\tau,m} \{\widehat{\Delta}^{(m)}(\tau; \boldsymbol{X}_{n+1}) - \Delta(\tau; \boldsymbol{X}_{n+1})\} \Big|^2 \\
&\leq 2 \Big(\Big| \sum_{m=1}^{M_0} \widehat{w}_{\tau,m} \{\widehat{\Delta}^{(m)}(\tau; \boldsymbol{X}_{n+1}) - \Delta^{*(m)}(\tau; \boldsymbol{X}_{n+1})\} \Big|^2 \\
&\qquad + \Big| \sum_{m=1}^{M_0} \widehat{w}_{\tau,m} \{\Delta^{*(m)}(\tau; \boldsymbol{X}_{n+1}) - \Delta(\tau; \boldsymbol{X}_{n+1})\} \Big|^2 \Big) \\
&\equiv 2(\Psi_{11} + \Psi_{12}).
\end{aligned} \tag{S.101}
$$

Similar to (S.100), we can show that

$$
\Psi_{11} = O_p(\frac{1}{n}). \tag{S.102}
$$

By the triangle inequality and Assumption S.3(iii), for fixed p, we have, for any $m \in \{1, \ldots, M_0\}$,

$$
\begin{aligned}
&[\Delta^{*(m)}(\tau; \boldsymbol{X}_{n+1}) - \Delta(\tau; \boldsymbol{X}_{n+1})]^2 \\
&\leq 2 \left\{ [\Delta^{*(m)}(\tau; \boldsymbol{X}_{n+1})]^2 + [\Delta(\tau; \boldsymbol{X}_{n+1})]^2 \right\} \\
&\leq C \left\{ ||\boldsymbol{X}_{n+1}^{(m)}||^2 (||\boldsymbol{\beta}_1^{*(m)}(\tau)||^2 + ||\boldsymbol{\beta}_0^{*(m)}(\tau)||^2) + [Q_1(\tau; \boldsymbol{X}_{n+1})]^2 + [Q_0(\tau; \boldsymbol{X}_{n+1})]^2 \right\} \\
&= O_p(1).
\end{aligned} \tag{S.103}
$$

Hence, by Cauchy-Schwarz inequality, Assumption S.3 (iii) for fixed p, (S.97), and (S.103), we obtain

$$\begin{aligned}
\Psi_{12} &= \Big| \sum_{m=1}^{M_0} \widehat{w}_{\tau,m} \{\Delta^{*(m)}(\tau; \boldsymbol{X}_{n+1}) - \Delta(\tau; \boldsymbol{X}_{n+1})\} \Big|^2 \\
&\leq \Big(\sum_{m=1}^{M_0} \widehat{w}_{\tau,m}^2 \Big) \Big(\sum_{m=1}^{M_0} [\Delta^{*(m)}(\tau; \boldsymbol{X}_{n+1}) - \Delta(\tau; \boldsymbol{X}_{n+1})]^2 \Big) \\
&= O_p(||\widehat{\boldsymbol{w}}_\tau^U||^2), \\
&= O_p(n^{-1/p} + n^{-1/2}). \qquad \text{(S.104)}
\end{aligned}$$

Combining (S.98), (S.100), (S.101), (S.102), and (S.104), we have

$$\begin{aligned}
|\widehat{\Delta}(\tau; \boldsymbol{X}_{n+1}, \widehat{\boldsymbol{w}}_\tau) - \Delta(\tau; \boldsymbol{X}_{n+1})|^2 &= O_p(n^{-1} + n^{-1/p} + n^{-1/2}) \\
&= O_p(n^{-1/p} + n^{-1/2}). \qquad \text{(S.105)}
\end{aligned}$$

This completes the proof of Theorem 6. □

C Additional simulation studies

This section provides additional simulation results that are omitted from the main text for space reasons. Section C.1 presents the detailed calculation of R^2 for the four simulation designs, and Section C.2 reports the predictive performance for Scenarios 1, 2, and 4.

C.1 Calculation of R^2 in the simulation designs

Scenario 1: High-dimensional nested models. In Scenario 1, the data generating process of observed outcome is

$$\begin{aligned} Y_i = D_i & \left[\upsilon \left(1 + \sum_{j=1}^{100} \frac{1}{j} X_{i,j} \right) + \left(1 + \sum_{j=1}^{100} \frac{1}{2j} X_{i,j} \right) \epsilon_i(1) \right] \\ & + (1 - D_i) \left[\upsilon \left(1 + \sum_{j=1}^{100} \frac{1}{2j} X_{i,j} \right) + \left(1 + \sum_{j=1}^{100} \frac{1}{4j} X_{i,j} \right) \epsilon_i(0) \right], \ i = 1, \ldots, n. \end{aligned}$$

Therefore, R^2 is defined as

$$R^2 = (Var(Y_i) - Var(\epsilon_i))/Var(Y_i),$$

where $\epsilon_i = D_i \left(1 + \sum_{j=1}^{100} \frac{1}{2j} X_{i,j} \right) \epsilon_i(1) + (1 - D_i) \left(1 + \sum_{j=1}^{100} \frac{1}{4j} X_{i,j} \right) \epsilon_i(0)$.

Scenario 2: Low-dimensional nested models. In Scenario 2, the generation of outcome is defined as

$$\begin{aligned} Y_i = D_i & \left[\upsilon(1 + \boldsymbol{X}_i^\top \boldsymbol{\gamma}_1) + (1 + \boldsymbol{X}_i^\top \boldsymbol{\iota}_1) \epsilon_i(1) \right] \\ & + (1 - D_i) \left[\upsilon(1 + \boldsymbol{X}_i^\top \boldsymbol{\gamma}_0) + (1 + \boldsymbol{X}_i^\top \boldsymbol{\iota}_0) \epsilon_i(0) \right], \ i = 1, \ldots, n, \end{aligned}$$

where $\boldsymbol{\gamma}_1 = (5, -5, 3, -1, 1)^\top$, $\boldsymbol{\iota}_1 = (0.5, 0.4, 0.3, 0.2, 0.1)^\top$, $\boldsymbol{\gamma}_0 = \frac{1}{2}\boldsymbol{\gamma}_1$, and $\boldsymbol{\iota}_0 = \frac{1}{2}\boldsymbol{\iota}_1$. The definition of R^2 is the same as in Scenario 1, $R^2 = (Var(Y_i) - Var(\epsilon_i))/Var(Y_i)$, where $\epsilon_i = D_i(1 + \boldsymbol{X}_i^\top \boldsymbol{\iota}_1)\epsilon_i(1) + (1 - D_i)(1 + \boldsymbol{X}_i^\top \boldsymbol{\iota}_0)\epsilon_i(0)$.

Scenario 3: Low-dimensional non-nested models. The definition of R^2 of simulation Scenario 3 is the same as in Scenario 2.

Scenario 4: Nonlinear outcome structure In Scenario 4, the observed outcome is generated as

$$Y_i = D_i \left[\upsilon \left(1 + \sum_{j=1}^{4} X_{i,j} + 0.5 X_{i,5}^2 \right) + \left(0.1 + \sum_{j=1}^{5} 0.1 X_{i,j} \right) \epsilon_i(1) \right] + (1 - D_i) \left[\upsilon \left(0.5 + \sum_{j=1}^{4} 0.5 X_{i,j} + 0.25 X_{i,5}^2 \right) + \left(0.05 + \sum_{j=1}^{5} 0.05 X_{i,j} \right) \epsilon_i(0) \right], \ i = 1, \ldots, n.$$

Therefore, R^2 is defined as

$$R^2 = (Var(Y_i) - Var(\epsilon_i))/Var(Y_i),$$

where $\epsilon_i = D_i \left(0.1 + \sum_{j=1}^{5} 0.1 X_{i,j} \right) \epsilon_i(1) + (1 - D_i) \left(0.05 + \sum_{j=1}^{5} 0.05 X_{i,j} \right) \epsilon_i(0)$.

C.2 Additional figures for Simulation Scenarios

Figures S1, S2, and S3 present the simulation results for Scenarios 1, 2, and 4, respectively. Across most configurations, MAQTE attains the lowest predictive risk and compares favorably with the competing methods overall. Taken together, the simulation results provide consistent support for MAQTE, with particularly strong performance at lower and upper quantiles and in smaller samples.

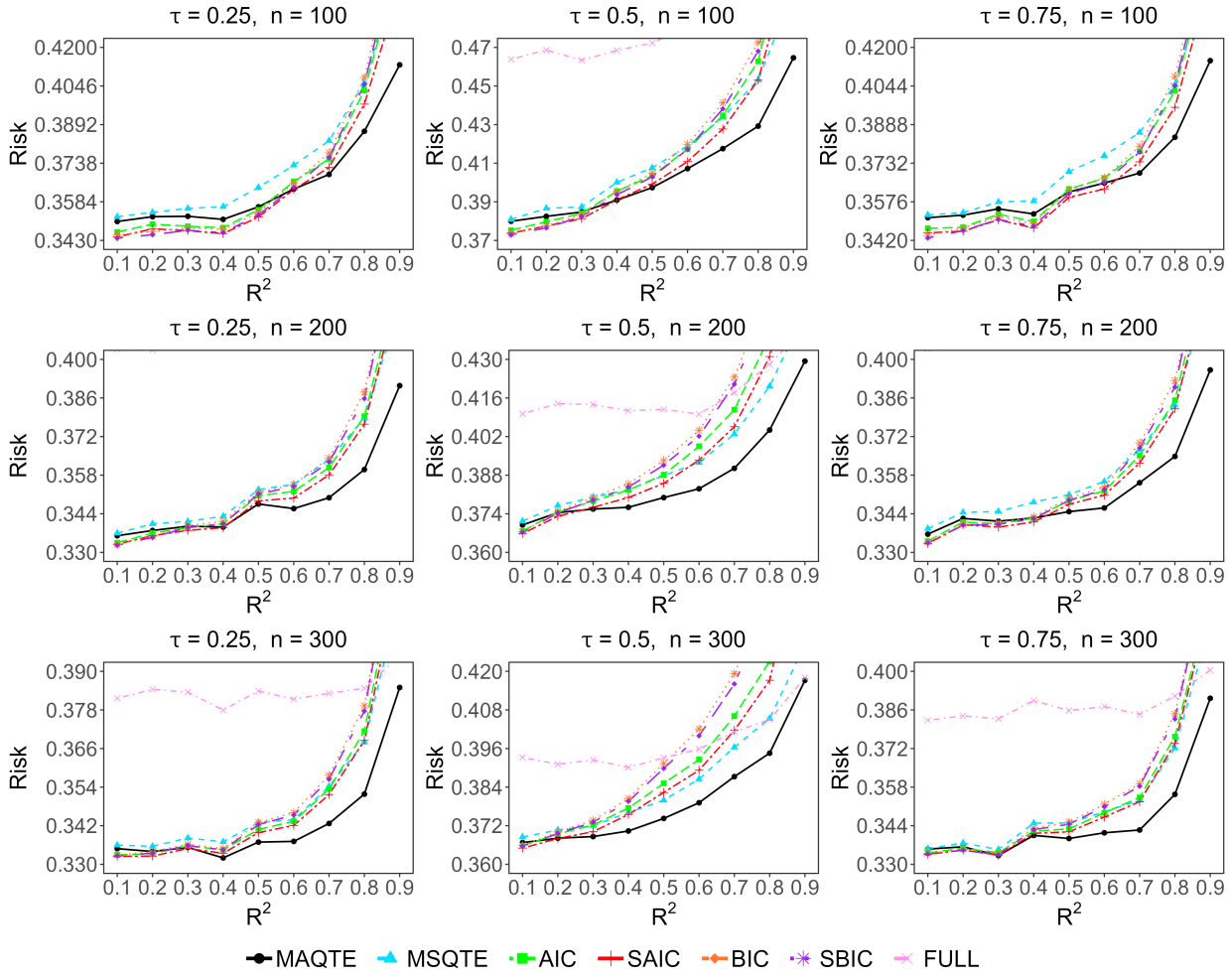


Figure S1: Simulation Results for Scenario 1.

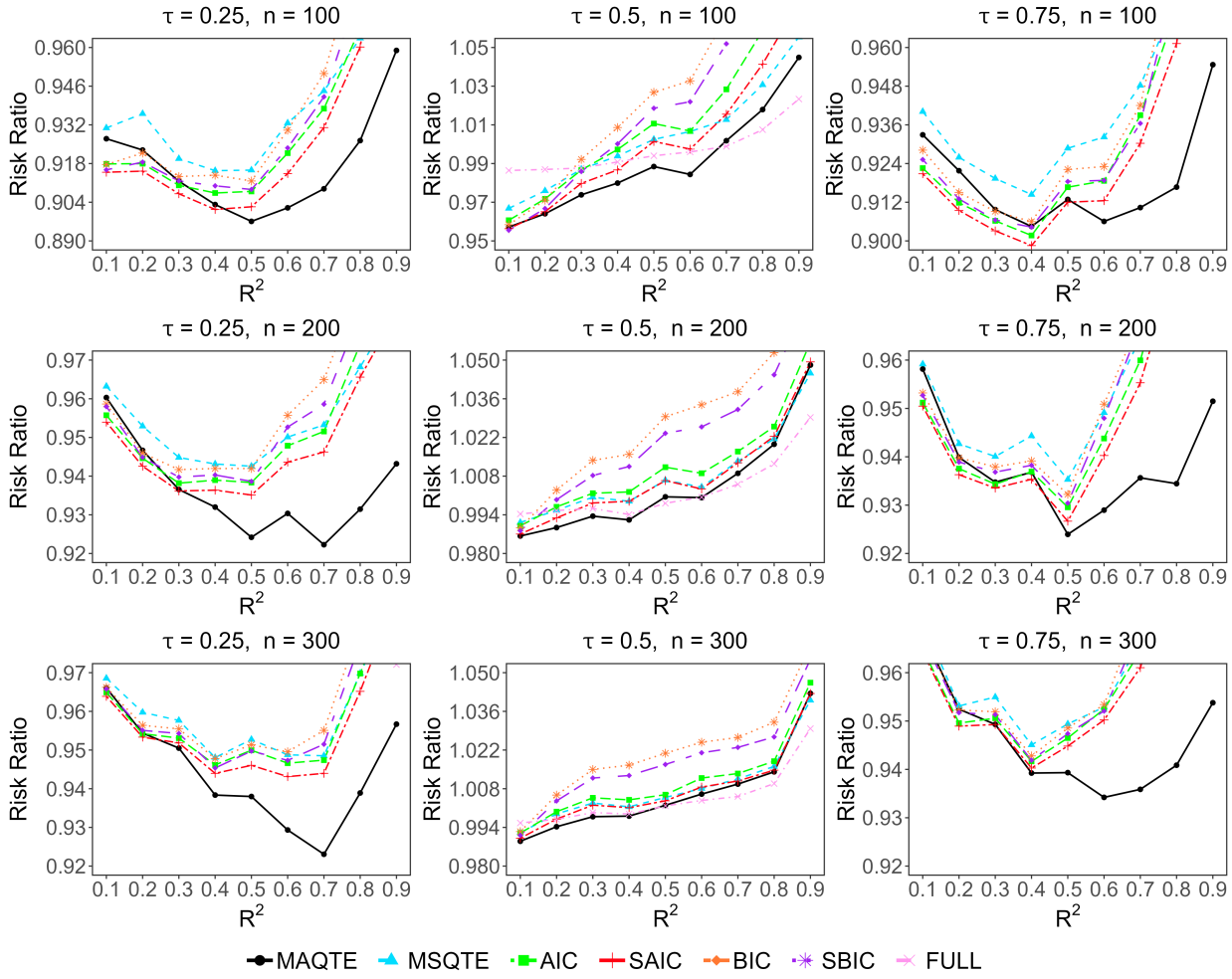


Figure S2: Simulation Results for Scenario 2.

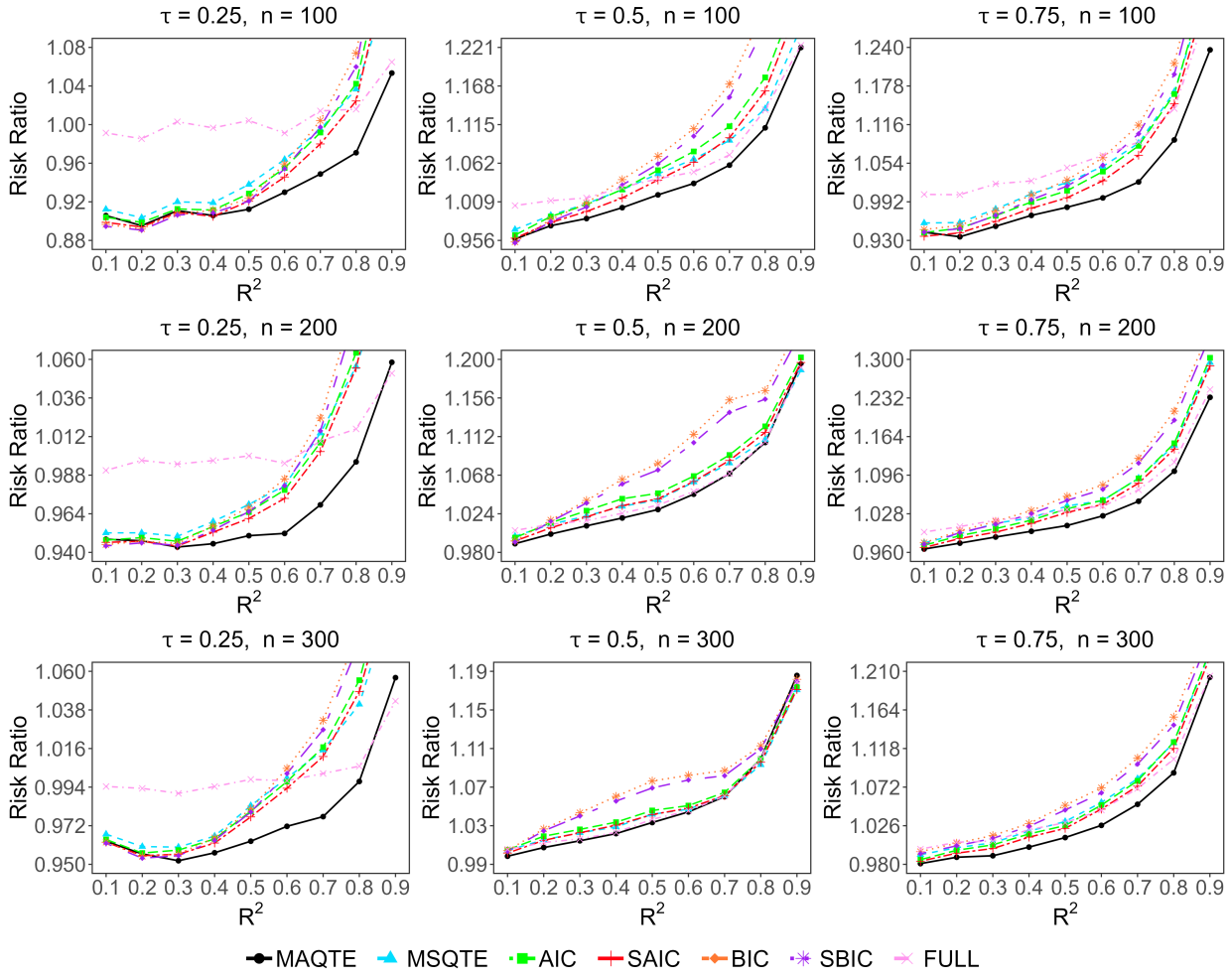


Figure S3: Simulation Results for Scenario 4.

D Additional results for the real data analysis

Table S1 reports the correlations between each covariate and the outcome, computed separately within the treatment and control groups, as well as the sum of their absolute values across the two groups. To exclude clearly irrelevant variables, we retain only the covariates for which this combined measure exceeds 1.5.

Table S2 reports descriptive statistics for all covariates. For continuous variables, we report the mean, variance, median, the first and third quartiles, and the minimum and maximum values. For categorical variables, we report the proportion of observations in each category.

Tables S3–S4 report the prediction error results for $n_{train} = 600$ and $n_{train} = 900$, respectively. The results show that MAQTE consistently performs at or near the top across specifications, achieving the lowest prediction error in most cases, and provides a favorable balance between predictive accuracy and stability in this dataset.

Figures S4–S12 present scatterplots and boxplots of the predicted CQTEs for all individuals in the original sample, grouped by each covariate category, based on the MAQTE estimates obtained from the full sample. The patterns are broadly similar to those shown in Figure 5 in the main text. Gender and childhood welfare exposure appear to be most useful for identifying subgroups with systematically stronger expected gains overall. By contrast, age and education appear to be more informative for understanding where in the earnings distribution the intervention is likely to be most effective.

Table S1: Correlation screening results

Treatment Group		Control Group		Combined (Sum of Abs)	
Variable	Correlation	Variable	Correlation	Variable	Correlation
age	0.1954	female	-0.1888	female	0.3676
female	-0.1788	educ	0.1556	age	0.3157
educ	0.1482	welfarechild	-0.1402	educ	0.3038
white	0.1477	age	0.1204	welfarechild	0.2622
hhsize	-0.1225	haschild	-0.1053	white	0.2472
welfarechild	-0.1220	white	0.0995	hhsize	0.2015
hsdegree	0.1090	smoke	-0.0952	hsdegree	0.1828
black	-0.0914	hhsize	-0.0789	black	0.1584
health	-0.0737	health	-0.0776	health	0.1513
cohabmarried	0.0719	alcohol	-0.0775	smoke	0.1418
geddegree	0.0653	hsdegree	0.0738	haschild	0.1315
everwkd	-0.0530	black	-0.0670	geddegree	0.1241
hispanic	-0.0513	educdad	0.0640	cohabmarried	0.1104
mwearn	0.0479	geddegree	0.0588	mwearn	0.1047
smoke	-0.0466	mwearn	0.0568	educdad	0.1040
educmum	0.0424	educmum	0.0472	educmum	0.0896
educdad	0.0400	cohabmarried	0.0386	alcohol	0.0870
haschild	0.0263	everwkd	-0.0290	everwkd	0.0820
english	-0.0212	english	-0.0095	hispanic	0.0544
alcohol	0.0095	hispanic	-0.0031	english	0.0307

Table S2: Descriptive statistics of variables

Continuous Variables							
Variable	**Min**	Q_{25}	**Median**	Q_{75}	**Max**	**Mean**	**Variance**
Age	16.000	17.000	18.000	20.000	24.000	18.528	4.601
Years of education	1.000	9.000	10.000	11.000	15.000	10.187	2.180
Household size	0.000	2.000	3.000	4.000	15.000	3.248	3.608
Categorical Variables							
Variable	**0**	**1**	**2**	**3**	**4**		
Indicator for Female	66.7%	33.3%					
Welfare receipt during childhood		56.1%	22.4%	7.5%	13.9%		
Indicator for White	48.5%	51.5%					
Indicator for High School Degree	80.6%	19.4%					
Indicator for Black	72.3%	27.7%					
Health		42.6%	45.1%	11.8%	0.4%		

Table S3: Out of sample prediction errors for the JC dataset ($n_{train} = 600$)

τ	R^2	Methods						
		MAQTE	SAIC	SBIC	MSQTE	AIC	BIC	FULL
	0.3	0.3549	**0.3548**	0.3553	0.3559	0.3550	0.3555	0.3778
0.25	0.5	**0.2347**	0.2350	0.2357	0.2363	0.2351	0.2359	0.2532
	0.7	**0.1557**	0.1574	0.1586	0.1584	0.1577	0.1588	0.1684
	0.3	**0.2656**	0.2669	0.2676	0.2676	0.2674	0.2679	0.2686
0.50	0.5	**0.1760**	0.1770	0.1786	0.1776	0.1773	0.1790	0.1773
	0.7	**0.1175**	0.1182	0.1199	0.1184	0.1184	0.1202	0.1176
	0.3	0.2078	**0.2069**	0.2074	0.2087	0.2070	0.2076	0.2126
0.75	0.5	0.1366	**0.1362**	0.1371	0.1373	0.1363	0.1371	0.1398
	0.7	**0.0889**	0.0892	0.0897	0.0897	0.0894	0.0899	0.0910

Table S4: Out of sample prediction errors for the JC dataset ($n_{train} = 900$)

τ	R^2	Methods						
		MAQTE	SAIC	SBIC	MSQTE	AIC	BIC	FULL
	0.3	0.3570	**0.3567**	0.3569	0.3575	0.3569	0.3570	0.3787
0.25	0.5	**0.2345**	0.2353	0.2357	0.2360	0.2354	0.2357	0.2495
	0.7	**0.1557**	0.1579	0.1585	0.1583	0.1581	0.1587	0.1669
	0.3	**0.2636**	0.2647	0.2657	0.2651	0.2650	0.2660	0.2654
0.50	0.5	**0.1745**	0.1752	0.1767	0.1754	0.1754	0.1771	0.1750
	0.7	0.1173	0.1177	0.1185	0.1178	0.1179	0.1187	**0.1172**
	0.3	0.2075	**0.2067**	0.2073	0.2080	0.2068	0.2074	0.2106
0.75	0.5	0.1356	**0.1355**	0.1359	0.1362	0.1356	0.1360	0.1379
	0.7	**0.0881**	0.0885	0.0889	0.0888	0.0885	0.0890	0.0896

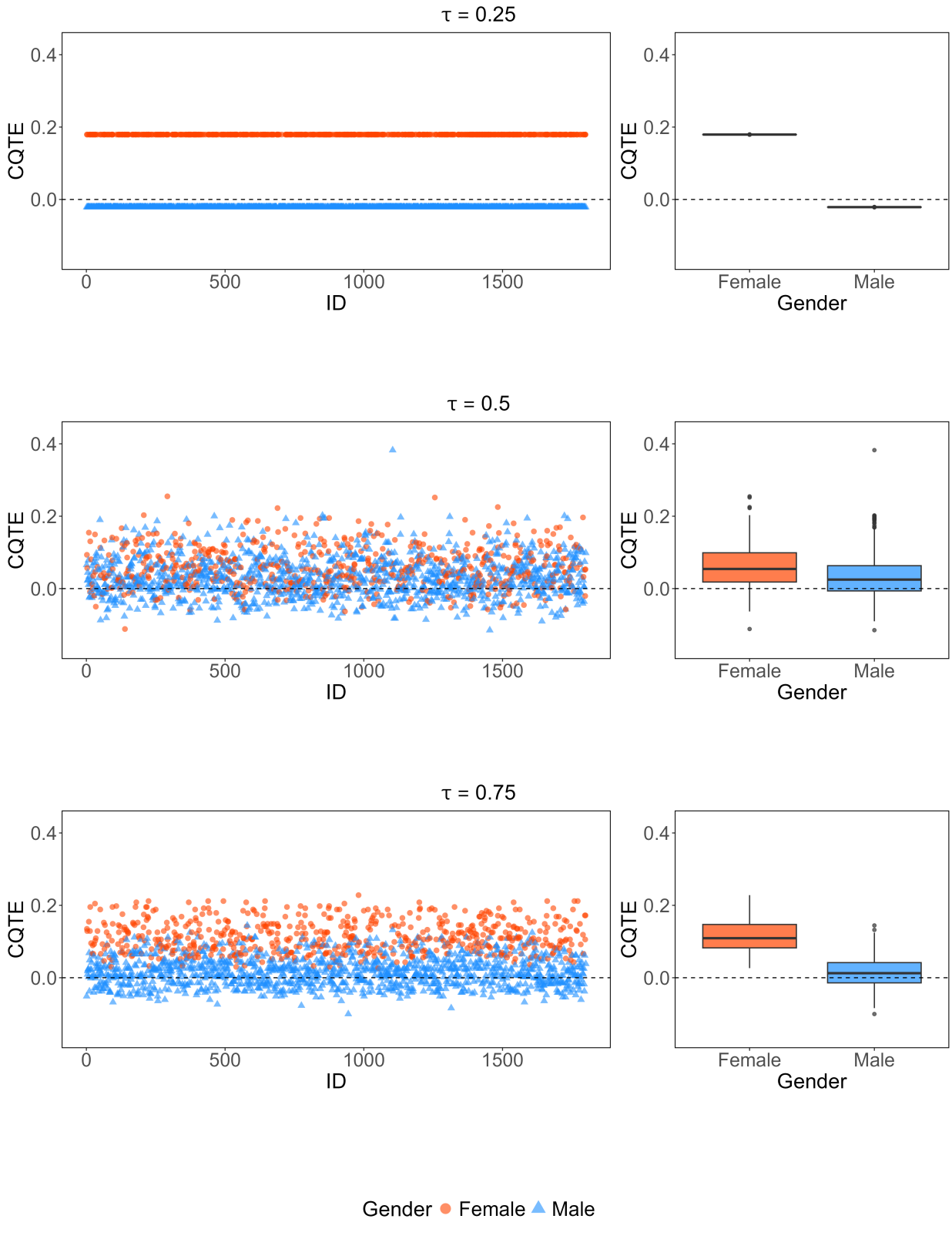


Figure S4: Model averaging estimators of CQTE by Gender (Female vs. Male).

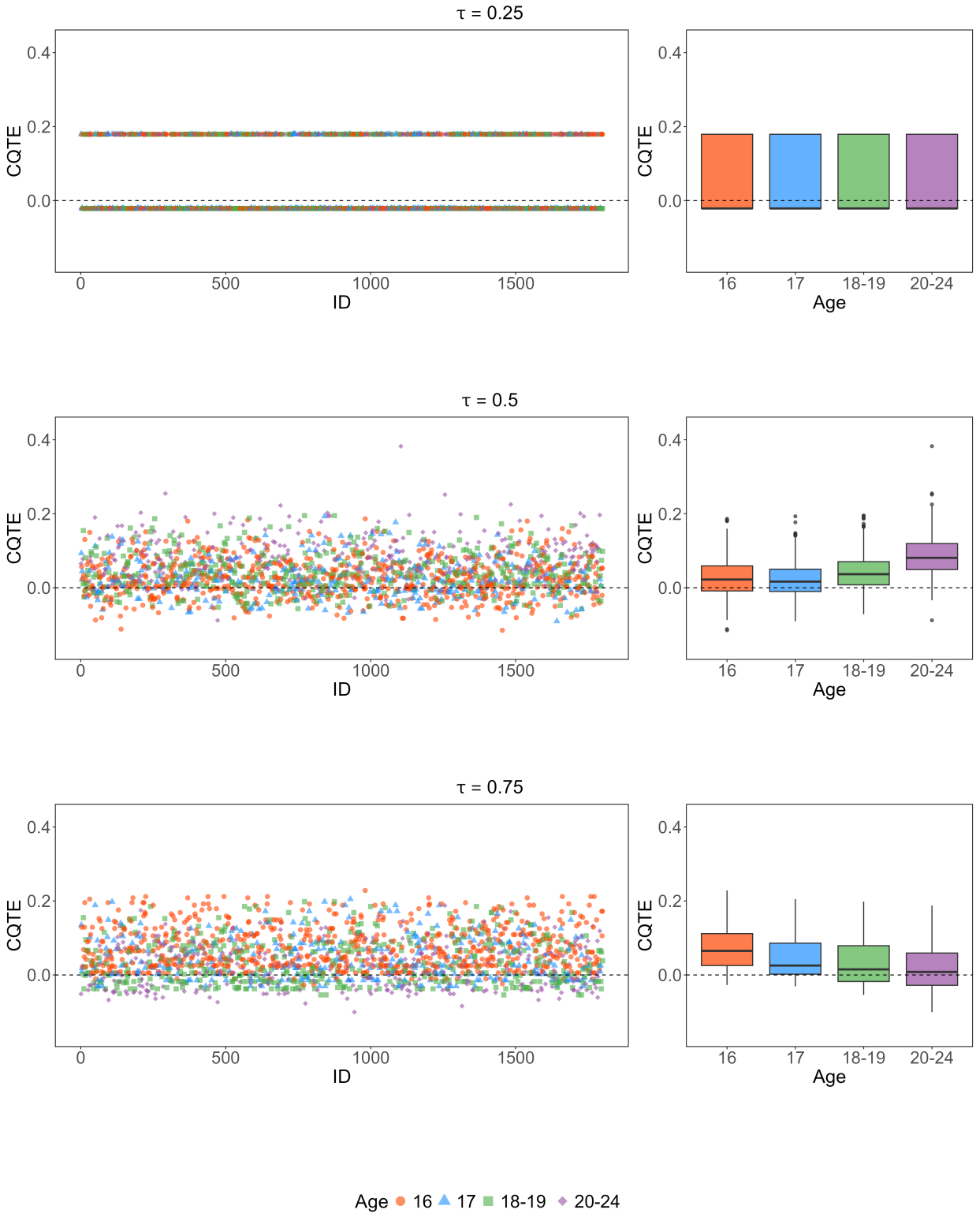


Figure S5: Model averaging estimators of CQTE by Age.

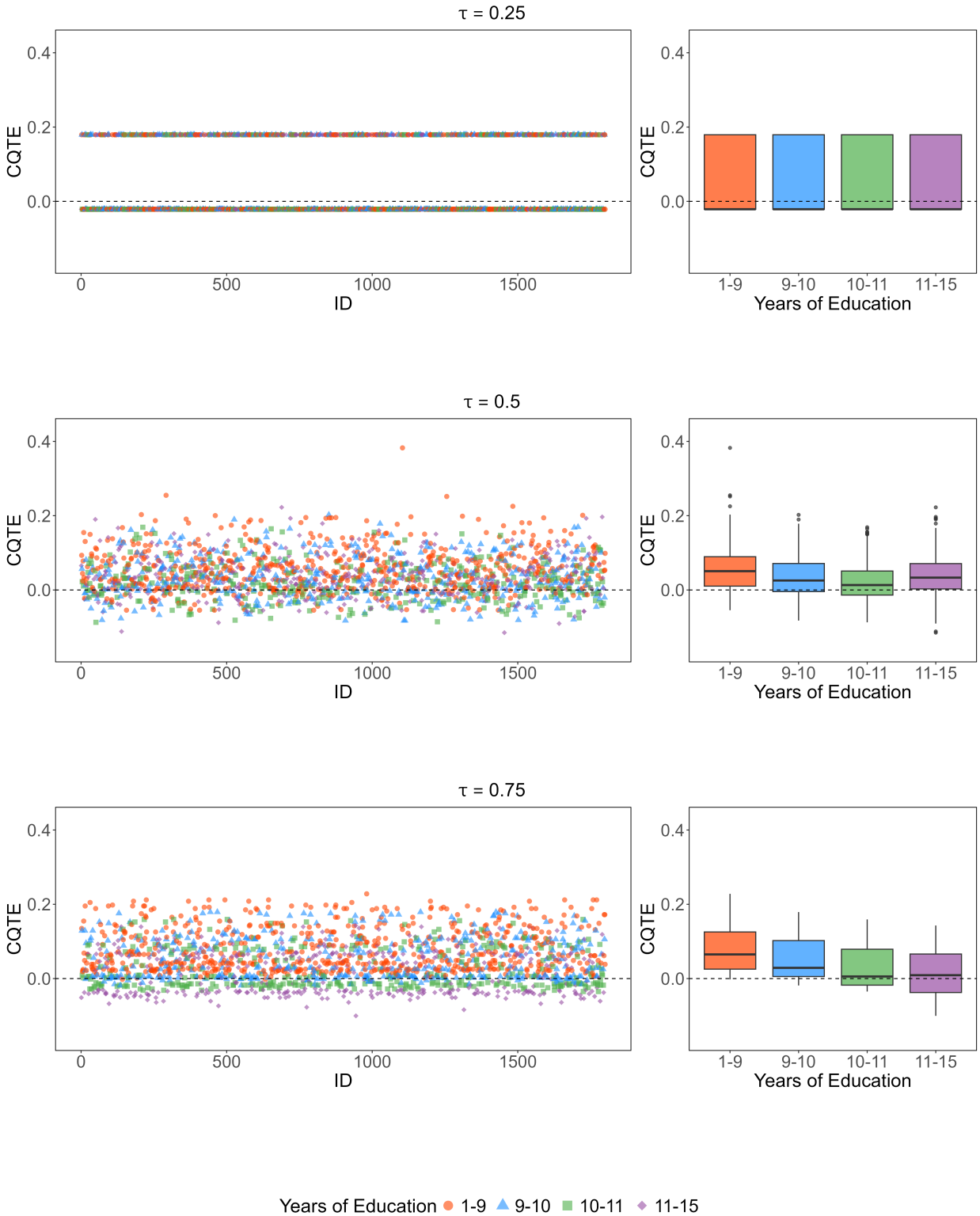


Figure S6: Model averaging estimators of CQTE by Years of Education.

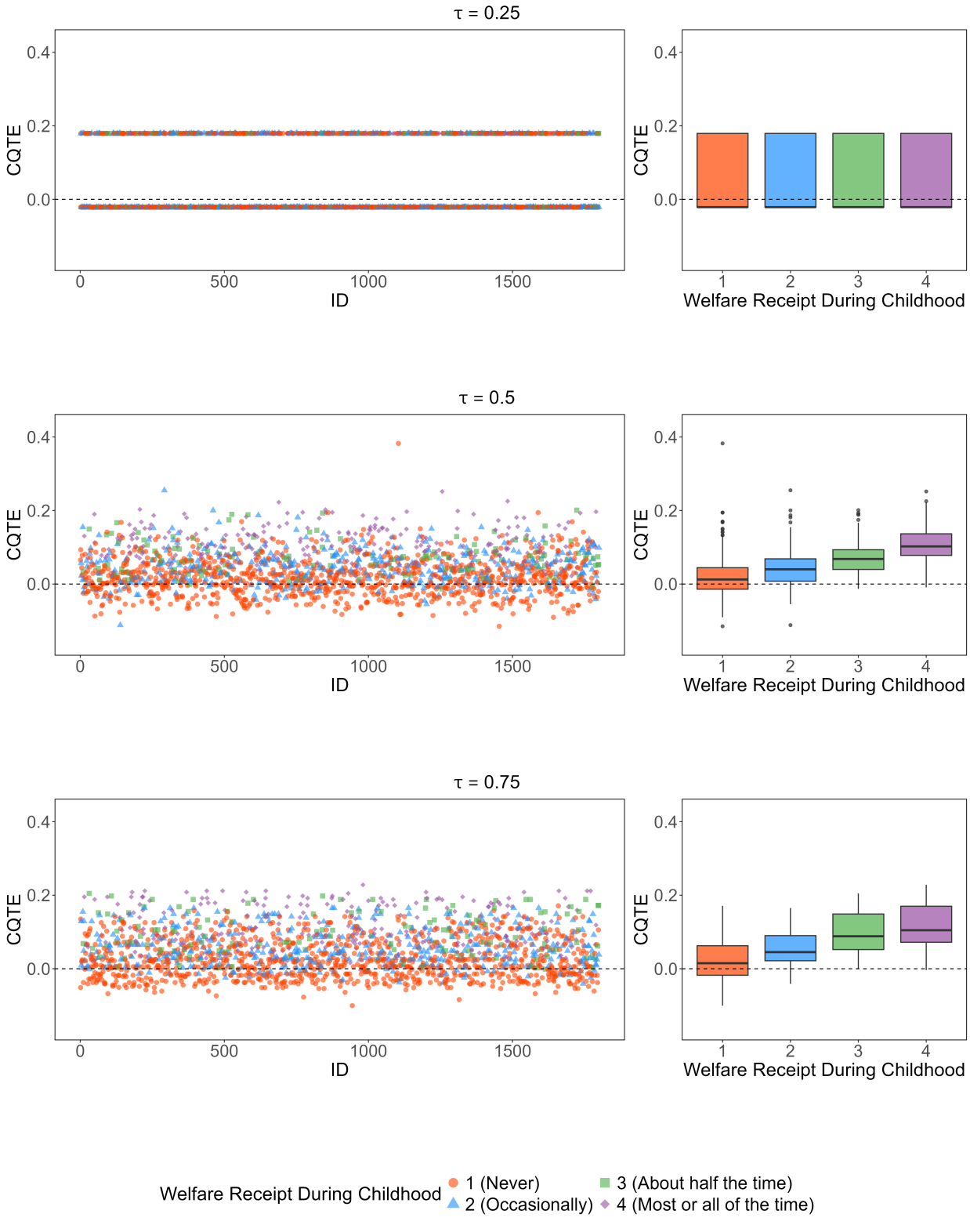


Figure S7: Model averaging estimators of CQTE by Welfare receipt during childhood.

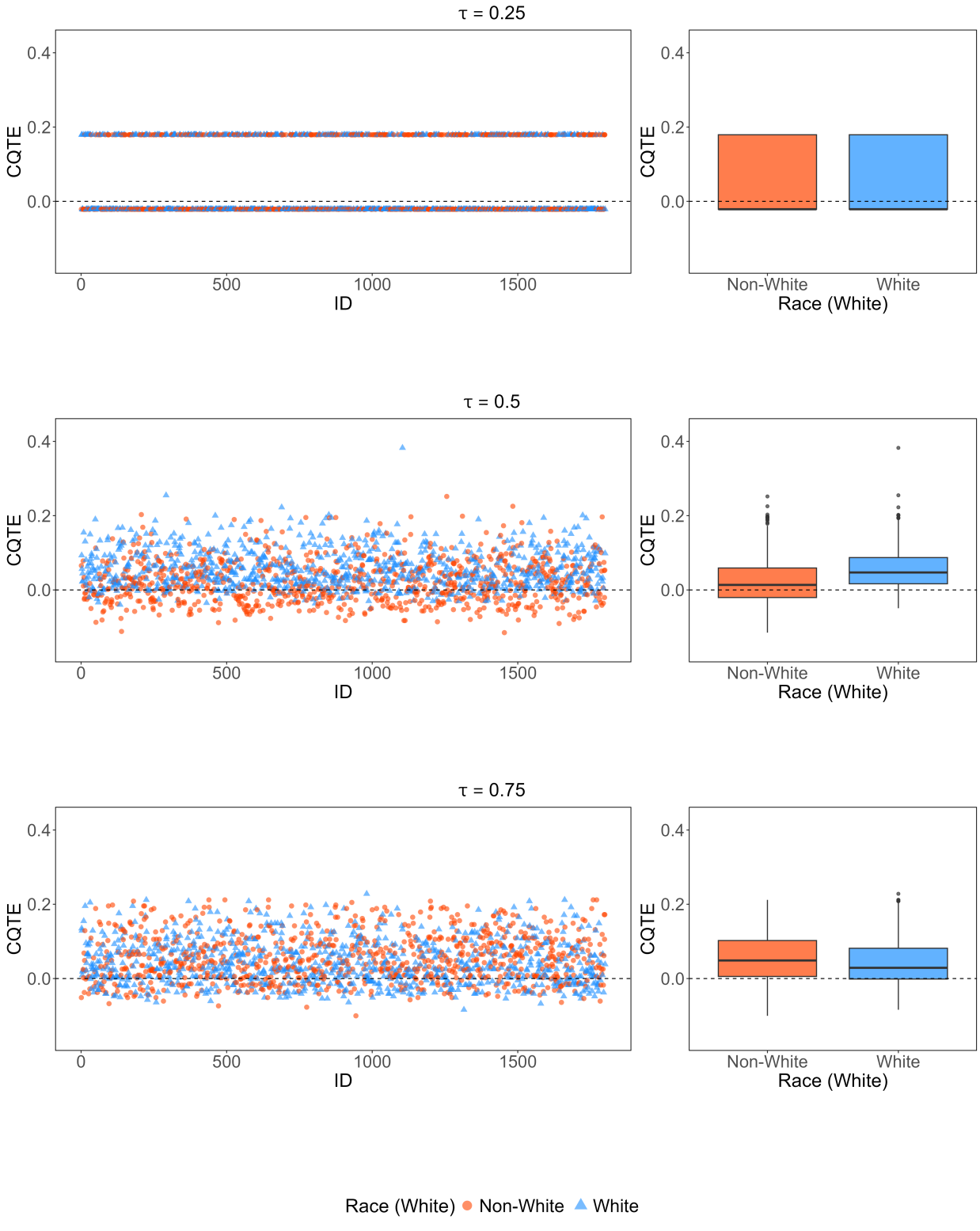


Figure S8: Model averaging estimators of CQTE by Race (White vs. Non-White).

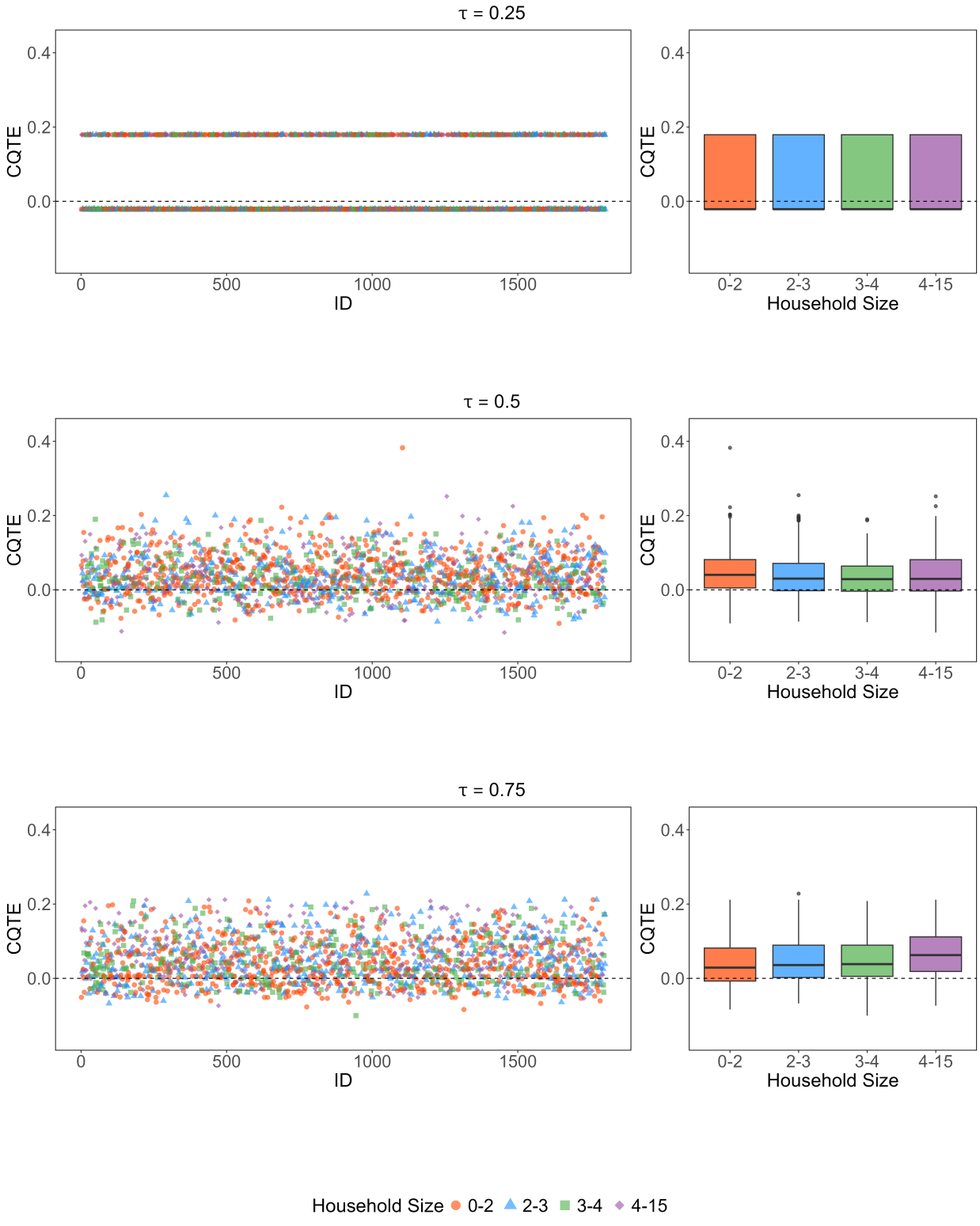


Figure S9: Model averaging estimators of CQTE by Household size.

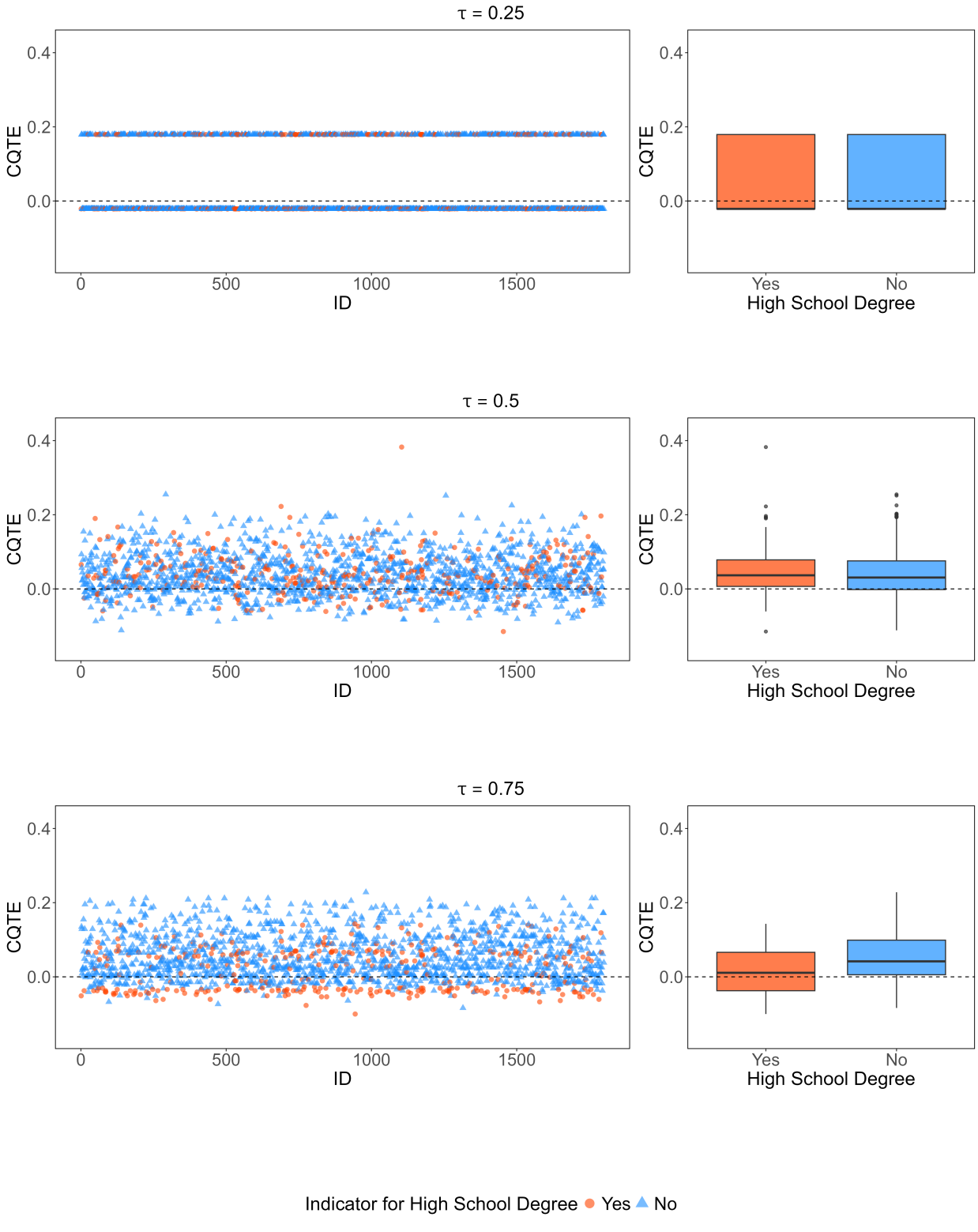


Figure S10: Model averaging estimators of CQTE by High School Degree (Yes vs. No).

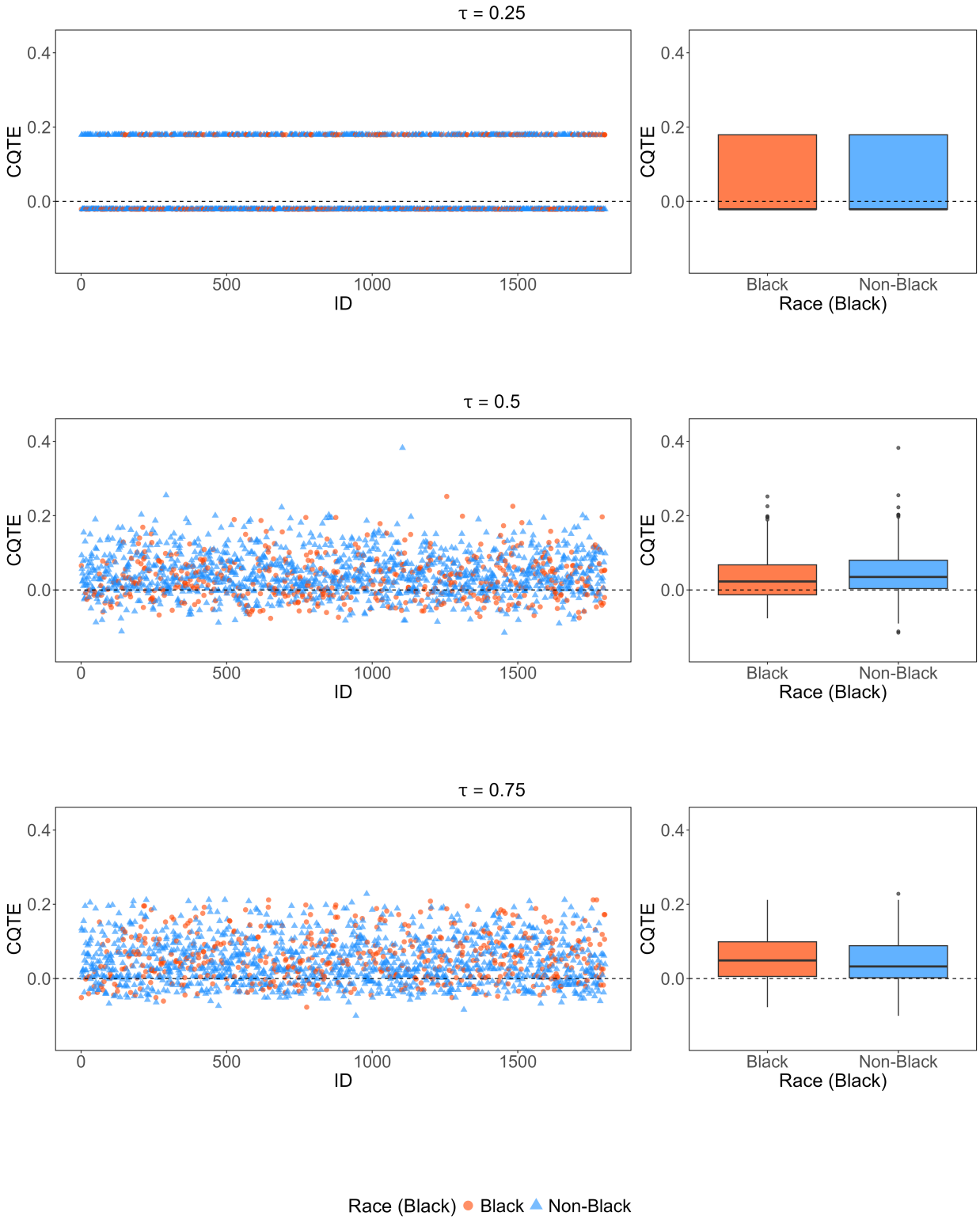


Figure S11: Model averaging estimators of CQTE by Race (Black vs. Non-Black).

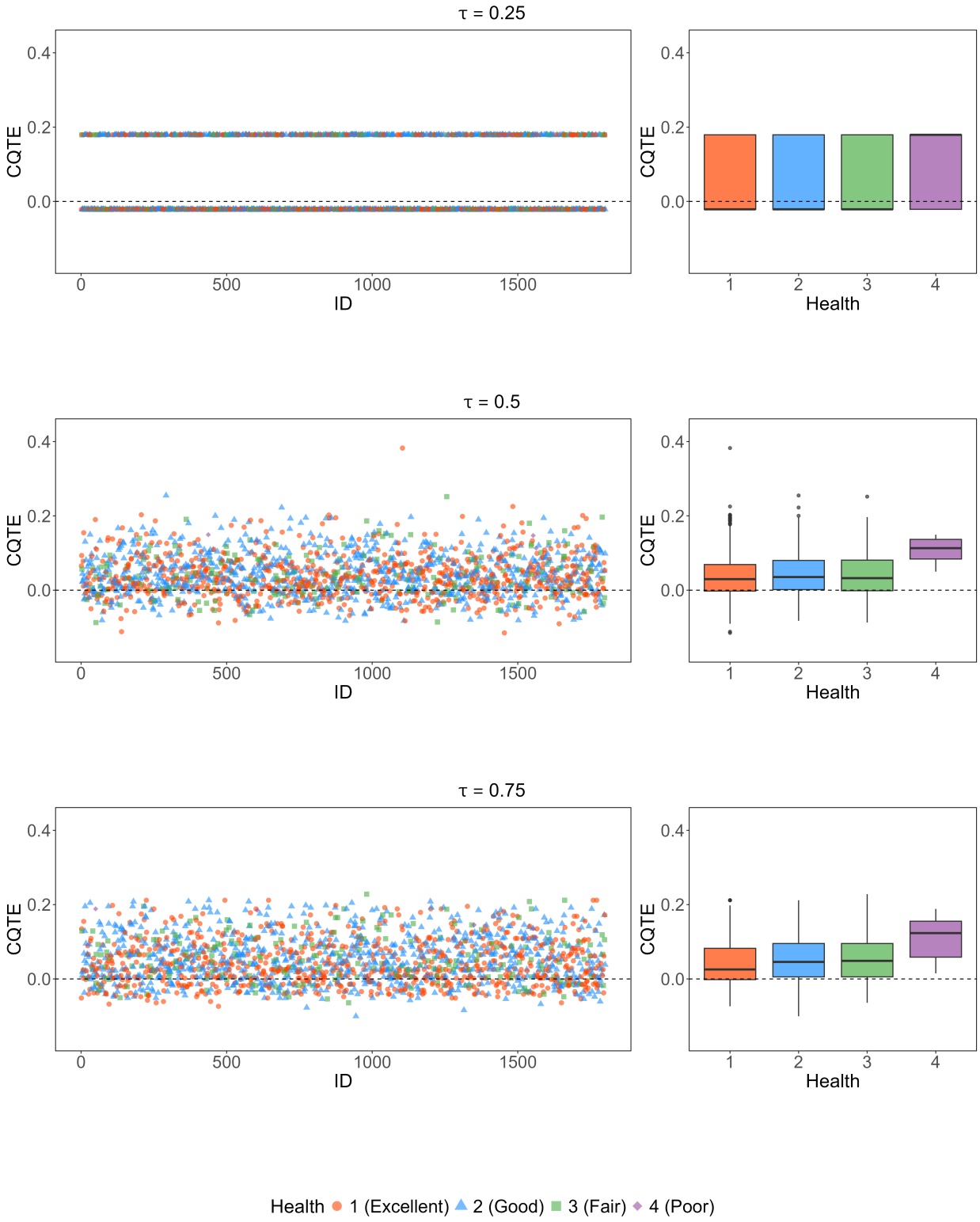


Figure S12: Model averaging estimators of CQTE by Health.

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